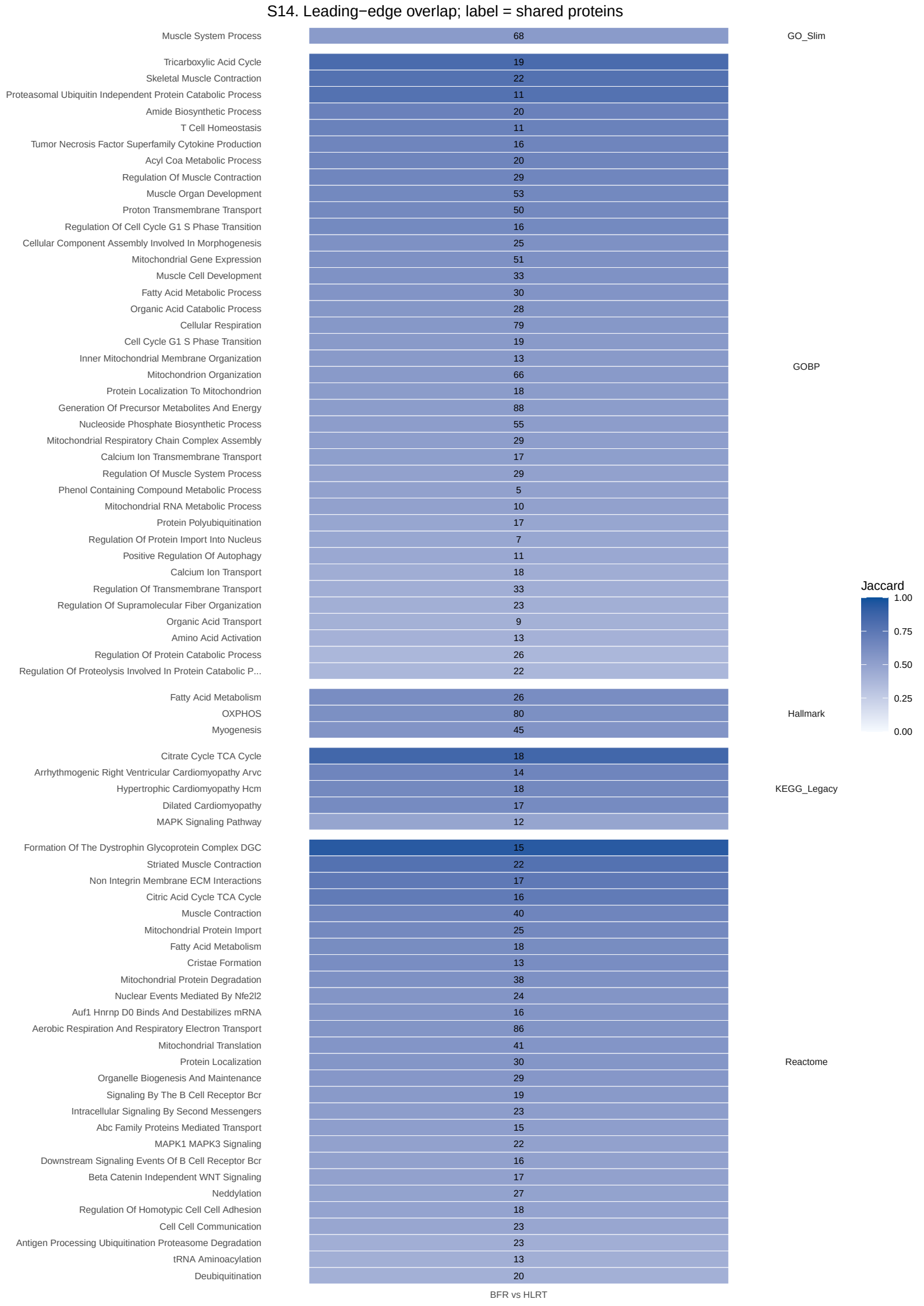
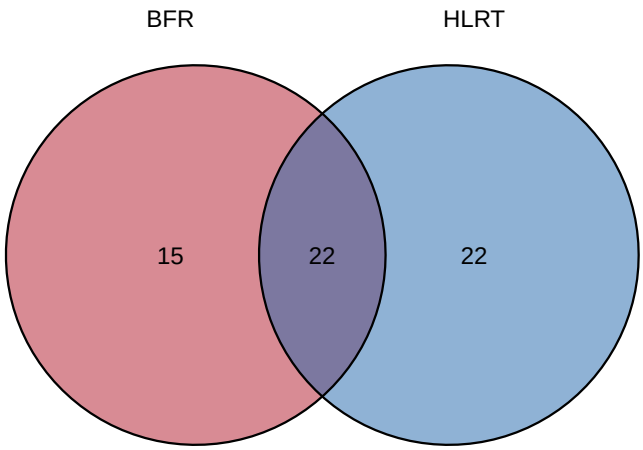

S14. Leading−edge overlap; label = shared proteins
Muscle System Process
68
GO_Slim
Tricarboxylic Acid Cycle
19
Skeletal Muscle Contraction
22
Proteasomal Ubiquitin Independent Protein Catabolic Process
11
Amide Biosynthetic Process
20
T Cell Homeostasis
11
Tumor Necrosis Factor Superfamily Cytokine Production
16
Acyl Coa Metabolic Process
20
Regulation Of Muscle Contraction
29
Muscle Organ Development
53
Proton Transmembrane Transport
50
Regulation Of Cell Cycle G1 S Phase Transition
16
Cellular Component Assembly Involved In Morphogenesis
25
Mitochondrial Gene Expression
51
Muscle Cell Development
33
Fatty Acid Metabolic Process
30
Organic Acid Catabolic Process
28
Cellular Respiration
79
Cell Cycle G1 S Phase Transition
19
Inner Mitochondrial Membrane Organization
13
GOBP
Mitochondrion Organization
66
Protein Localization To Mitochondrion
18
Generation Of Precursor Metabolites And Energy
88
Nucleoside Phosphate Biosynthetic Process
55
Mitochondrial Respiratory Chain Complex Assembly
29
Calcium Ion Transmembrane Transport
17
Regulation Of Muscle System Process
29
Phenol Containing Compound Metabolic Process
5
Mitochondrial RNA Metabolic Process
10
Protein Polyubiquitination
17
Regulation Of Protein Import Into Nucleus
7
Positive Regulation Of Autophagy
11
Calcium Ion Transport
18
Regulation Of Transmembrane Transport
33
Regulation Of Supramolecular Fiber Organization
23
Organic Acid Transport
9
Amino Acid Activation
13
Regulation Of Protein Catabolic Process
26
Regulation Of Proteolysis Involved In Protein Catabolic P...
22
Jaccard
1.00
0.75
0.50
0.25
0.00
Fatty Acid Metabolism
26
OXPHOS
80
Hallmark
Myogenesis
45
Citrate Cycle TCA Cycle
18
Arrhythmogenic Right Ventricular Cardiomyopathy Arvc
14
Hypertrophic Cardiomyopathy Hcm
18
KEGG_Legacy
Dilated Cardiomyopathy
17
MAPK Signaling Pathway
12
Formation Of The Dystrophin Glycoprotein Complex DGC
15
Striated Muscle Contraction
22
Non Integrin Membrane ECM Interactions
17
Citric Acid Cycle TCA Cycle
16
Muscle Contraction
40
Mitochondrial Protein Import
25
Fatty Acid Metabolism
18
Cristae Formation
13
Mitochondrial Protein Degradation
38
Nuclear Events Mediated By Nfe2l2
24
Auf1 Hnrnp D0 Binds And Destabilizes mRNA
16
Aerobic Respiration And Respiratory Electron Transport
86
Mitochondrial Translation
41
Protein Localization
30
Reactome
Organelle Biogenesis And Maintenance
29
Signaling By The B Cell Receptor Bcr
19
Intracellular Signaling By Second Messengers
23
Abc Family Proteins Mediated Transport
15
MAPK1 MAPK3 Signaling
22
Downstream Signaling Events Of B Cell Receptor Bcr
16
Beta Catenin Independent WNT Signaling
17
Neddylation
27
Regulation Of Homotypic Cell Cell Adhesion
18
Cell Cell Communication
23
Antigen Processing Ubiquitination Proteasome Degradation
23
tRNA Aminoacylation
13
Deubiquitination
20
BFR vs HLRT

S15. Leading edges by leg

Regulation Of Proteolysis Involved In Protein Catabolic Process

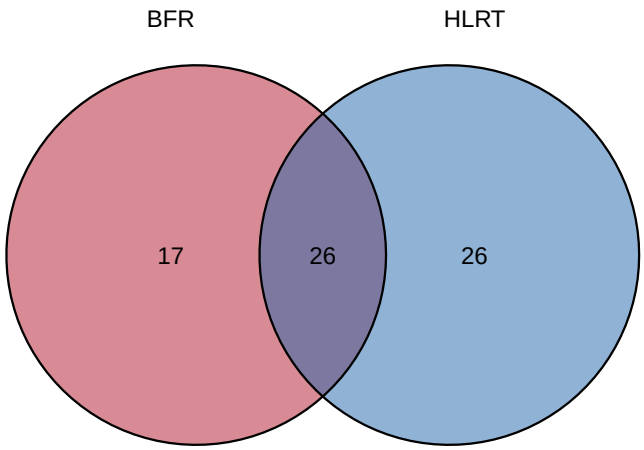
BFR: NES −2.00, padj 5.4e−05
HLRT: NES −1.79, padj 0.002
shared 22, expected 14.7



Shared: CDK5RAP3, DNAJB2, GABARAPL2, GPX1, PARK7, PRKN, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, PSMD10, RAD23A, UBQLN1

Regulation Of Protein Catabolic Process

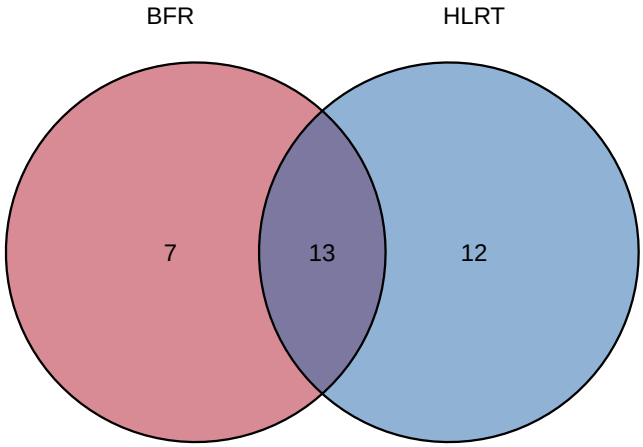
BFR: NES −1.81, padj 0.00028
HLRT: NES −1.56, padj 0.019
shared 26, expected 15.4



Shared: ASB11, CDK5RAP3, CDKN1B, DNAJB2, GABARAPL2, GPX1, PARK7, PRKN, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, PSMD10, RAD23A, RDX, UBQLN1, VPS28

Amino Acid Activation

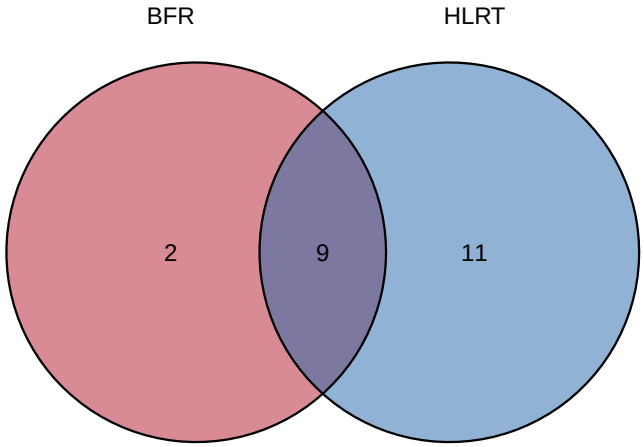
BFR: NES 2.31, padj 3.1e−05
HLRT: NES 2.22, padj 0.00015
shared 13, expected 12.8



Shared: DARS1, DARS2, HARS1, IARS1, IARS2, LARS1, LARS2, NARS1, QARS1, SARS1, SARS2, TARS1, TARS3

Organic Acid Transport

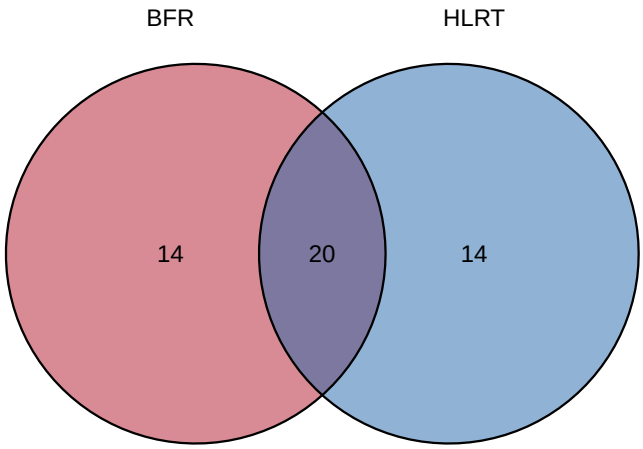
BFR: NES 1.80, padj 0.0067
HLRT: NES 1.70, padj 0.026
shared 9, expected 4.7



Shared: ACSL1, AKT2, GOT2, MPC1, RBP1, SFXN1, SLC16A3, SLC25A11, SLC25A12

Deubiquitination

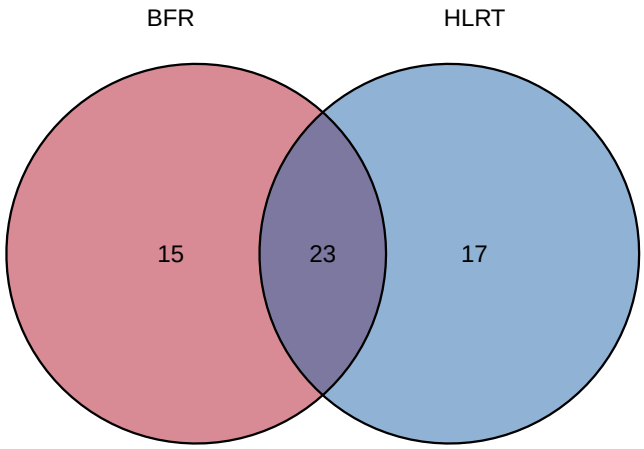
BFR: NES −1.86, padj 0.00067
HLRT: NES −1.59, padj 0.027
shared 20, expected 13.9



Shared: OTUB1, PRKN, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, RAD23A, UCHL3, USP14, USP24

Regulation Of Supramolecular Fiber Organization

BFR: NES −1.79, padj 0.00074
HLRT: NES −1.54, padj 0.02
shared 23, expected 12.4

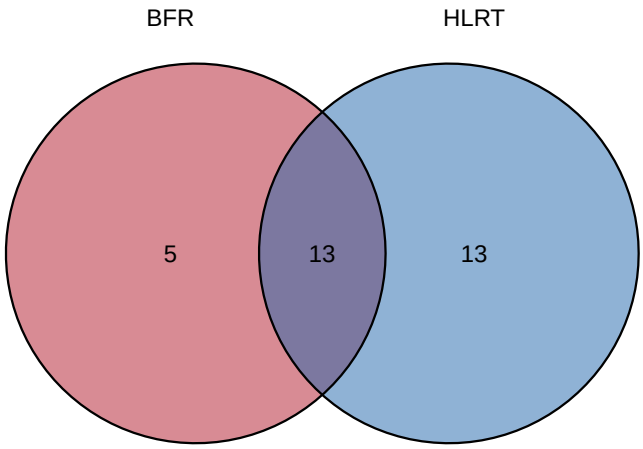


Shared: CAPZB, CDKN1B, CFL2, CLIP1, EML2, GPX1, GRB2, HSPG2, LMOD1, MECP2, MYLK3, PARK7, PFDN4, PFDN5, PFN2, PIK3R1, PPP2CA, PRKN, PRUNE1, RDX, RNH1, SYNPO, SYNPO2

S15. Leading edges by leg

tRNA Aminoacylation

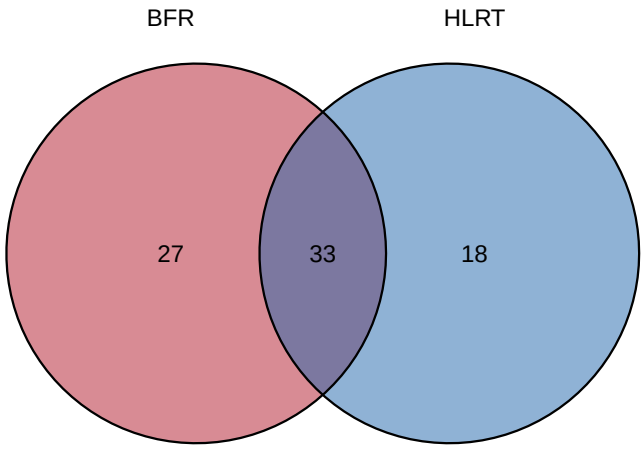
BFR: NES 2.21, padj 7.3e−05
HLRT: NES 2.30, padj 0.0001
shared 13, expected 13.0



Shared: DARS1, DARS2, HARS1, IARS1, IARS2, LARS1, LARS2, NARS1, PPA2, QARS1, SARS1, SARS2, TARS1

Regulation Of Transmembrane Transport

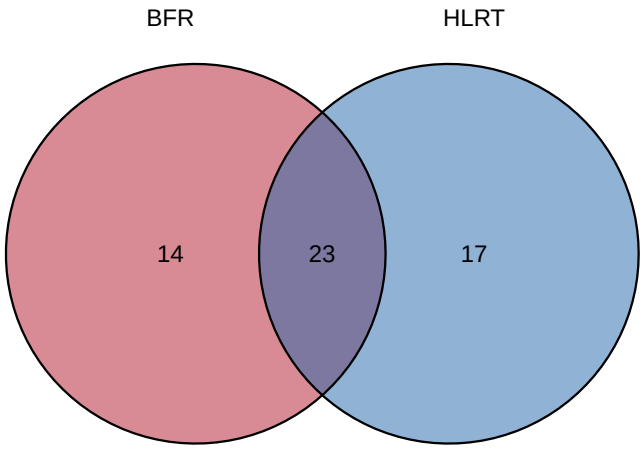
BFR: NES −1.82, padj 0.0007
HLRT: NES −1.44, padj 0.048
shared 33, expected 24.1



Shared: ATP2A1, BCL2, CAB39, CACNA2D1, CACNB1, CACNG1, CASQ1, CHP1, DHRS7C, DMD, GSTM2, HRC, JPH1, LCN2, MAPK14, PIK3R1, PPP3CA, PPP3CB, PPP3R1, PSMC2, PSMC5, PSMC6, RTN2, RYR2, SNTA1, SORBS1, STAC3, STIM1, STXBP3, THBS1, TMEM38B, UBQLN1, YWHAE

Antigen Processing Ubiquitination Proteasome Degradation

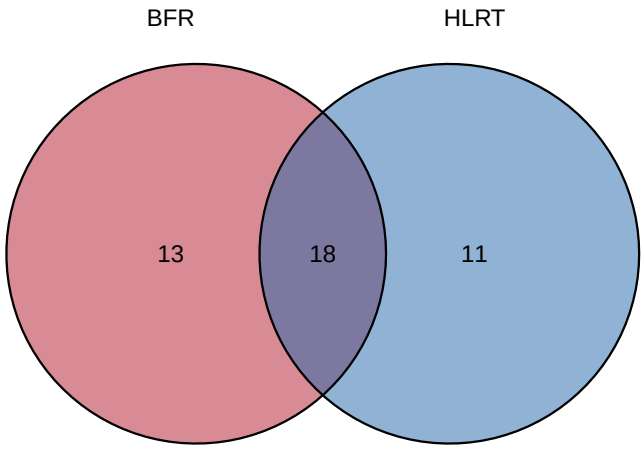
BFR: NES −2.11, padj 2.5e−05
HLRT: NES −2.05, padj 6.8e−05
shared 23, expected 15.7



Shared: ARIH2, ASB11, BLMH, CUL5, HUWE1, NPEPPS, PRKN, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, SKP1, UBE2B, UBE2V2

Calcium Ion Transport

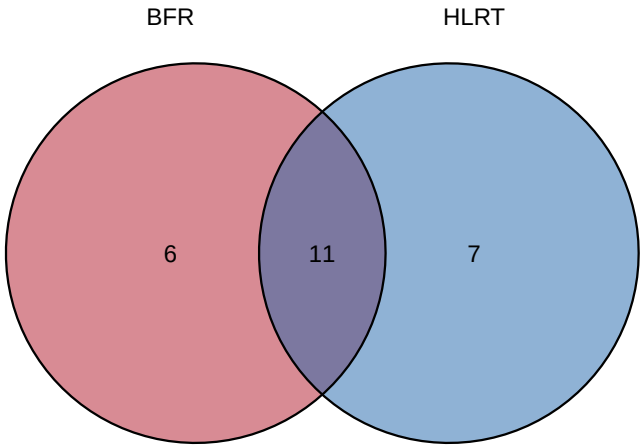
BFR: NES −1.98, padj 0.00018
HLRT: NES −1.66, padj 0.0097
shared 18, expected 9.1



Shared: ATP2A1, ATP2A2, ATP2B2, BCL2, CACNA1S, CACNA2D1, CACNB1, CASQ1, CUL5, DHRS7C, DMD, GSTM2, JPH1, PPP3CB, PPP3R1, RYR1, TMCO1, TPT1

Positive Regulation Of Autophagy

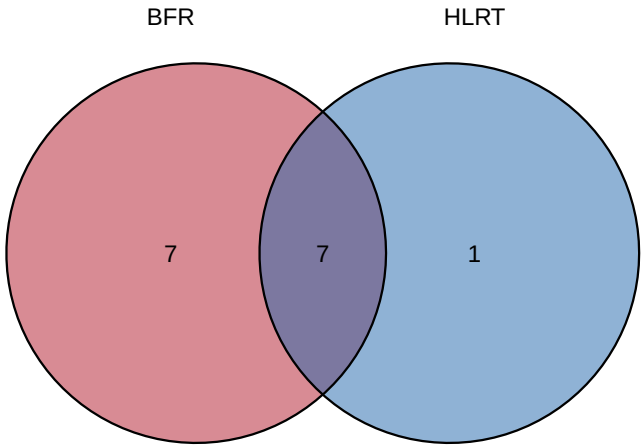
BFR: NES −1.59, padj 0.048
HLRT: NES −1.66, padj 0.032
shared 11, expected 5.7



Shared: BAG3, CDK5RAP3, DEPTOR, FYCO1, HUWE1, MAP2K1, PARK7, PLEKHF1, PRKN, SH3GLB1, SQSTM1

Regulation Of Protein Import Into Nucleus

BFR: NES −1.91, padj 0.0078
HLRT: NES −2.05, padj 0.0038
shared 7, expected 5.6

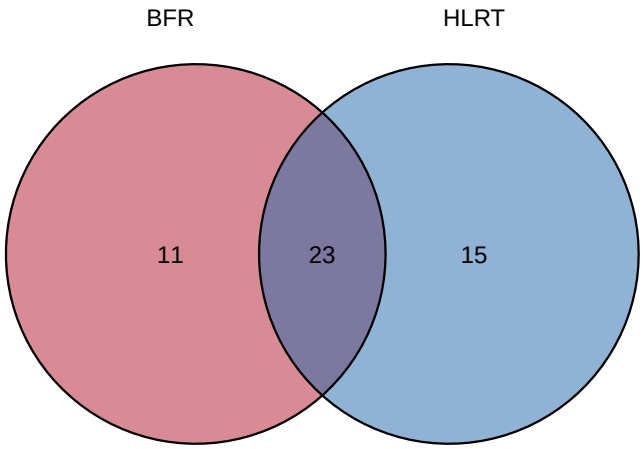


Shared: BAG3, CHP1, MAPK14, PIK3R1, RAN, TPR, YWHAE

S15. Leading edges by leg

Cell Cell Communication

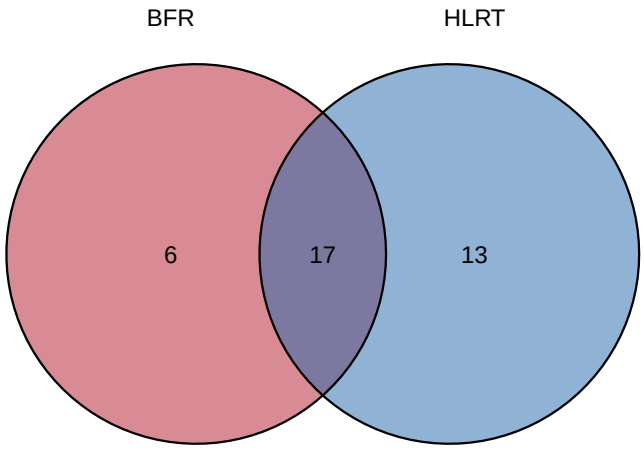
BFR: NES −2.05, padj 4.2e−05
HLRT: NES −1.52, padj 0.029
shared 23, expected 12.8



Shared: AGO2, CTNNA1, GRB2, LIMS1, NCK2, PARVB, PIK3R1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, RBBP7, RPN2, STT3A

Protein Polyubiquitination

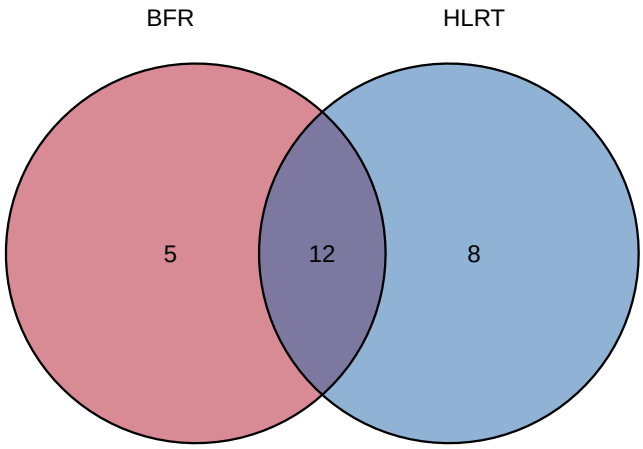
BFR: NES −1.91, padj 0.0011
HLRT: NES −2.02, padj 0.00038
shared 17, expected 12.3



Shared: ARIH2, ASB11, BCL2, BLMH, CUL5, HUWE1, MKRN2, NPEPPS, PIK3R1, PLAA, PRKN, RNF126, RNF170, SKP1, UBE2B, UBE2V2, WDR77

MAPK Signaling Pathway

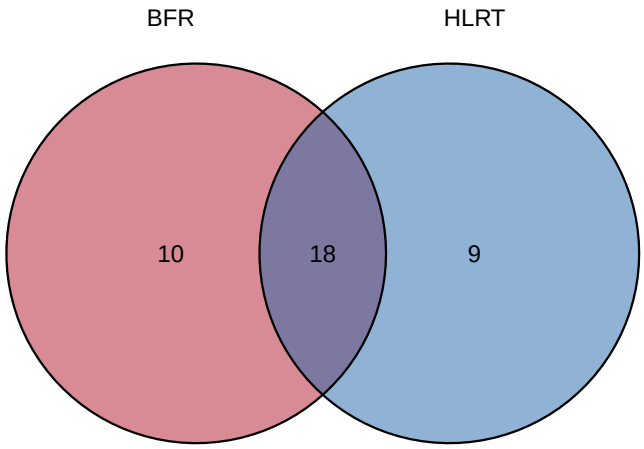
BFR: NES −1.90, padj 0.00095
HLRT: NES −1.61, padj 0.03
shared 12, expected 5.6



Shared: CACNA1S, CACNA2D1, CACNB1, CACNG1, CHP1, GRB2, HSPA2, MAP2K1, MAPK12, PPM1B, PPP3CB, PPP3R1

Regulation Of Homotypic Cell Cell Adhesion

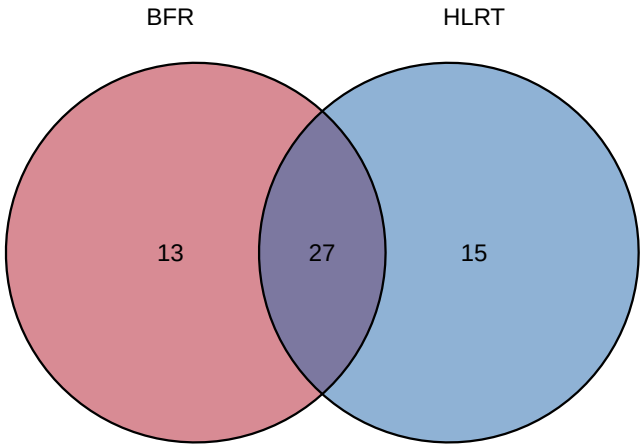
BFR: NES −2.16, padj 2.6e−05
HLRT: NES −1.66, padj 0.025
shared 18, expected 11.0



Shared: AGO2, CTNNA1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, RBBP7, RPN2, STT3A

Neddylation

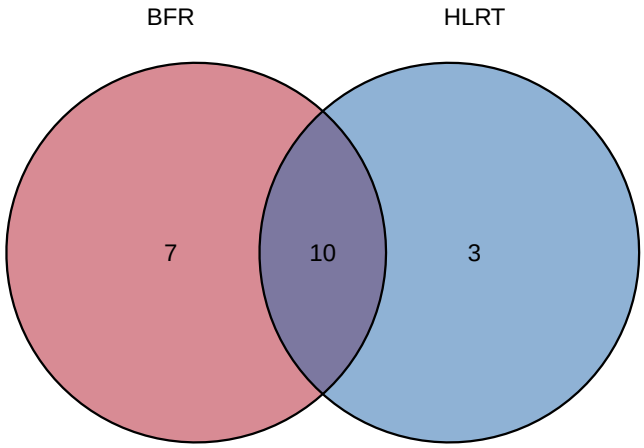
BFR: NES −2.16, padj 2.6e−05
HLRT: NES −2.16, padj 1.8e−05
shared 27, expected 20.5



Shared: ASB11, COPS2, COPS4, COPS7A, COPS7B, COPS8, CUL1, CUL5, NPLOC4, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, RBBP7, SKP1, SQSTM1, UCHL3

Mitochondrial RNA Metabolic Process

BFR: NES 2.34, padj 8.2e−05
HLRT: NES 2.10, padj 0.0015
shared 10, expected 10.0

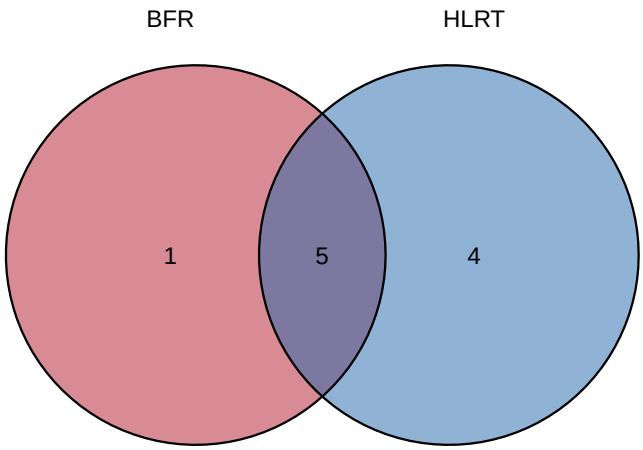


Shared: DARS2, GRSF1, HSD17B10, LRPPRC, MRPL12, PNPT1, PRKAA1, SARS2, SLIRP, SUPV3L1

S15. Leading edges by leg

Phenol Containing Compound Metabolic Process

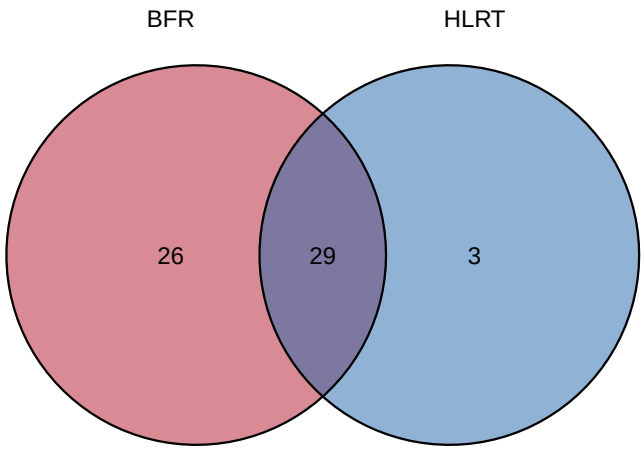
BFR: NES −1.84, padj 0.016
HLRT: NES −1.76, padj 0.046
shared 5, expected 3.0



Shared: ALDH9A1, BCL2, CRYM, PARK7, PRKN

Regulation Of Muscle System Process

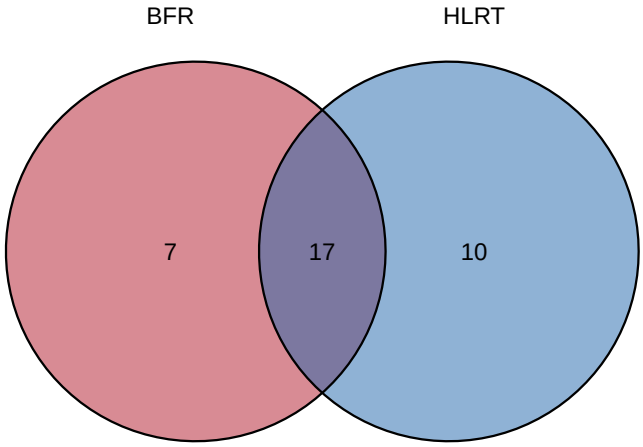
BFR: NES −2.25, padj 3.4e−06
HLRT: NES −2.07, padj 3.3e−05
shared 29, expected 20.0



Shared: ACTN3, ATP2A1, ATP2A2, CACNA1S, CACNA2D1, CACNB1, CACNG1, CASQ1, DAG1, DMD, ENO1, GSTM2, GSTO1, JPH1, MAP2K1, MLIP, MYH7, MYL2, MYL3, MYLK2, PLN, PRKAG3, RYR2, TMEM38B, TNNC1, TNNC2, TNNT1, TNNT3, TPM1

Beta Catenin Independent WNT Signaling

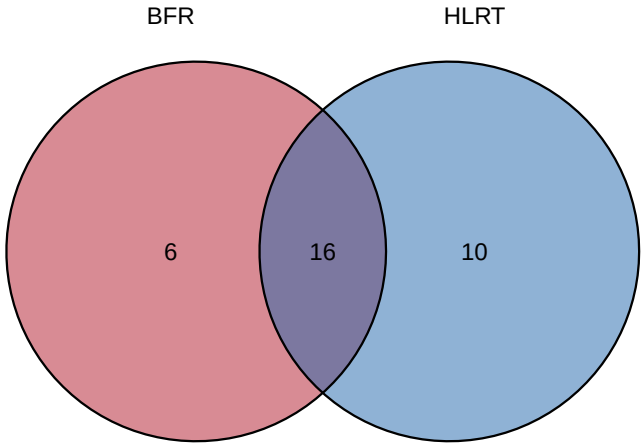
BFR: NES −2.12, padj 6.9e−05
HLRT: NES −1.96, padj 0.0013
shared 17, expected 11.8



Shared: AGO2, CAMK2A, PPP3CB, PPP3R1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

Downstream Signaling Events Of B Cell Receptor Bcr

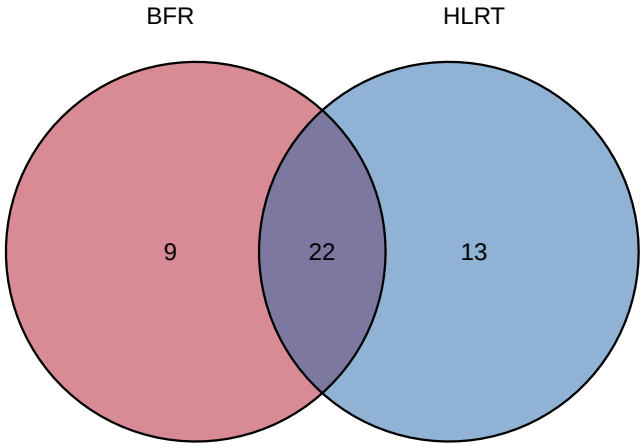
BFR: NES −2.24, padj 3.1e−05
HLRT: NES −2.15, padj 0.00028
shared 16, expected 13.3



Shared: PPP3CB, PPP3R1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, SKP1

MAPK1 MAPK3 Signaling

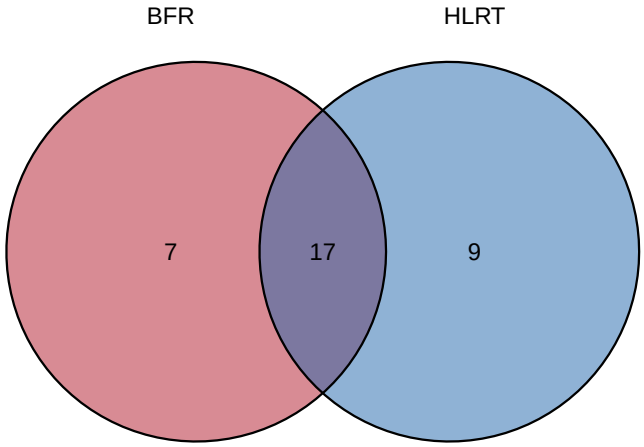
BFR: NES −2.08, padj 4e−05
HLRT: NES −1.58, padj 0.029
shared 22, expected 13.1



Shared: CAMK2A, FNTA, GRB2, MAP2K1, MAPK12, PEBP1, PIK3R1, PPP2CA, PPP2R5A, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

Calcium Ion Transmembrane Transport

BFR: NES −1.80, padj 0.0026
HLRT: NES −1.82, padj 0.0032
shared 17, expected 8.1

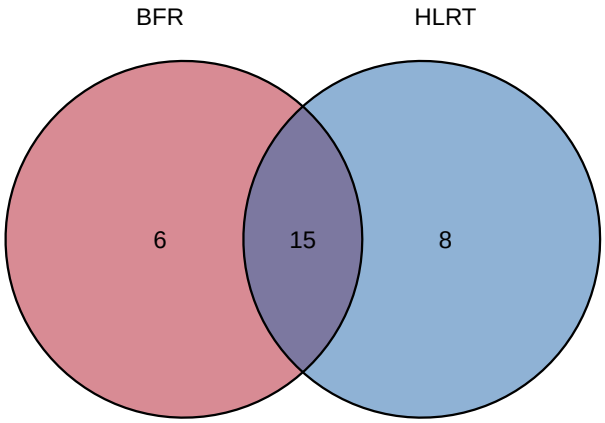


Shared: ATP2A1, ATP2A2, ATP2B2, BCL2, CACNA1S, CACNA2D1, CACNB1, CASQ1, CUL5, DHRS7C, DMD, GSTM2, JPH1, PPP3CB, PPP3R1, RYR1, TMCO1

S15. Leading edges by leg

Abc Family Proteins Mediated Transport

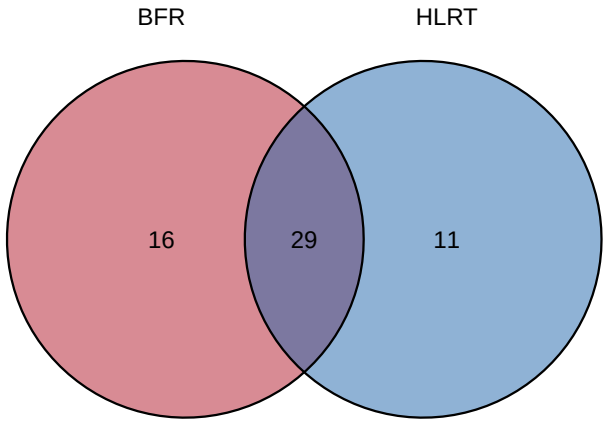
BFR: NES −1.99, padj 0.0014
HLRT: NES −2.01, padj 0.0025
shared 15, expected 10.1



Shared: EIF2S2, ERLIN2, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

Mitochondrial Respiratory Chain Complex Assembly

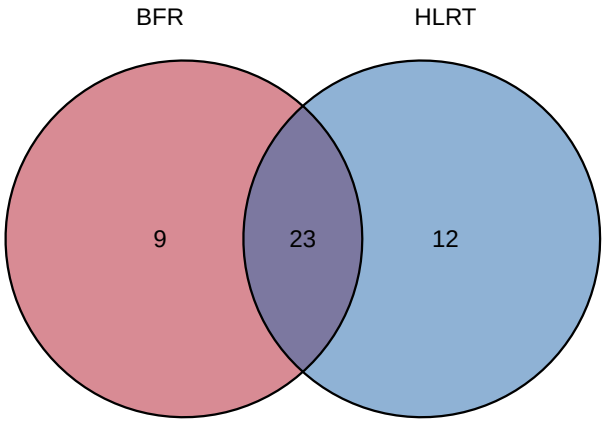
BFR: NES 2.83, padj 1.6e−12
HLRT: NES 2.37, padj 1e−06
shared 29, expected 27.3



Shared: ACAD9, AIFM1, ATP5F1D, ATPAF1, COA4, COA6, COA7, COX17, COX20, COX7A1, COX7A2L, LYRM7, MT−ND1, NDUFAF2, NDUFAF3, NDUFAF4, NDUFS1, NDUFS3, NDUFS4, NDUFS6, NDUFS8, OXA1L, SDHAF2, SMIM20, TACO1, TIMM21, TMEM70, UQCC1, UQCRFS1

Intracellular Signaling By Second Messengers

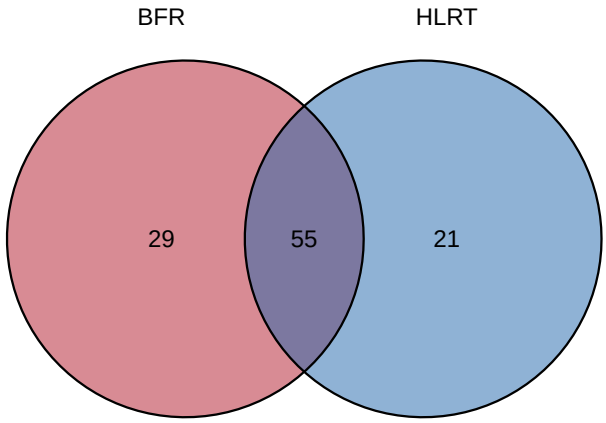
BFR: NES −2.11, padj 5.3e−05
HLRT: NES −1.56, padj 0.035
shared 23, expected 13.7



Shared: AGO2, CAMK2A, CDKN1B, GRB2, LAMTOR1, PIK3R1, PPP2CA, PPP2R5A, PRKAR2A, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, RBBP7

Nucleoside Phosphate Biosynthetic Process

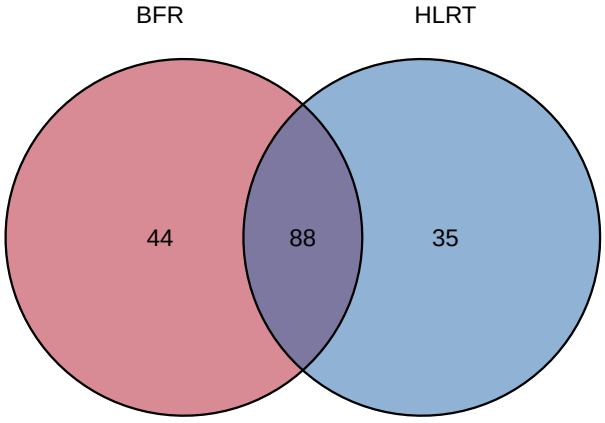
BFR: NES 2.15, padj 2.5e−07
HLRT: NES 1.95, padj 3.3e−05
shared 55, expected 36.7



Shared: ACSF3, ACSL1, ACSS1, AK2, AK3, APRT, ATP5F1A, ATP5F1C, ATP5F1D, ATP5IF1, ATP5ME, ATP5MG, ATP5PB, ATP5PD, BCKDK, CBR4, DLAT, DLST, GFPT1, HK1, HSD17B12, MMUT, NADK2, NAMPT, NDUFA10, NDUFA9, NDUFB10, NDUFB11, NDUFB4, NDUFB5, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS6, NDUFS7, NDUFS8, NDUFV1, NDUFV2, OGDH, PCCA, PDHA1, PDHB, PDHX, PDK2, PDK4, PDP1, PDPR, PPCS, SDHA, SDHC, SDHD, STOML2, UGDH

Generation Of Precursor Metabolites And Energy

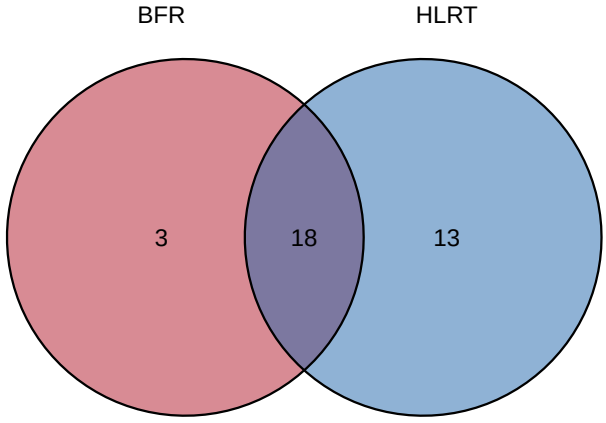
BFR: NES 2.67, padj 2.4e−19
HLRT: NES 2.51, padj 8.9e−17
shared 88, expected 61.5



Shared: ACO2, AKT2, ATP5F1A, ATP5F1C, ATP5F1D, ATP5IF1, ATP5ME, ATP5MG, ATP5PB, ATP5PD, BCKDK, CHCHD2, CISD1, COQ10A, COX5B, COX6B1, COX7A2, COX7A2L, COX7B, CS, CYC1, CYCS, DLAT, DLST, ETFA, ETFB, FH, GAA, GBE1, GFPT1, GHITM, GYS1, HCCS, HK1, HMGCL, IDH3A, IDH3B, IDH3G, LYRM7, MDH2, MTCH2, NDUFA10, NDUFA4, NDUFA9, NDUFB10, NDUFB11, NDUFB5, NDUFB7, NDUFB9, NDUFC2, NDUFS1, NDUFS6, NDUFS8, NDUFV1, NDUFV2, NNT, OGDH, OXA1L, PDHA1, PDHB, PDHX, PDK2, PDK4, PDP1, PDPR, PFKFB1, PHKA1, PHKB, PHKG1, PNPT1, PPIF, PRKAA1, SDHA, SDHC, SDHD, SHMT2, SIRT3, SLC25A4, SOD2, STOML2, SUCLA2, SUCLG1, SUCLG2, TRAP1, UQCRC1, UQCRC2, UQCRFS1, UQCRH

Protein Localization To Mitochondrion

BFR: NES 2.08, padj 0.00074
HLRT: NES 2.03, padj 0.0015
shared 18, expected 14.5

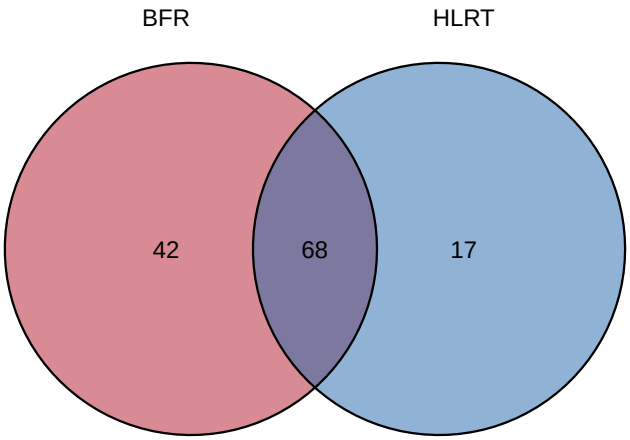


Shared: AIFM1, GRPEL1, HK1, HSPD1, MTCH2, MTX1, OXA1L, ROMO1, SAMM50, TIMM13, TIMM21, TIMM22, TIMM44, TIMM50, TIMM8A, TOMM40, TOMM40L, TSPO

S15. Leading edges by leg

Muscle System Process

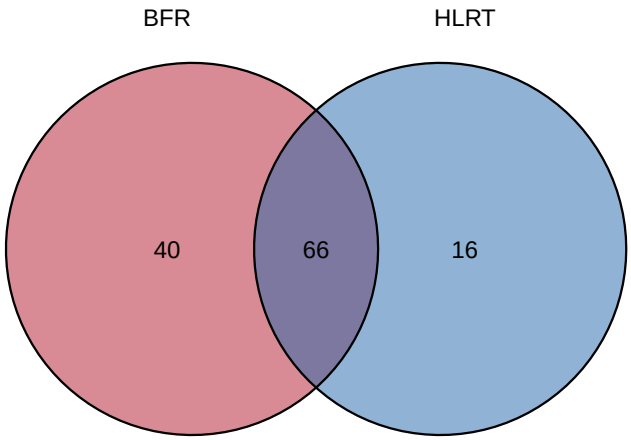
BFR: NES −1.99, padj 5.6e−07
HLRT: NES −1.65, padj 0.0015
shared 68, expected 42.9



Shared: ACTA1, ACTN3, ATP2A1, ATP2A2, CA3, CACNA1S, CACNA2D1, CACNB1, CACNG1, CASQ1, CERT1, DAG1, DMD, DTNA, EMD, ENO1, GSTM2, GSTO1, HRC, JPH1, LMNA, LMOD1, MAP2K1, MLIP, MYH1, MYH13, MYH14, MYH7, MYH8, MYL1, MYL2, MYL3, MYL4, MYL5, MYL6B, MYLK2, MYOM2, MYOZ2, PDLIM5, PGAM2, PLN, PPP3CA, PRKAG3, PVALB, RPL38, RYR1, RYR2, SCN4A, SGCA, SGCD, SNTA1, SSPN, STAC3, SYNM, TCAP, TMEM38A, TMEM38B, TMOD4, TNNC1, TNNC2, TNNI1, TNNI2, TNNT1, TNNT3, TPM1, TPM3, TRIM72, TTN

Mitochondrion Organization

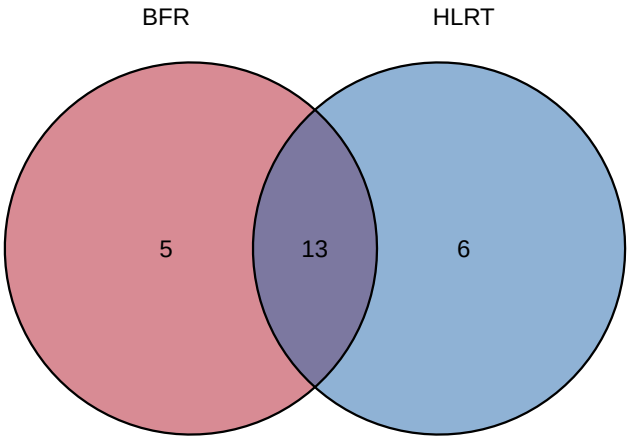
BFR: NES 2.60, padj 5.3e−15
HLRT: NES 2.14, padj 1.3e−08
shared 66, expected 43.5



Shared: ACAD9, AFG3L2, AIFM1, ATP5F1D, ATP5IF1, ATPAF1, CCDC93, CHCHD10, CHCHD2, CHCHD3, CLPP, CLUH, COA4, COA6, COA7, COX17, COX20, COX7A2L, DAP3, ENDOG, GHITM, HAX1, HMGCL, HSD17B10, HSPA9, HSPD1, HTRA2, LETM1, LONP1, LYRM7, MICOS13, MTCH2, MTX1, NDUFAF2, NDUFAF3, NDUFAF4, NDUFC2, NDUFS1, NDUFS6, NDUFS8, OPA1, OPA3, OXA1L, PHB2, PNPT1, POLDIP2, PPIF, PRDX3, SIRT5, SLC25A4, SLIRP, SMIM20, SOD2, SSBP1, STOML2, SUPV3L1, TACO1, TIMM13, TIMM21, TIMM50, TIMM8A, TMEM11, TMEM70, TRIAP1, UQCRFS1, VDAC2

Inner Mitochondrial Membrane Organization

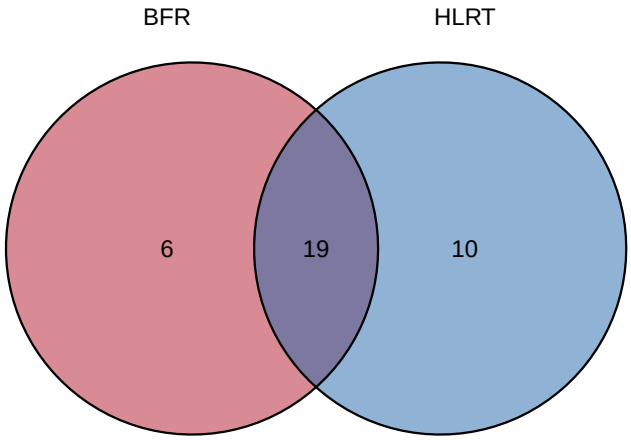
BFR: NES 2.24, padj 0.00013
HLRT: NES 2.26, padj 0.00025
shared 13, expected 10.1



Shared: AFG3L2, CHCHD10, CHCHD3, GHITM, HSPA9, LETM1, MICOS13, MTX1, OPA1, OXA1L, TIMM13, TIMM8A, TMEM11

Cell Cycle G1 S Phase Transition

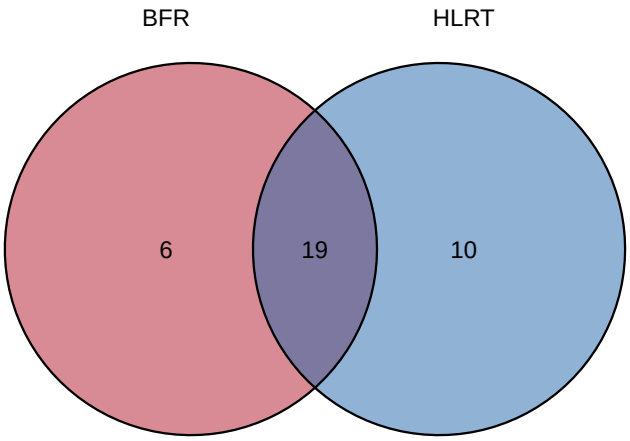
BFR: NES −2.02, padj 0.00057
HLRT: NES −2.03, padj 0.00088
shared 19, expected 13.4



Shared: BCL2, CAMK2A, CDKN1B, CUL5, EIF4G1, PPP2CA, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, RDX, SKP1

Signaling By The B Cell Receptor Bcr

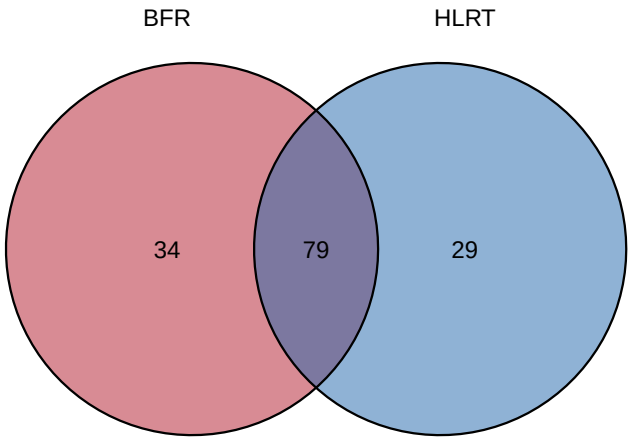
BFR: NES −2.28, padj 3.8e−05
HLRT: NES −2.22, padj 0.00017
shared 19, expected 15.1



Shared: GRB2, PIK3R1, PPP3CB, PPP3R1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, SKP1, STIM1

Cellular Respiration

BFR: NES 2.74, padj 3.3e−18
HLRT: NES 2.67, padj 4.4e−17
shared 79, expected 60.7

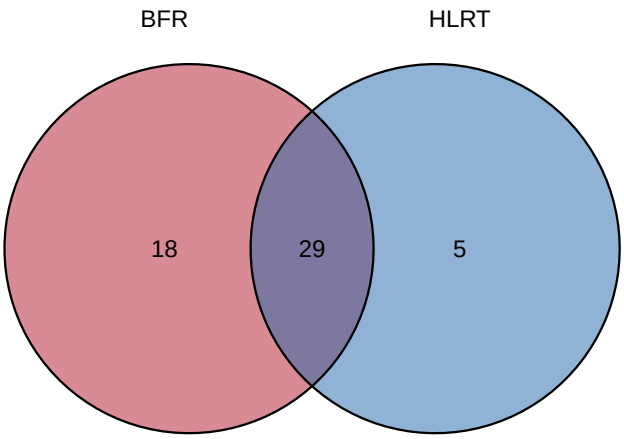


Shared: ACO2, ATP5F1A, ATP5F1C, ATP5F1D, ATP5IF1, ATP5ME, ATP5MG, ATP5PB, ATP5PD, BCKDK, CHCHD2, CISD1, COQ10A, COX5B, COX6B1, COX7A2, COX7A2L, COX7B, CS, CYC1, CYCS, DLAT, DLST, ETFA, ETFB, FH, GHITM, HCCS, HK1, IDH3A, IDH3B, IDH3G, LYRM7, MDH2, MTCH2, NDUFA10, NDUFA4, NDUFA9, NDUFB10, NDUFB11, NDUFB5, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS6, NDUFS8, NDUFV1, NDUFV2, NNT, OGDH, OXA1L, PDHA1, PDHB, PDHX, PDK2, PDK4, PDP1, PDPR, PFKFB1, PNPT1, PPIF, PRKAA1, SDHA, SDHC, SDHD, SHMT2, SIRT3, SOD2, STOML2, SUCLA2, SUCLG1, SUCLG2, TRAP1, UQCRC1, UQCRC2, UQCRFS1, UQCRH

S15. Leading edges by leg

Organelle Biogenesis And Maintenance

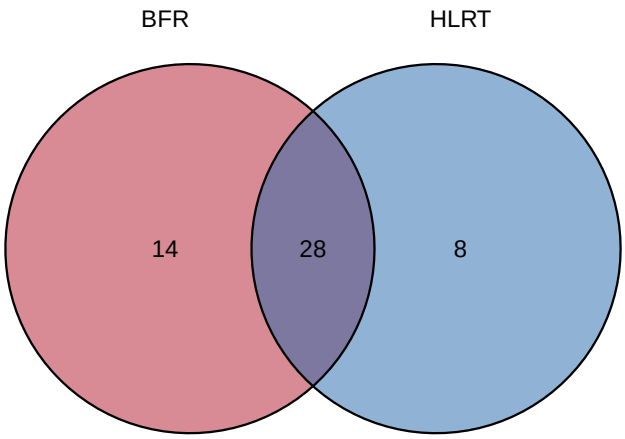
BFR: NES 1.90, padj 0.00073
HLRT: NES 1.95, padj 0.00044
shared 29, expected 19.7



Shared: ARF4, ATP5F1A, ATP5F1C, ATP5F1D, ATP5ME, ATP5MG, ATP5PB, ATP5PD, ATP5PO, CHCHD3, CYCS, DYNC1H1, DYNC1I2, DYNLRB1, GLUD1, HSPA9, IDH2, MICOS13, MTX1, PRKAB2, PRKAG1, SIRT3, SIRT5, SOD2, SSBP1, TMEM11, TUBA8, TUBB2B, TUBB6

Organic Acid Catabolic Process

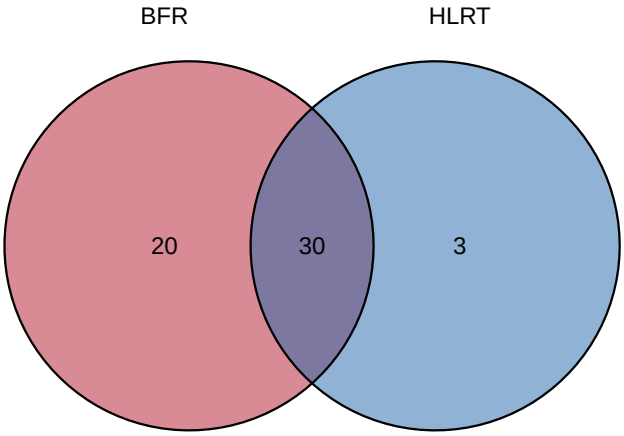
BFR: NES 2.01, padj 5.3e−05
HLRT: NES 1.52, padj 0.041
shared 28, expected 14.0



Shared: ACSF3, AKT2, ALDH5A1, AUH, BCKDK, CARNS1, CRAT, DECR1, DLAT, DLST, ECH1, ECI1, ECI2, ETFA, ETFB, GLUD1, GOT2, HIBCH, HSD17B10, ILVBL, LDHD, MCCC2, NUDT19, PCCA, PCCB, SLC16A1, SLC16A3, TST

Protein Localization

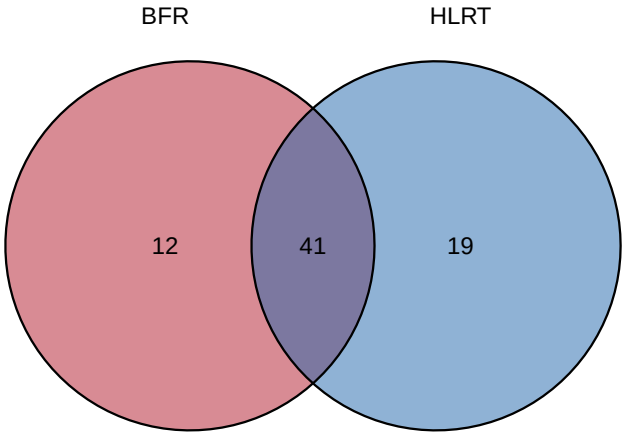
BFR: NES 2.65, padj 1.3e−11
HLRT: NES 2.10, padj 3.7e−05
shared 30, expected 17.9



Shared: ACO2, ATP5F1A, CHCHD10, CHCHD2, CHCHD3, CMC2, COA4, COA6, COX17, CRAT, CS, CYC1, ECH1, ECI2, GSTK1, HSPA9, HSPD1, IDH3G, LDHD, MTX1, NUDT19, PITRM1, PMPCA, SLC25A12, SLC25A4, TIMM13, TIMM21, TIMM44, TIMM50, TIMM8A

Mitochondrial Translation

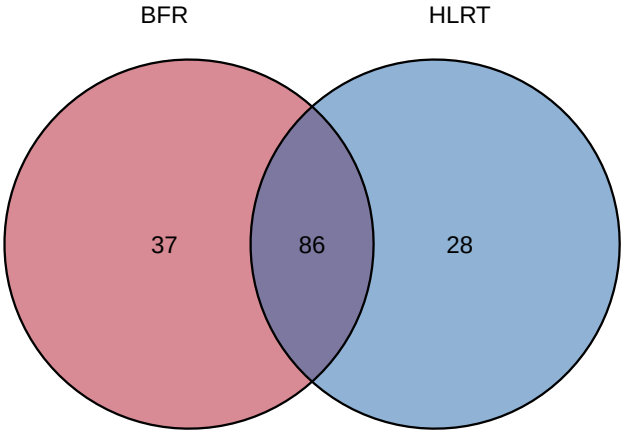
BFR: NES 2.86, padj 1.3e−13
HLRT: NES 2.76, padj 5.5e−14
shared 41, expected 38.3



Shared: DAP3, GFM1, MALSU1, MRPL1, MRPL10, MRPL11, MRPL12, MRPL14, MRPL15, MRPL2, MRPL21, MRPL22, MRPL33, MRPL38, MRPL39, MRPL44, MRPL46, MRPL48, MRPL50, MRPL53, MRPL9, MRPS11, MRPS16, MRPS18A, MRPS18B, MRPS21, MRPS23, MRPS24, MRPS27, MRPS28, MRPS30, MRPS35, MRPS9, MRRF, MT−CO3, MT−CYB, MT−ND1, OXA1L, PTCD3, TSFM, TUFM

Aerobic Respiration And Respiratory Electron Transport

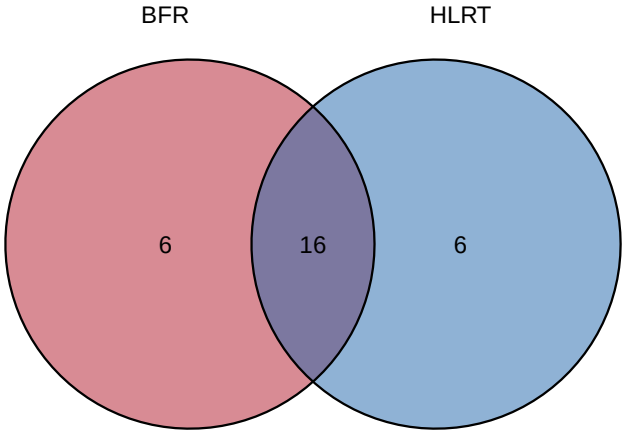
BFR: NES 3.22, padj 4.6e−30
HLRT: NES 2.89, padj 6.3e−22
shared 86, expected 75.0



Shared: ACAD9, ACO2, ATP5F1A, ATP5F1C, ATP5F1D, ATP5ME, ATP5MG, ATP5PB, ATP5PD, COQ10A, COX17, COX20, COX5B, COX6B1, COX7A1, COX7A2, COX7A2L, COX7B, CS, CYC1, CYCS, DLAT, DLST, ETFA, ETFB, FAHD1, FH, GOT2, HCCS, HSPA9, IDH3A, IDH3B, IDH3G, L2HGDH, LETM1, LYRM7, MDH2, ME2, MPC1, NDUFA10, NDUFA4, NDUFA9, NDUFAF2, NDUFAF3, NDUFAF4, NDUFB10, NDUFB11, NDUFB5, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS4, NDUFS6, NDUFS8, NDUFV1, NDUFV2, NNT, OGDH, OXA1L, PDHA1, PDHB, PDHX, PDK2, PDK4, PDP1, PDPR, SDHA, SDHC, SDHD, SIRT3, SLC25A11, SLC25A12, SLC25A4, SMIM20, SUCLA2, SUCLG1, SUCLG2, TACO1, TIMM21, TRAP1, UQCRC1, UQCRC2, UQCRFS1, UQCRH

Auf1 Hnrnp D0 Binds And Destabilizes mRNA

BFR: NES −2.40, padj 2.9e−06
HLRT: NES −1.98, padj 0.0022
shared 16, expected 13.1

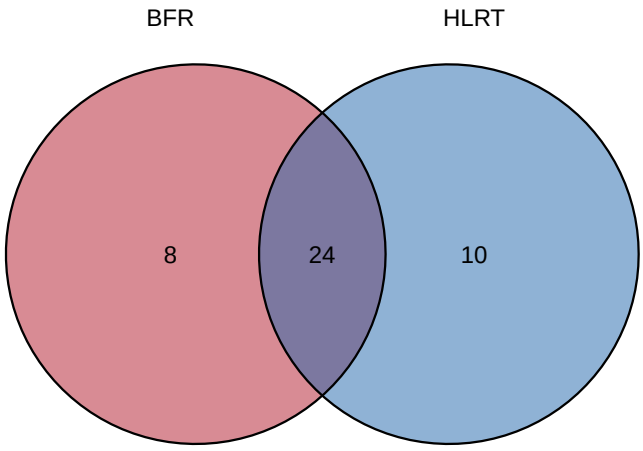


Shared: EIF4G1, HSPA8, HSPB1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

S15. Leading edges by leg

Nuclear Events Mediated By Nfe2l2

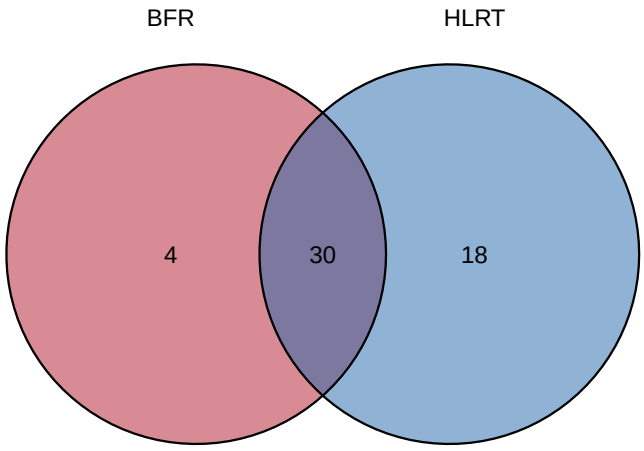
BFR: NES −2.30, padj 4.2e−06
HLRT: NES −2.35, padj 1.8e−06
shared 24, expected 19.4



Shared: BCL2, CUL1, GCLM, IDH1, ME1, PRDX1, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, SKP1, SOD3, SQSTM1, TALDO1

Fatty Acid Metabolic Process

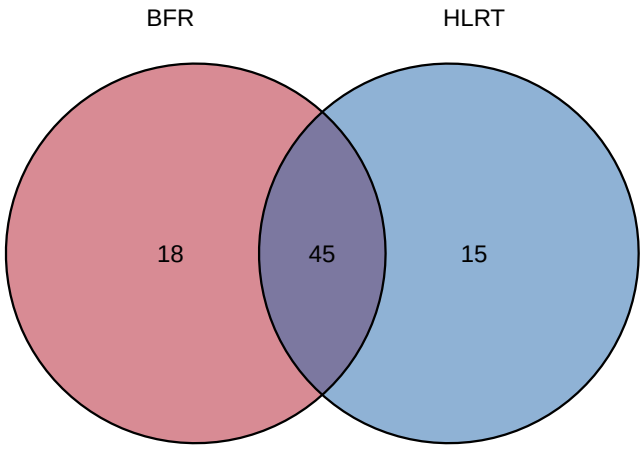
BFR: NES 1.87, padj 0.00019
HLRT: NES 1.74, padj 0.0025
shared 30, expected 11.9



Shared: ACAD9, ACOT9, ACSF3, ACSL1, ACSS1, AKR1C3, AKT2, AUH, BCKDK, CBR4, CRAT, DECR1, ECH1, ECI1, ECI2, ETFA, ETFB, HSD17B10, HSD17B12, HSD17B8, MECR, MMUT, NUDT19, PCCA, PCCB, PDK2, PDK4, PDP1, PRKAA1, PTGES2

Myogenesis

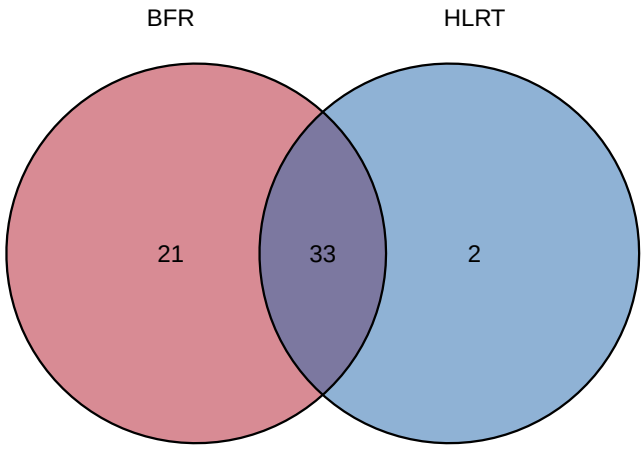
BFR: NES −1.68, padj 0.0058
HLRT: NES −1.79, padj 0.0015
shared 45, expected 32.3



Shared: ACTA1, ACTN3, AK1, ATP2A1, CACNG1, CASQ1, COL15A1, COL4A2, COL6A2, CTF1, DMD, DTNA, GABARAPL2, HRC, HSPB8, ITGA7, LAMA2, MAPK12, MYH1, MYH7, MYH8, MYL1, MYL2, MYL3, MYL4, MYL6B, MYOM2, PGAM2, RYR1, SGCA, SGCD, SGCG, SH3BGR, SOD3, SORBS1, SSPN, SVIL, SYNGR2, TCAP, TNNC1, TNNC2, TNNI2, TNNT1, TNNT3, TPD52L1

Muscle Cell Development

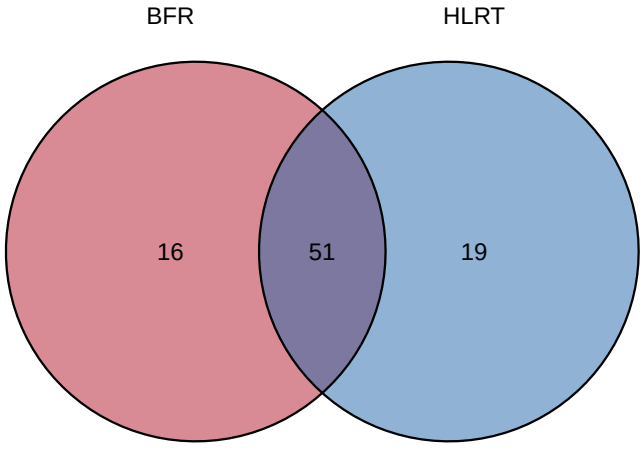
BFR: NES −2.15, padj 1.7e−05
HLRT: NES −1.88, padj 0.00095
shared 33, expected 21.0



Shared: ACTA1, ACTN3, ATP2A2, BCL2, CACNA1S, CASQ1, CFL2, DMD, GPX1, LMOD1, MYBPC1, MYBPC2, MYL2, MYLK3, MYOM2, MYOM3, NEB, PDLIM5, PGM5, PLEC, PPP3CB, RYR1, SGCB, SGCD, STAC3, SYNPO2L, SYPL2, TCAP, TMOD4, TNNT1, TNNT3, TPM1, TTN

Mitochondrial Gene Expression

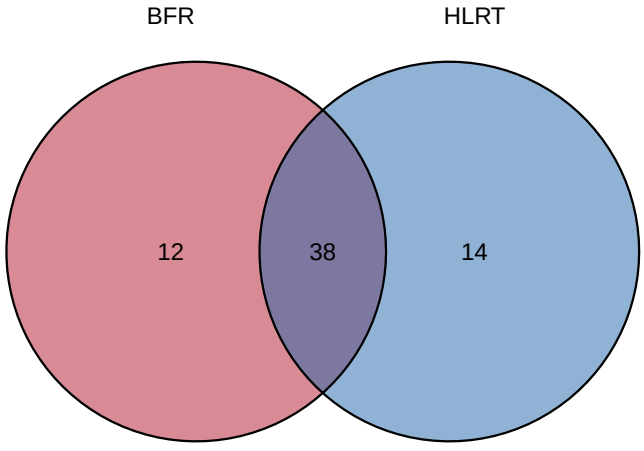
BFR: NES 3.06, padj 2.4e−19
HLRT: NES 2.95, padj 3.3e−17
shared 51, expected 48.9



Shared: DAP3, DARS2, GFM1, HSD17B10, IARS2, LARS2, LRPPRC, MALSU1, MRPL1, MRPL10, MRPL11, MRPL12, MRPL14, MRPL15, MRPL2, MRPL21, MRPL22, MRPL33, MRPL38, MRPL39, MRPL44, MRPL46, MRPL48, MRPL50, MRPL53, MRPL9, MRPS11, MRPS16, MRPS18A, MRPS18B, MRPS21, MRPS23, MRPS24, MRPS27, MRPS28, MRPS30, MRPS35, MRPS9, OXA1L, PNPT1, PRKAA1, PTCD3, SARS2, SHMT2, SLIRP, SUPV3L1, TACO1, TBRG4, TRMT10C, TSFM, TUFM

Mitochondrial Protein Degradation

BFR: NES 3.14, padj 3.2e−20
HLRT: NES 2.84, padj 1.6e−14
shared 38, expected 30.6

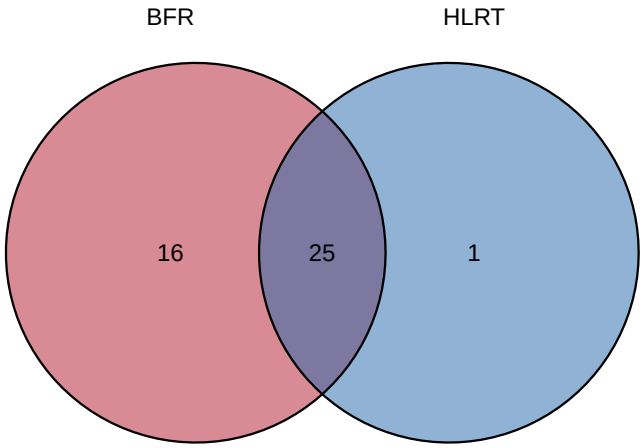


Shared: ACO2, AFG3L2, ALDH1B1, ATP5F1A, ATP5F1C, ATP5MG, ATP5PD, CHCHD2, CLPP, COX5B, CS, ECH1, ECI1, FH, GLUD1, HSD17B10, HSPA9, HSPD1, HTRA2, IARS2, IDH3A, LDHD, LONP1, MDH2, ME2, NADK2, NDUFS1, NDUFV1, OGDH, OPA1, PCCB, PDHA1, PDHB, PMPCA, SHMT2, SSBP1, SUCLG2, UQCRC2

S15. Leading edges by leg

Cellular Component Assembly Involved In Morphogenesis

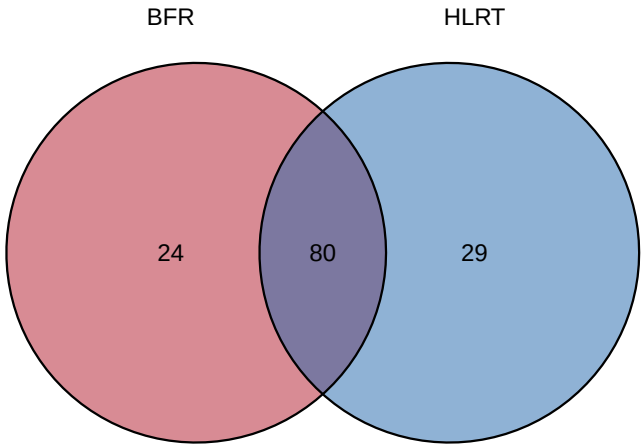
BFR: NES −2.07, padj 0.00014
HLRT: NES −1.93, padj 0.0025
shared 25, expected 18.4



Shared: ACTA1, AGFG1, CASQ1, CFL2, GNPAT, GPC1, LMOD1, MYBPC1, MYBPC2, MYL2, MYLK3, MYOM2, MYOM3, MYOZ1, NEB, PGM5, PLEC, PLN, SYNPO2L, TCAP, TMOD4, TNNT1, TNNT3, TPM1, TTN

OXPHOS

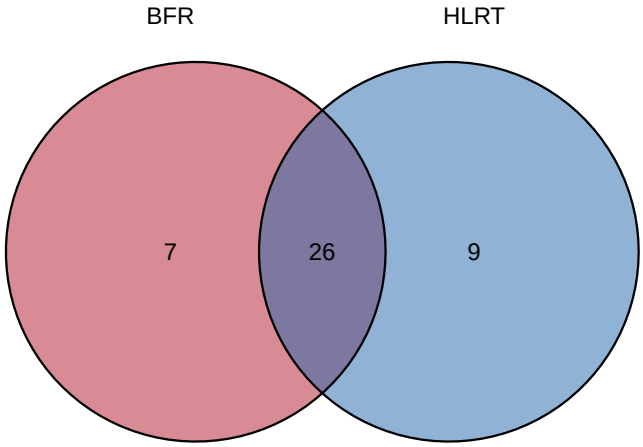
BFR: NES 3.14, padj 1.5e−26
HLRT: NES 2.94, padj 1e−22
shared 80, expected 65.1



Shared: ABCB7, ACO2, AFG3L2, AIFM1, ATP5F1A, ATP5F1C, ATP5F1D, ATP5ME, ATP5MG, ATP5PB, ATP5PD, BDH2, COX17, COX5B, COX6B1, COX7A2, COX7A2L, COX7B, CS, CYC1, CYCS, DECR1, DLAT, DLST, ECH1, ECI1, ETFA, ETFB, FH, GLUD1, GOT2, HCCS, HSD17B10, HSPA9, HTRA2, IDH3A, IDH3B, IDH3G, LRPPRC, MDH2, MPC1, MRPL11, MRPS30, NDUFA4, NDUFA9, NDUFB5, NDUFB7, NDUFS1, NDUFS6, NDUFS8, NDUFV1, NDUFV2, NNT, OGDH, OPA1, OXA1L, PDHA1, PDHB, PDHX, PDK4, PDP1, PHB2, PMPCA, PRDX3, SDHA, SDHC, SDHD, SLC25A12, SLC25A3, SLC25A4, SUCLA2, SUCLG1, SUPV3L1, TIMM13, TIMM50, UQCRC1, UQCRC2, UQCRFS1, UQCRH, VDAC2

Fatty Acid Metabolism

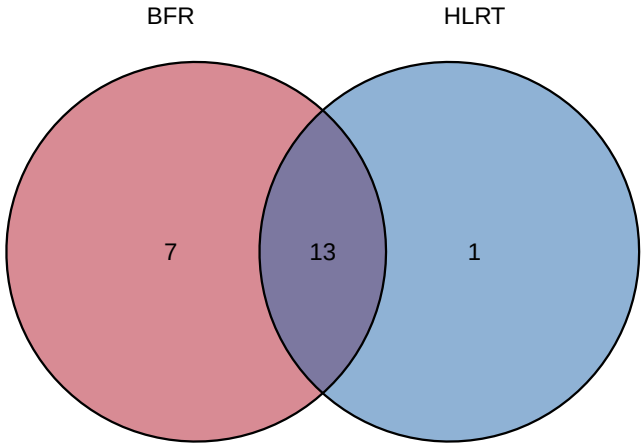
BFR: NES 2.00, padj 6.9e−05
HLRT: NES 1.61, padj 0.019
shared 26, expected 11.6



Shared: ACO2, ACSL1, ACSS1, AUH, BPHL, CPOX, CRAT, DECR1, DLST, ECH1, ECI1, ECI2, FH, HCCS, HIBCH, HSD17B10, HSDL2, IDH3B, IDH3G, MDH2, PDHA1, PDHB, SDHA, SUCLA2, SUCLG1, SUCLG2

Cristae Formation

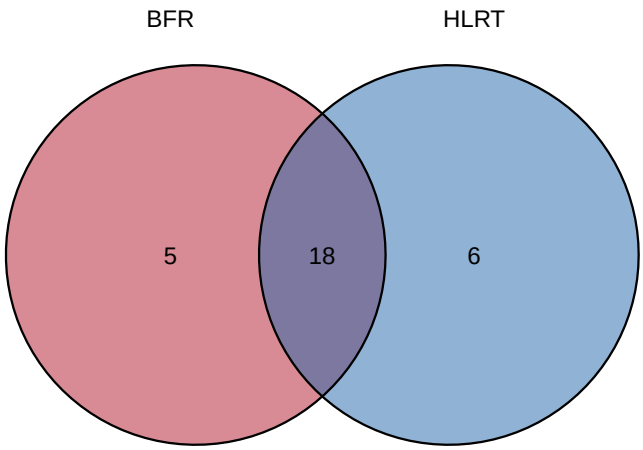
BFR: NES 2.20, padj 0.00049
HLRT: NES 2.03, padj 0.0018
shared 13, expected 10.8



Shared: ATP5F1A, ATP5F1C, ATP5F1D, ATP5ME, ATP5MG, ATP5PB, ATP5PD, ATP5PO, CHCHD3, HSPA9, MICOS13, MTX1, TMEM11

Fatty Acid Metabolism

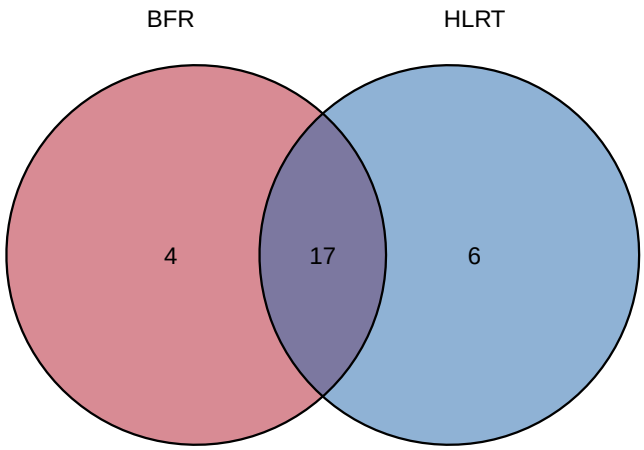
BFR: NES 2.09, padj 0.00014
HLRT: NES 1.56, padj 0.048
shared 18, expected 8.4



Shared: ACOT9, ACSF3, ACSL1, AKR1C3, CBR4, CRAT, DECR1, ECI1, ECI2, EPHX2, HSD17B12, HSD17B8, MECR, MMUT, NUDT19, PCCA, PCCB, PTGES2

Dilated Cardiomyopathy

BFR: NES −2.03, padj 0.00043
HLRT: NES −2.49, padj 4.6e−07
shared 17, expected 13.4

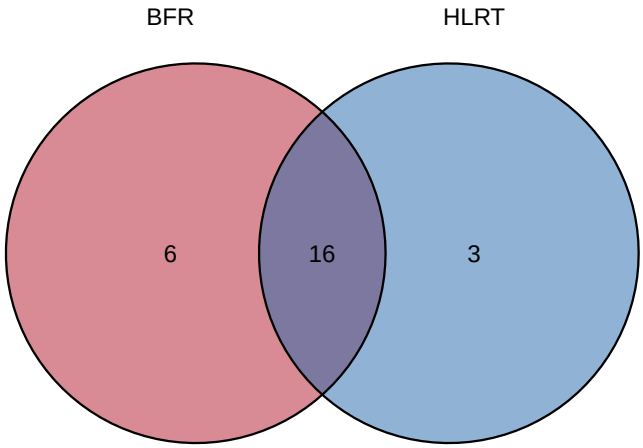


Shared: ATP2A2, CACNA1S, CACNA2D1, CACNB1, CACNG1, DAG1, DMD, ITGA7, LAMA2, MYH7, MYL2, MYL3, RYR2, SGCA, SGCB, SGCD, SGCG

S15. Leading edges by leg

Regulation Of Cell Cycle G1 S Phase Transition

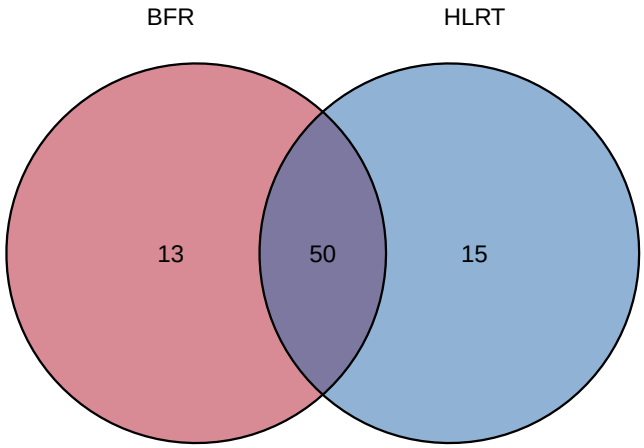
BFR: NES −2.19, padj 6.7e−05
HLRT: NES −1.95, padj 0.0055
shared 16, expected 11.0



Shared: BCL2, CDKN1B, EIF4G1, PPP2CA, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, RDX

Proton Transmembrane Transport

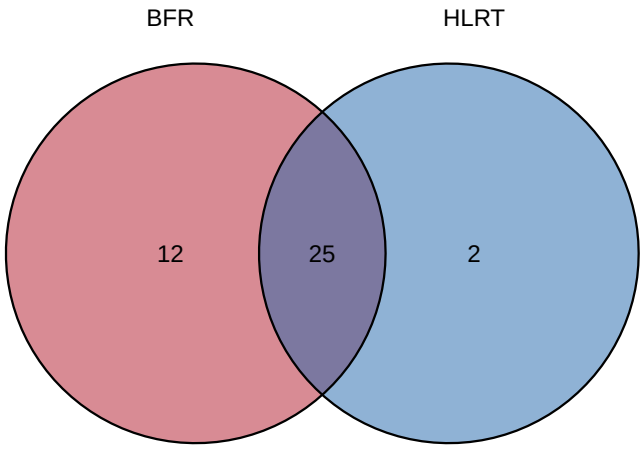
BFR: NES 2.67, padj 8.6e−12
HLRT: NES 2.67, padj 9.8e−12
shared 50, expected 44.5



Shared: ATP1A1, ATP5F1A, ATP5F1B, ATP5F1C, ATP5F1D, ATP5ME, ATP5MG, ATP5PB, ATP5PD, ATP5PO, COX4I1, COX5B, COX6B1, COX7A1, COX7A2L, COX7B, CYC1, GHITM, LETM1, MT−CO3, MT−CYB, MT−ND1, NDUFA10, NDUFA12, NDUFA4, NDUFA9, NDUFB10, NDUFB4, NDUFB5, NDUFB6, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS4, NDUFS6, NDUFS7, NDUFS8, NDUFV1, NDUFV2, NNT, PHB2, SLC16A1, SLC25A12, SLC25A3, SLC25A4, UQCRC1, UQCRFS1, UQCRH

Mitochondrial Protein Import

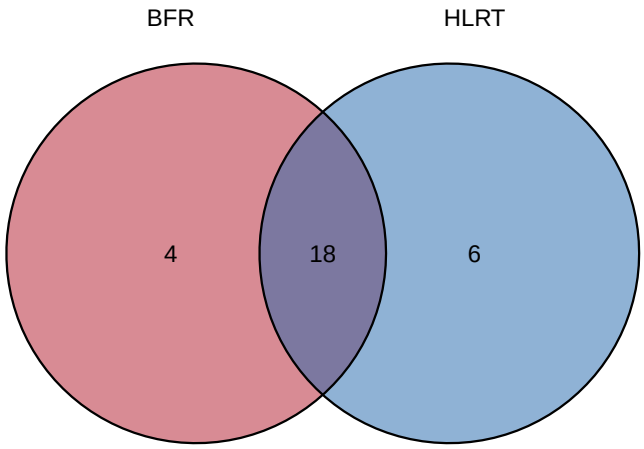
BFR: NES 2.94, padj 1.4e−13
HLRT: NES 2.54, padj 2.1e−08
shared 25, expected 19.2



Shared: ACO2, ATP5F1A, CHCHD10, CHCHD2, CHCHD3, CMC2, COA4, COA6, COX17, CS, CYC1, HSPA9, HSPD1, IDH3G, LDHD, MTX1, PITRM1, PMPCA, SLC25A12, SLC25A4, TIMM13, TIMM21, TIMM44, TIMM50, TIMM8A

Hypertrophic Cardiomyopathy Hcm

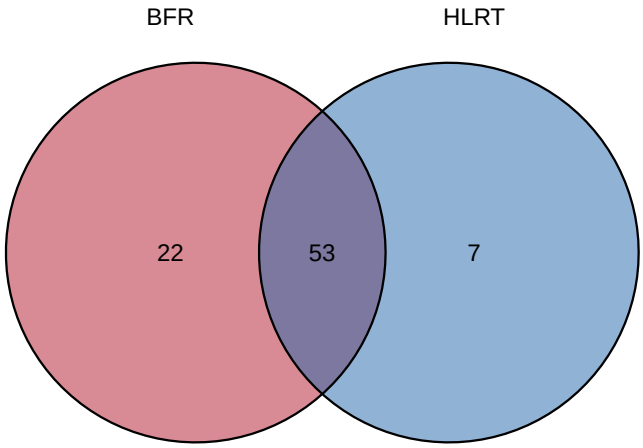
BFR: NES −1.96, padj 0.001
HLRT: NES −2.32, padj 3.7e−05
shared 18, expected 13.9



Shared: ATP2A2, CACNA1S, CACNA2D1, CACNB1, CACNG1, DAG1, DMD, ITGA7, LAMA2, MYH7, MYL2, MYL3, PRKAG3, RYR2, SGCA, SGCB, SGCD, SGCG

Muscle Organ Development

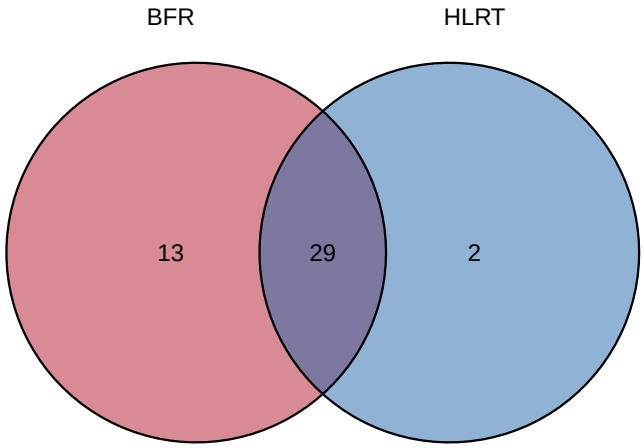
BFR: NES −2.38, padj 1.6e−08
HLRT: NES −1.98, padj 0.00013
shared 53, expected 38.8



Shared: ACTA1, ACTN3, BCL2, CACNA1S, CASQ1, CAV2, CAVIN4, CFL2, COL6A3, COPS2, CTF1, DAG1, DMD, EMD, FHL3, FXR1, GMPPA, GPC1, GPX1, HDGFL2, ITGA7, JPH1, LAMA2, LMNA, MAPK12, MAPK14, MYH14, MYH15, MYH7, MYL2, MYL3, MYL6B, MYLK2, MYOZ2, NEB, PLEC, PPP3CA, PPP3CB, RYR1, RYR2, SGCA, SGCB, SGCD, SGCG, SMTNL1, STAC3, SVIL, TCAP, TNNC1, TNNI1, TPM1, TRIM72, TTN

Regulation Of Muscle Contraction

BFR: NES −2.13, padj 5.4e−05
HLRT: NES −2.12, padj 0.00023
shared 29, expected 20.0

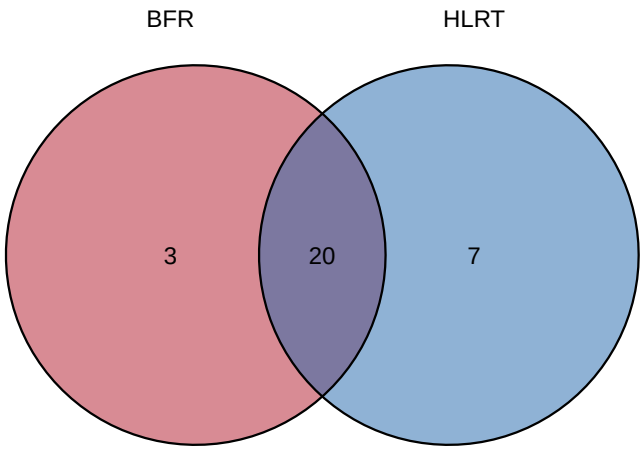


Shared: ACTN3, ATP2A1, ATP2A2, CACNA1S, CACNA2D1, CACNB1, CACNG1, CASQ1, DMD, ENO1, GSTM2, GSTO1, HRC, JPH1, MAP2K1, MYH7, MYL2, MYL3, MYL5, MYLK2, PLN, RYR2, SCN4A, TMEM38B, TNNC1, TNNC2, TNNT1, TNNT3, TPM1

S15. Leading edges by leg

Acyl Coa Metabolic Process

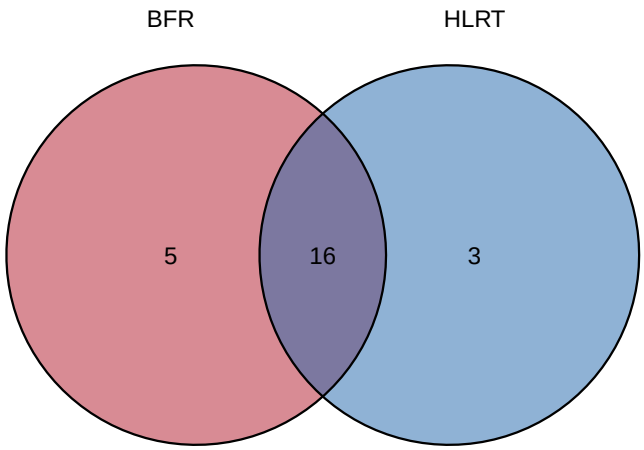
BFR: NES 2.33, padj 9.1e−06
HLRT: NES 2.21, padj 8.3e−05
shared 20, expected 11.5



Shared: ACOT9, ACSF3, ACSL1, ACSS1, BCKDK, CBR4, DLAT, DLST, HSD17B12, MMUT, NUDT19, OGDH, PCCA, PDHA1, PDK2, PDK4, PDP1, SUCLA2, SUCLG1, SUCLG2

Tumor Necrosis Factor Superfamily Cytokine Production

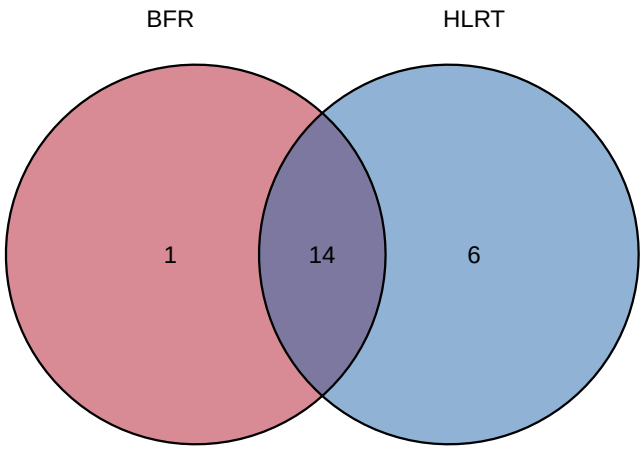
BFR: NES −2.17, padj 7.9e−05
HLRT: NES −1.80, padj 0.015
shared 16, expected 12.1



Shared: FXR1, HSPB1, NRDC, PARK7, PIK3R1, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, THBS1

Arrhythmogenic Right Ventricular Cardiomyopathy Arvc

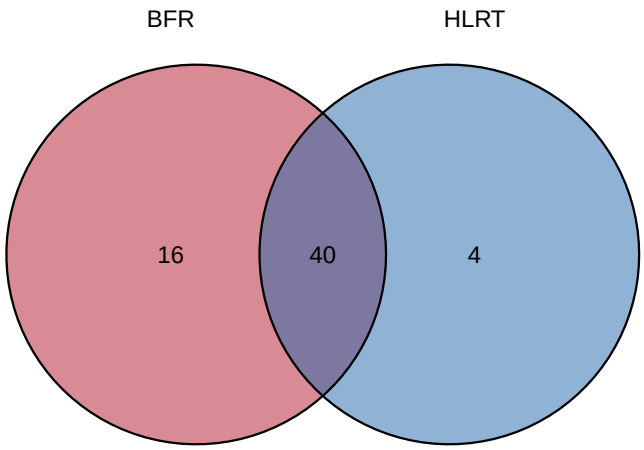
BFR: NES −2.17, padj 0.00014
HLRT: NES −2.25, padj 0.00026
shared 14, expected 10.3



Shared: ATP2A2, CACNA1S, CACNA2D1, CACNB1, CACNG1, CTNNA1, DAG1, DMD, EMD, ITGA7, SGCA, SGCB, SGCD, SGCG

Muscle Contraction

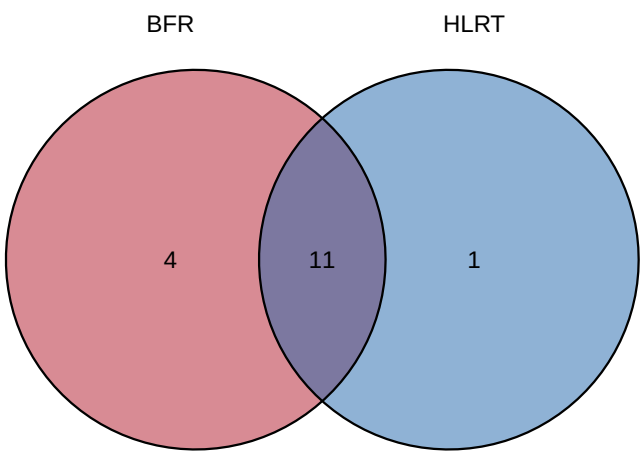
BFR: NES −2.15, padj 1.9e−05
HLRT: NES −1.68, padj 0.0064
shared 40, expected 28.7



Shared: ACTA1, ACTN3, ALDH2, ATP2A1, ATP2A2, ATP2B2, CACNB1, CAMK2A, CASQ1, DMD, LMOD1, MYBPC1, MYBPC2, MYH8, MYL1, MYL2, MYL3, MYL4, MYL5, MYL6B, NEB, PLN, PXN, RYR1, RYR2, SCN4A, SORBS1, STIM1, TCAP, TMOD4, TNNC1, TNNC2, TNNI1, TNNI2, TNNT1, TNNT3, TPM1, TPM3, TRIM72, TTN

T Cell Homeostasis

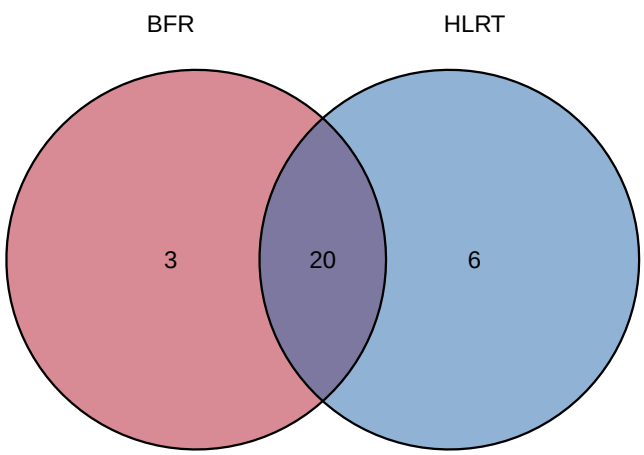
BFR: NES −2.17, padj 0.00021
HLRT: NES −2.25, padj 0.00023
shared 11, expected 8.2



Shared: BCL2, PPP2CA, PPP3CB, PSMA3, PSMA4, PSMA5, PSMB1, PSMB2, PSMB3, PSMB4, STAT5B

Amide Biosynthetic Process

BFR: NES 2.24, padj 3.1e−05
HLRT: NES 1.91, padj 0.0013
shared 20, expected 10.1

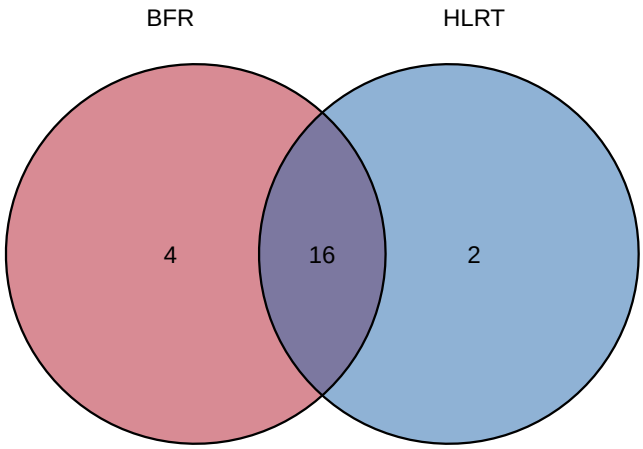


Shared: ACSF3, ACSL1, ACSS1, BCKDK, CARNS1, CBR4, DLAT, DLST, FH, HSD17B12, MECR, MMUT, OGDH, PCCA, PDHA1, PDK2, PDK4, PDP1, PRKAA1, SIRT3

S15. Leading edges by leg

Citric Acid Cycle TCA Cycle

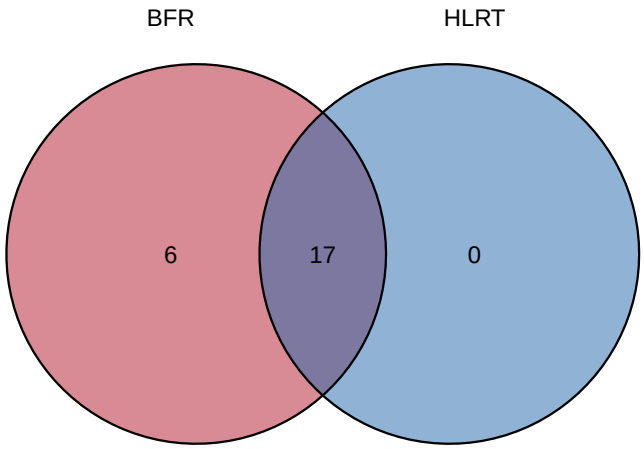
BFR: NES 2.71, padj 1.6e−08
HLRT: NES 2.35, padj 1.1e−05
shared 16, expected 12.4



Shared: ACO2, CS, DLST, FH, IDH3A, IDH3B, IDH3G, MDH2, NNT, OGDH, SDHA, SIRT3, SUCLA2, SUCLG1, SUCLG2, TRAP1

Non Integrin Membrane ECM Interactions

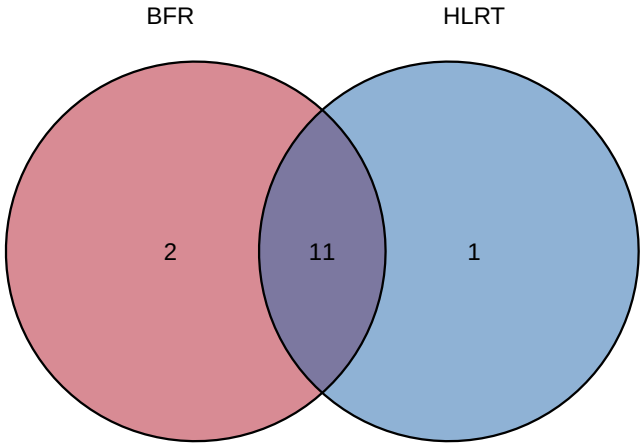
BFR: NES −2.09, padj 0.00051
HLRT: NES −2.36, padj 5.5e−05
shared 17, expected 13.5



Shared: ACTA1, COL4A1, COL4A2, DAG1, DMD, DTNA, HSPG2, LAMA2, LAMB2, LAMC1, SGCA, SGCB, SGCD, SGCG, SNTA1, SNTB2, SSPN

Proteasomal Ubiquitin Independent Protein Catabolic Process

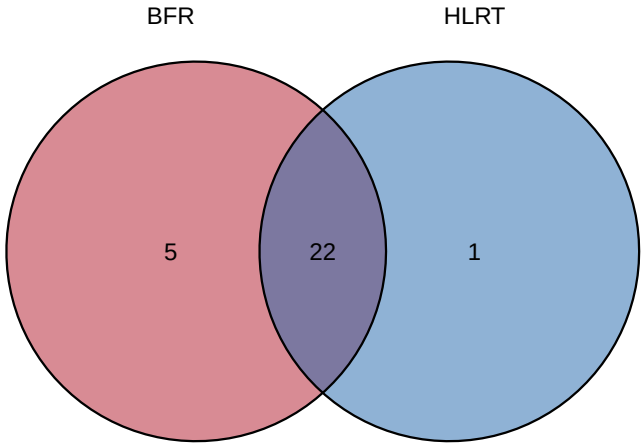
BFR: NES −2.15, padj 0.00043
HLRT: NES −2.13, padj 0.0013
shared 11, expected 10.4



Shared: PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6

Skeletal Muscle Contraction

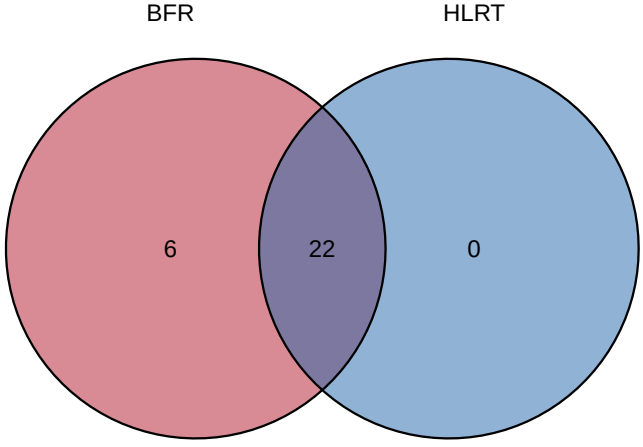
BFR: NES −1.78, padj 0.013
HLRT: NES −2.33, padj 6.2e−05
shared 22, expected 19.4



Shared: ACTN3, ATP2A1, CA3, CASQ1, DMD, FKBP1A, GSTM2, GSTO1, MYH14, MYH7, MYH8, RCSD1, SCN4A, STAC3, SYNM, TCAP, TNNC1, TNNC2, TNNI2, TNNT1, TNNT3, TTN

Striated Muscle Contraction

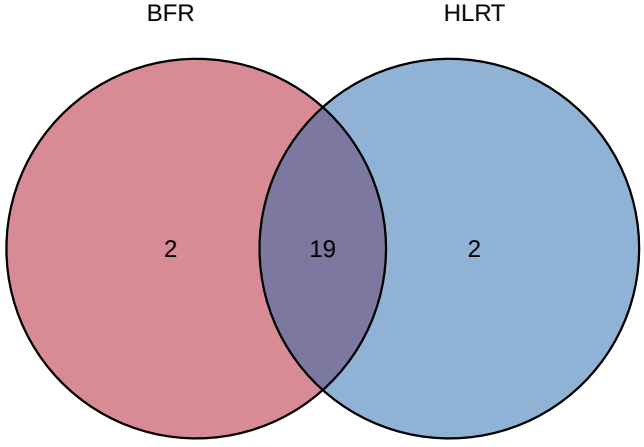
BFR: NES −1.80, padj 0.013
HLRT: NES −2.22, padj 0.00027
shared 22, expected 20.5



Shared: ACTA1, ACTN3, DMD, MYBPC1, MYBPC2, MYH8, MYL2, MYL3, MYL4, NEB, TCAP, TMOD1, TMOD4, TNNC1, TNNC2, TNNI2, TNNT1, TNNT3, TPM1, TPM2, TPM3, TTN

Tricarboxylic Acid Cycle

BFR: NES 2.70, padj 9e−09
HLRT: NES 2.17, padj 0.00034
shared 19, expected 15.8

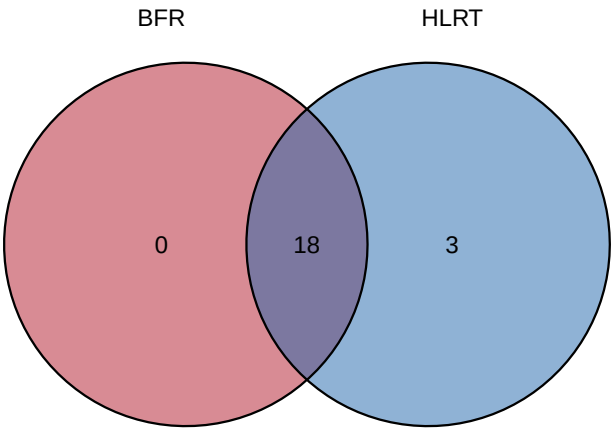


Shared: ACO2, CS, DLAT, DLST, FH, IDH3A, IDH3B, IDH3G, MDH2, NNT, OGDH, PDHA1, PDHB, SDHA, SDHC, SDHD, SUCLA2, SUCLG1, SUCLG2

S15. Leading edges by leg

Citrate Cycle TCA Cycle

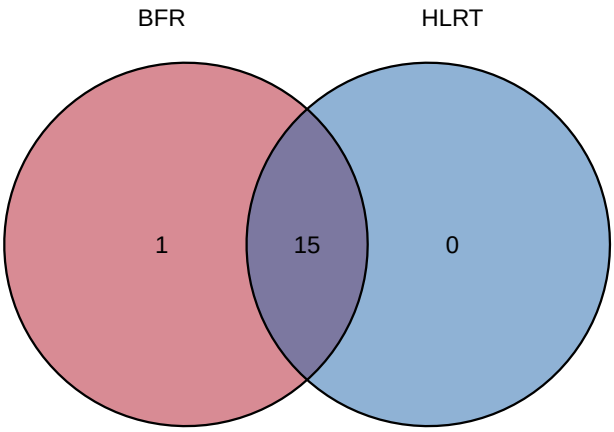
BFR: NES 2.59, padj 5.4e−07
HLRT: NES 2.20, padj 0.00017
shared 18, expected 14.5



Shared: ACO2, CS, DLAT, DLST, FH, IDH3A, IDH3B, IDH3G, MDH2, OGDH, PDHA1, PDHB, SDHA, SDHC, SDHD, SUCLA2, SUCLG1, SUCLG2

Formation Of The Dystrophin Glycoprotein Complex DGC

BFR: NES −2.00, padj 0.0026
HLRT: NES −2.32, padj 6.8e−05
shared 15, expected 13.3



Shared: ACTA1, DAG1, DMD, DTNA, HSPG2, LAMA2, LAMB2, LAMC1, SGCA, SGCB, SGCD, SGCG, SNTA1, SNTB2, SSPN

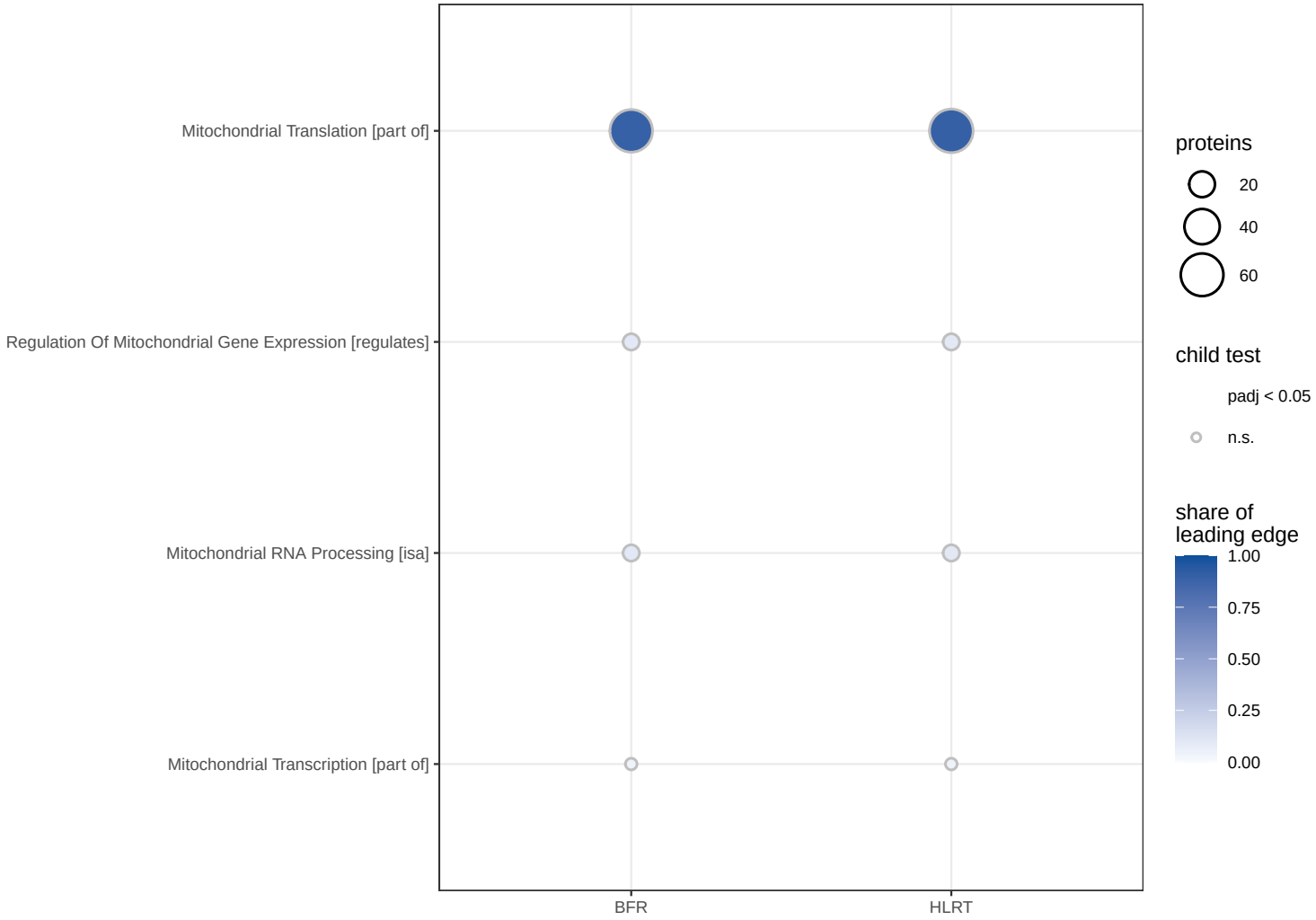
S16. Pathway NES, BFR against HLRT

1990 pathways, Spearman rho 0.78



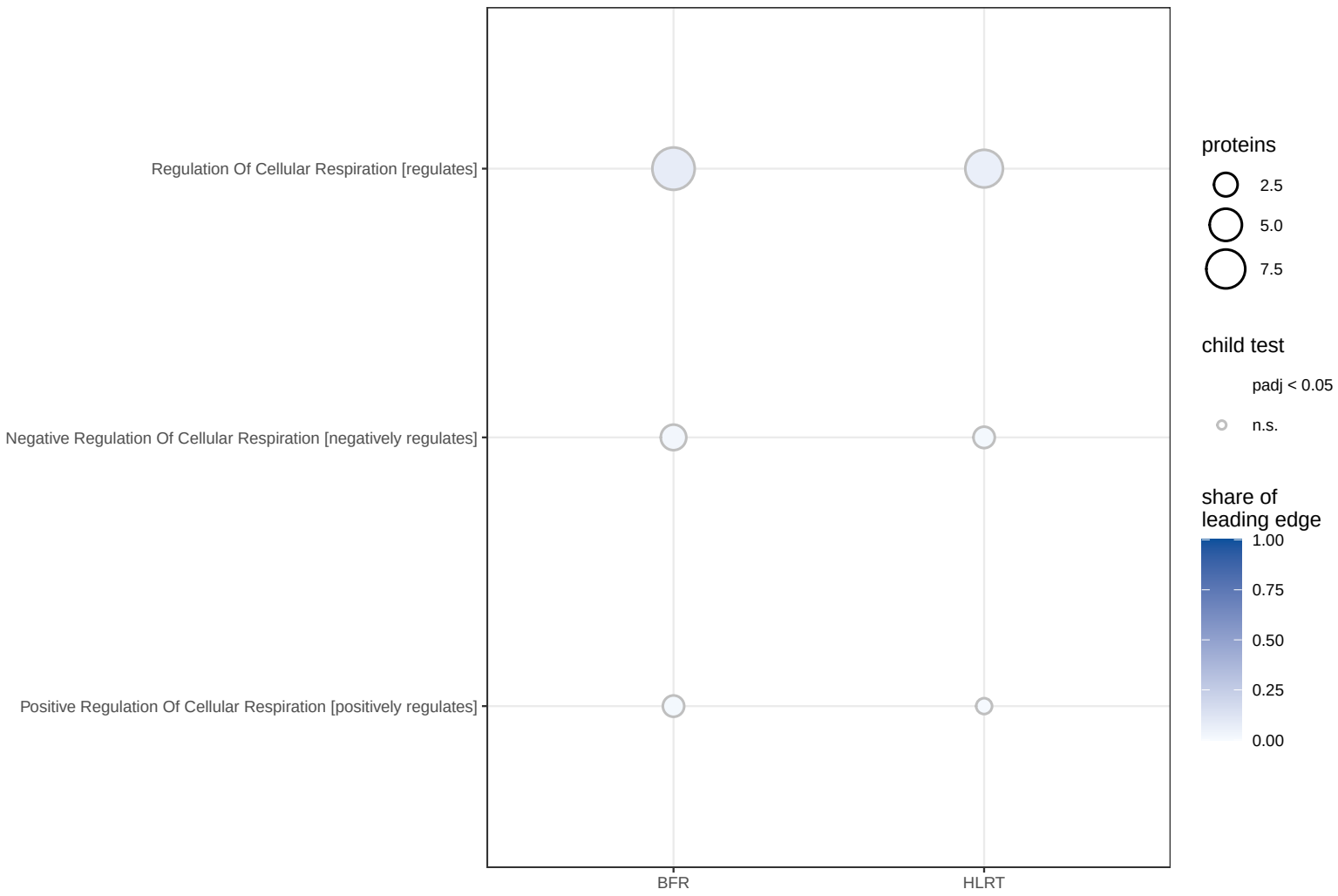
S17. Mitochondrial Gene Expression

leading edge: BFR 67, HLRT 70; in significant descendants: BFR 90%, HLRT 90%



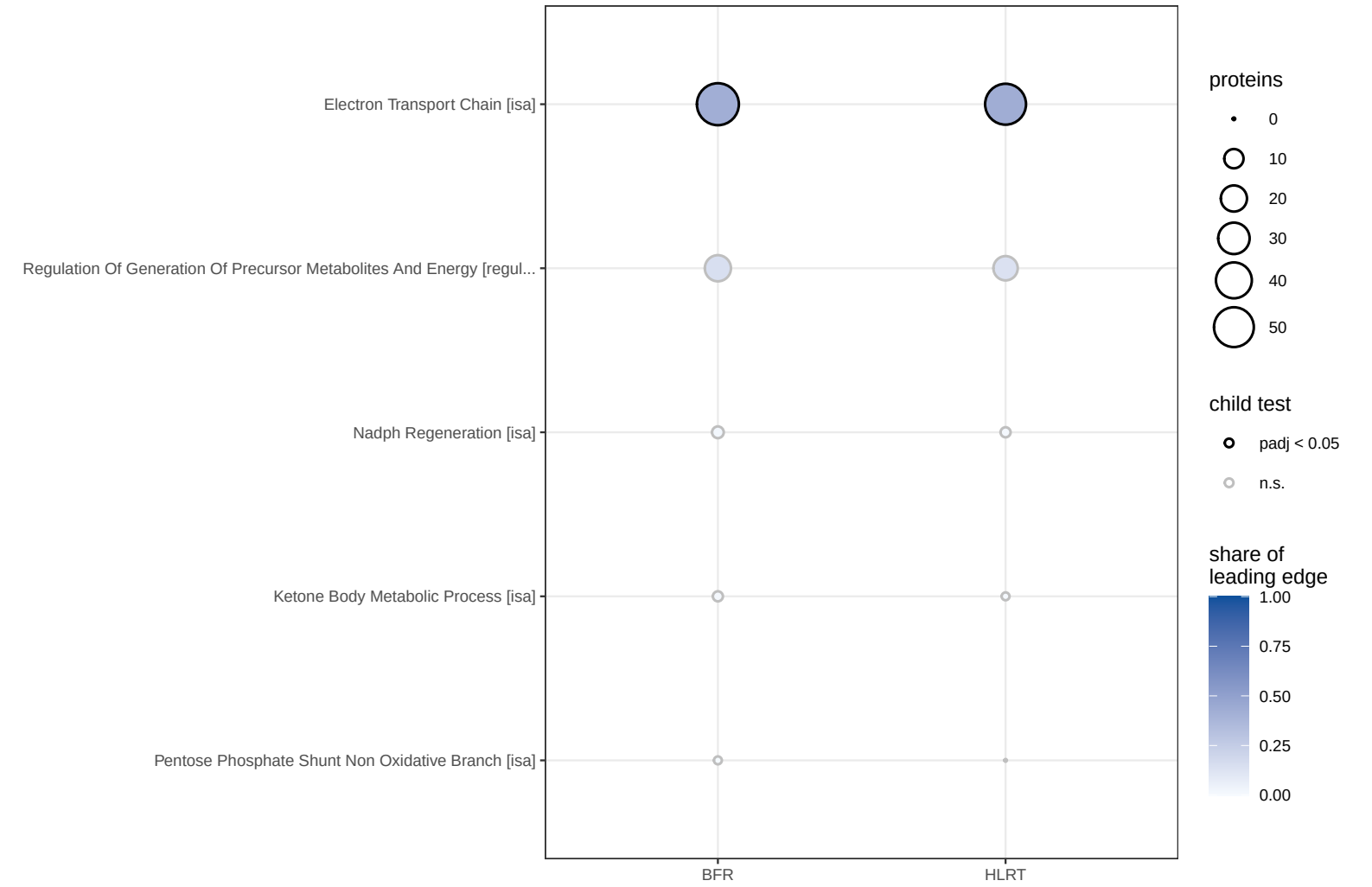
S17. Cellular Respiration

leading edge: BFR 113, HLRT 108; in significant descendants: BFR 74%, HLRT 77%



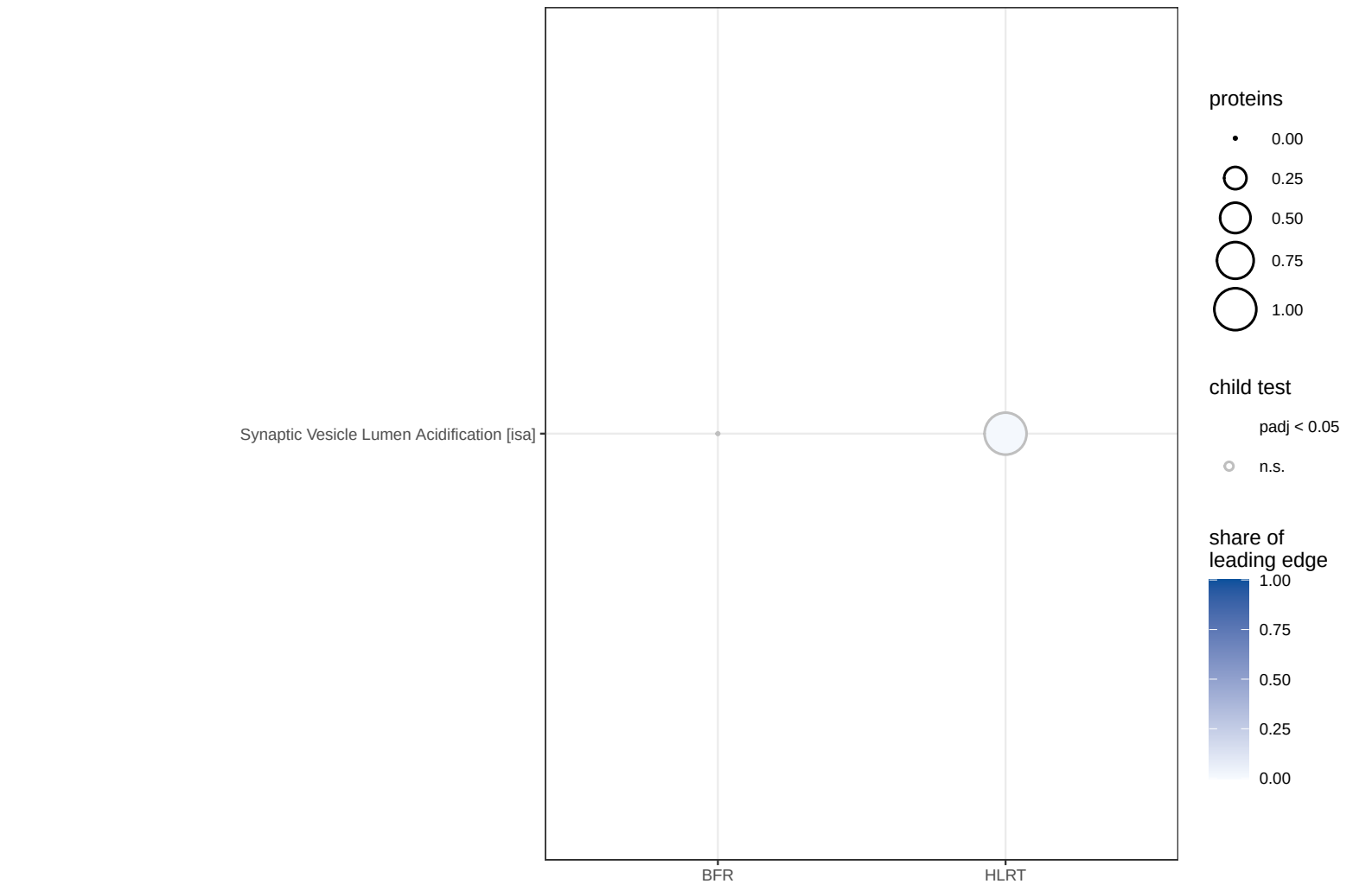
S17. Generation Of Precursor Metabolites And Energy

leading edge: BFR 132, HLRT 123; in significant descendants: BFR 86%, HLRT 87%



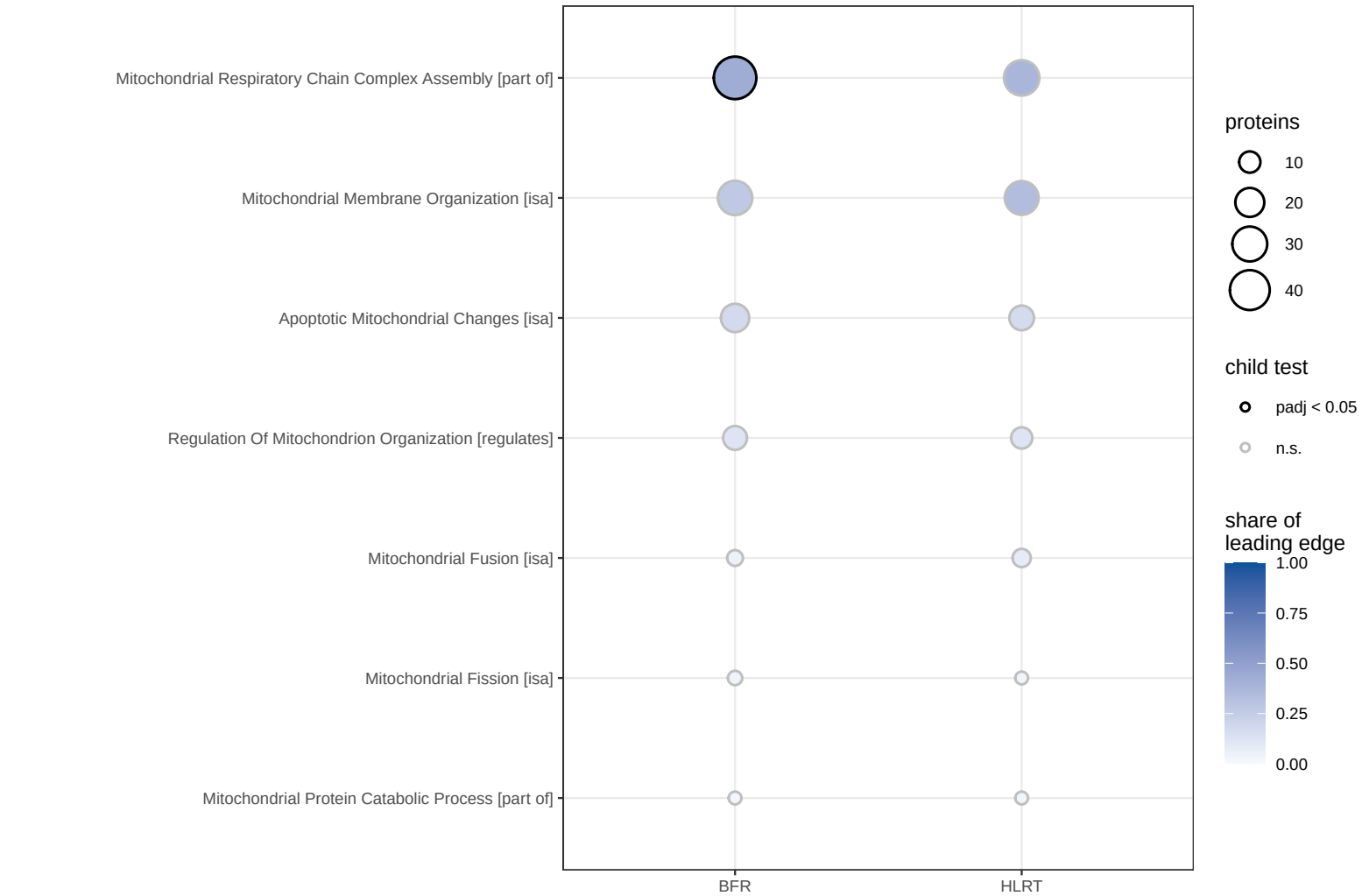
S17. Proton Transmembrane Transport

leading edge: BFR 63, HLRT 65; in significant descendants: BFR 0%, HLRT 0%



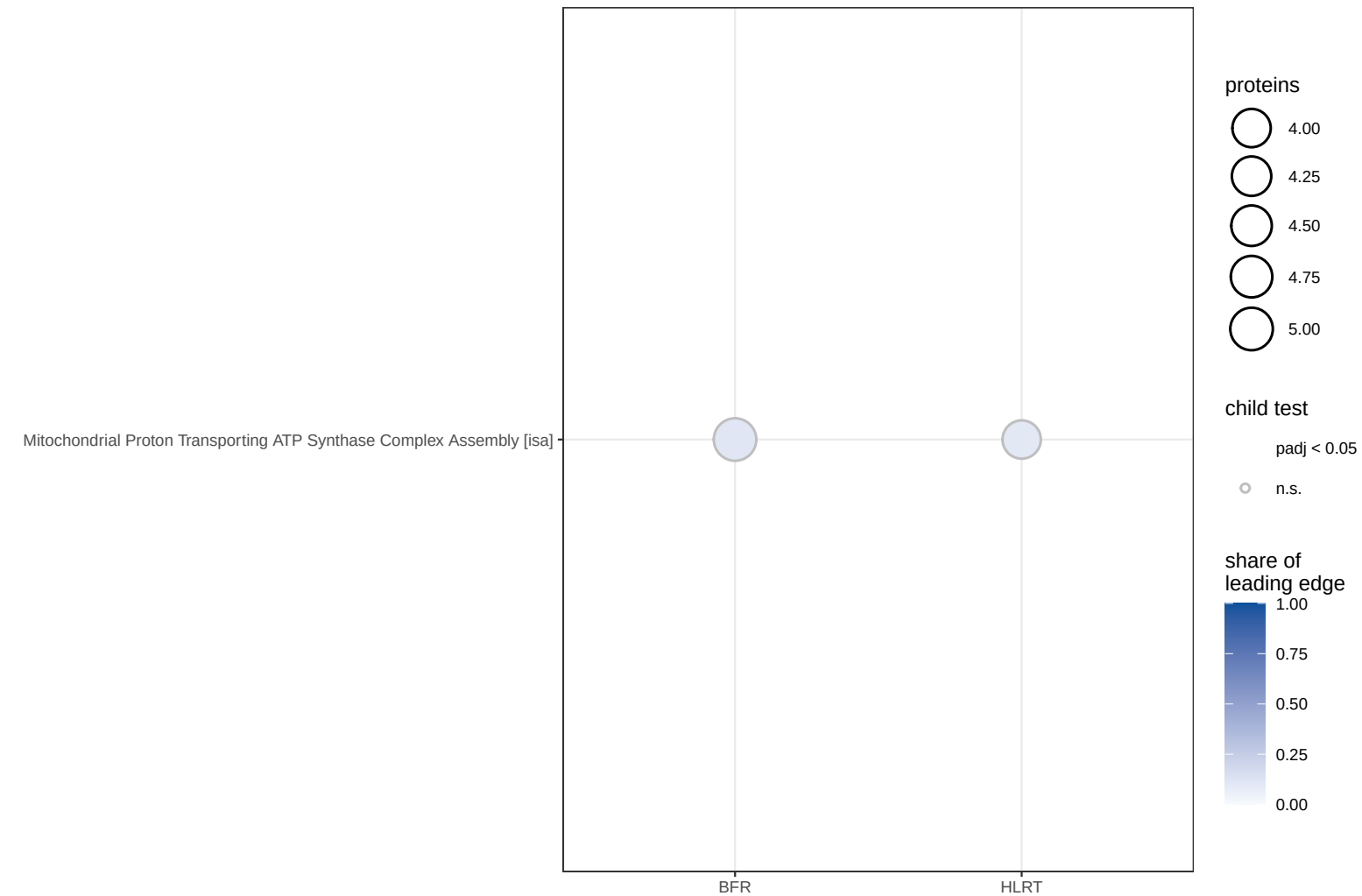
S17. Mitochondrion Organization

leading edge: BFR 106, HLRT 82; in significant descendants: BFR 68%, HLRT 68%



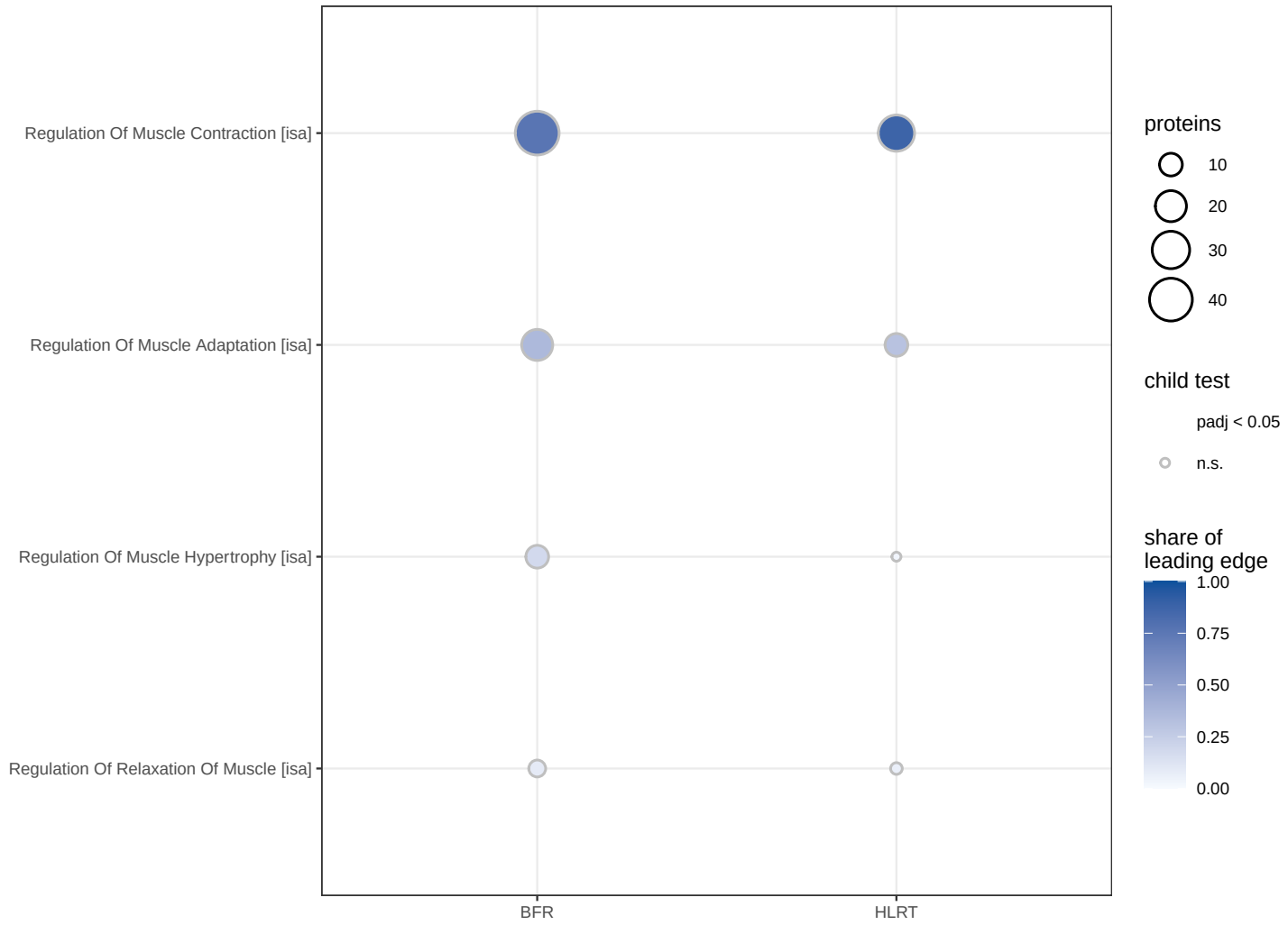
S17. Mitochondrial Respiratory Chain Complex Assembly

leading edge: BFR 45, HLRT 40; in significant descendants: BFR 0%, HLRT 0%



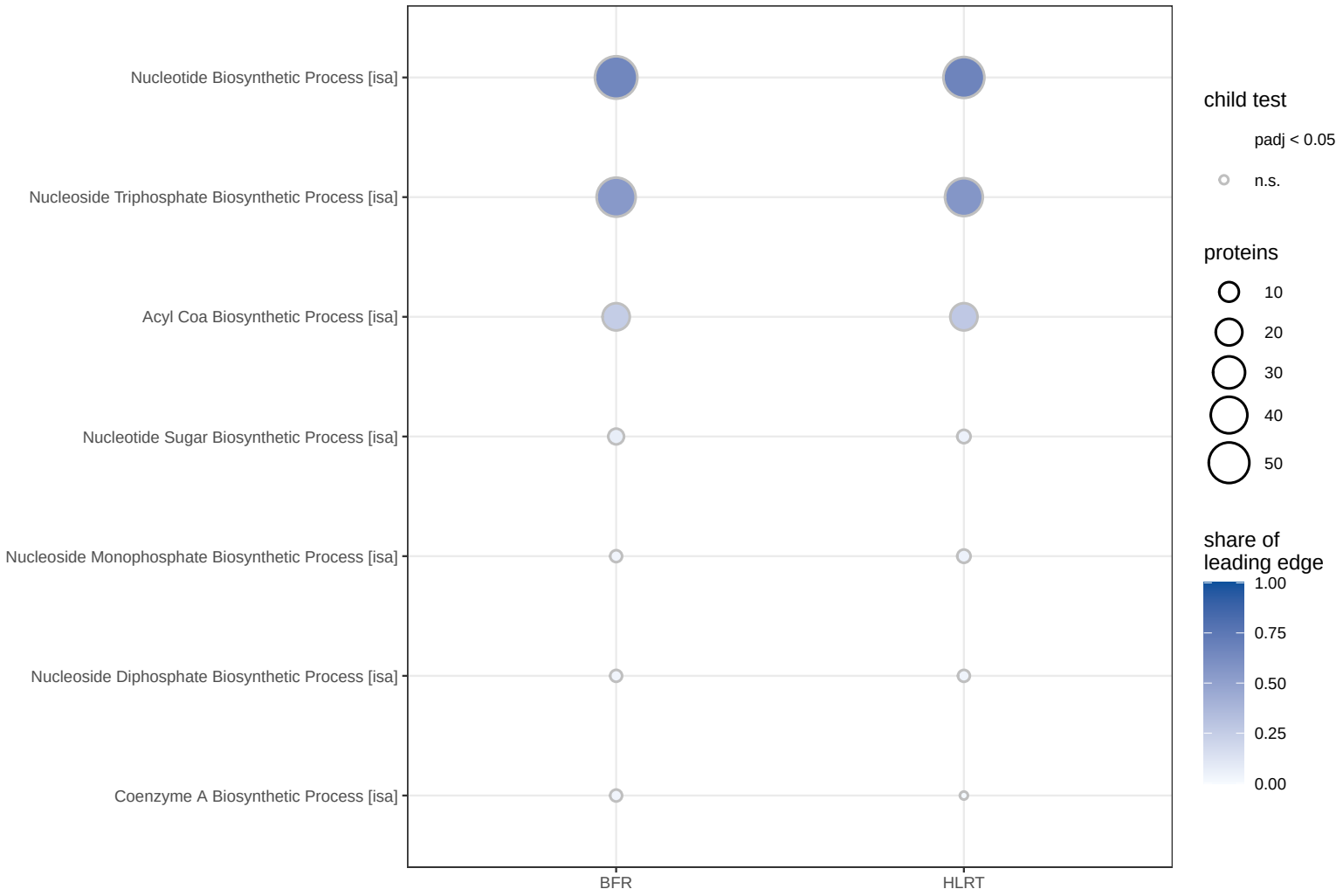
S17. Regulation Of Muscle System Process

leading edge: BFR 55, HLRT 32; in significant descendants: BFR 98%, HLRT 88%



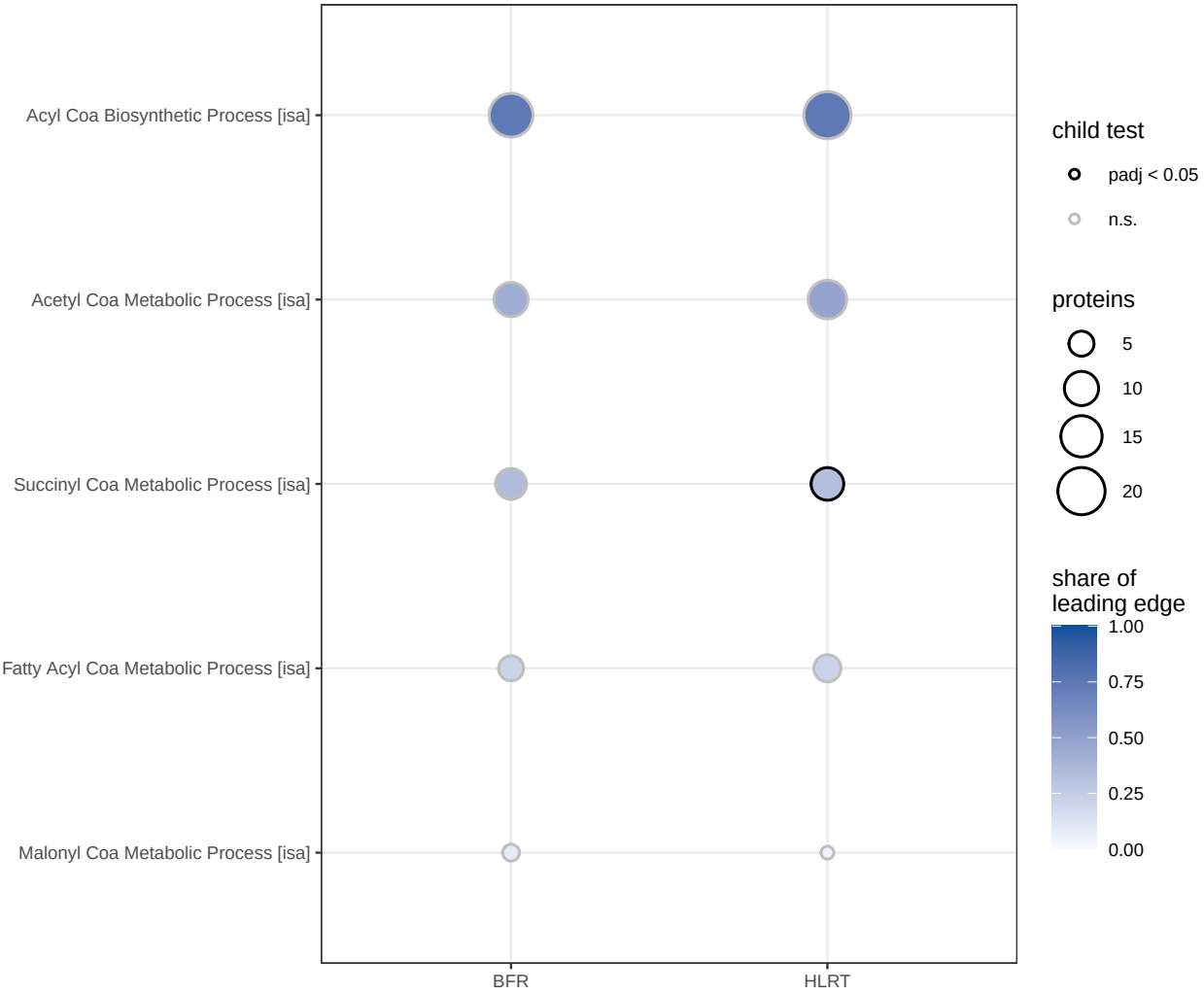
S17. Nucleoside Phosphate Biosynthetic Process

leading edge: BFR 84, HLRT 76; in significant descendants: BFR 89%, HLRT 95%



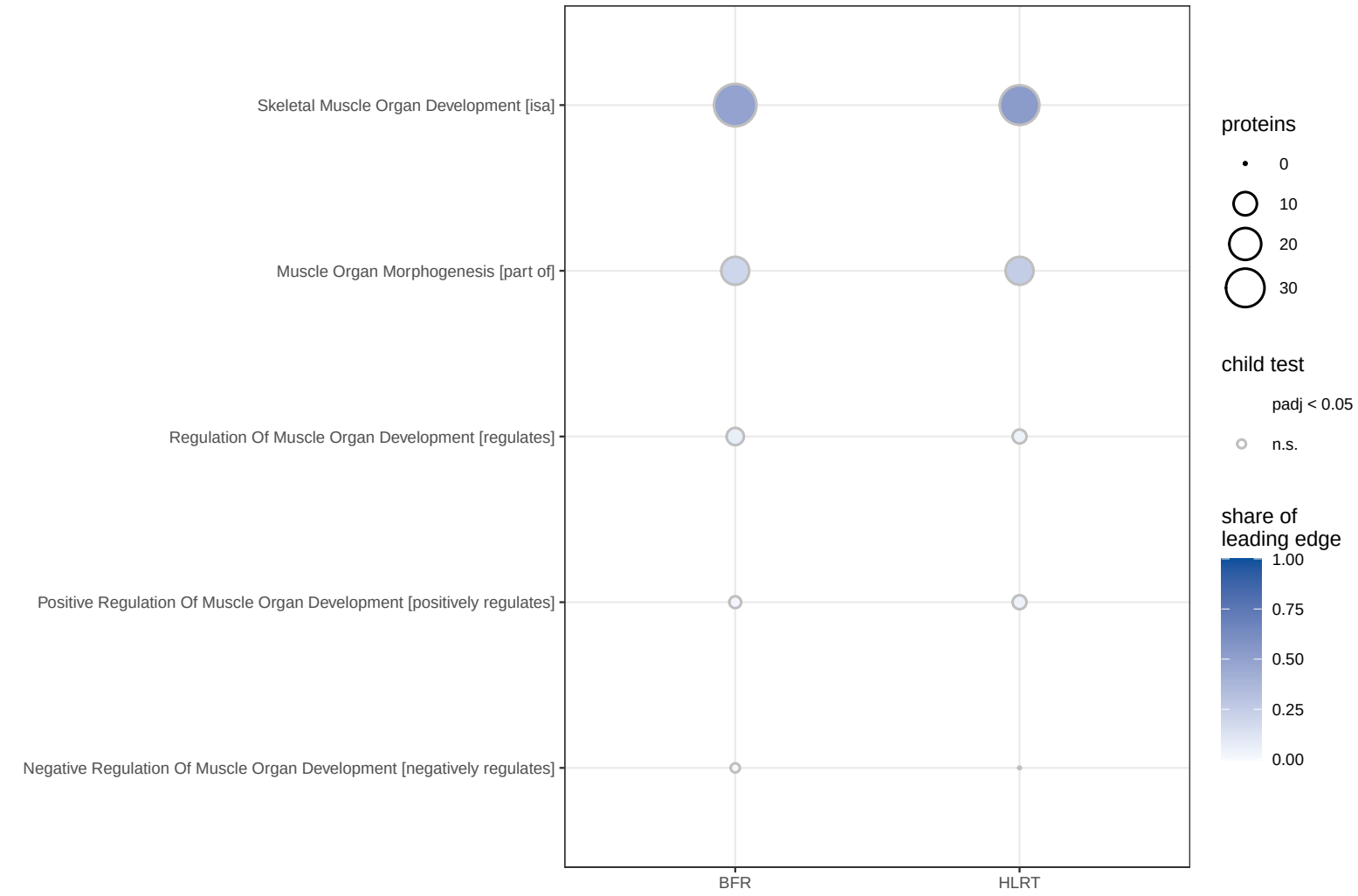
S17. Acyl Coa Metabolic Process

leading edge: BFR 23, HLRT 27; in significant descendants: BFR 74%, HLRT 78%



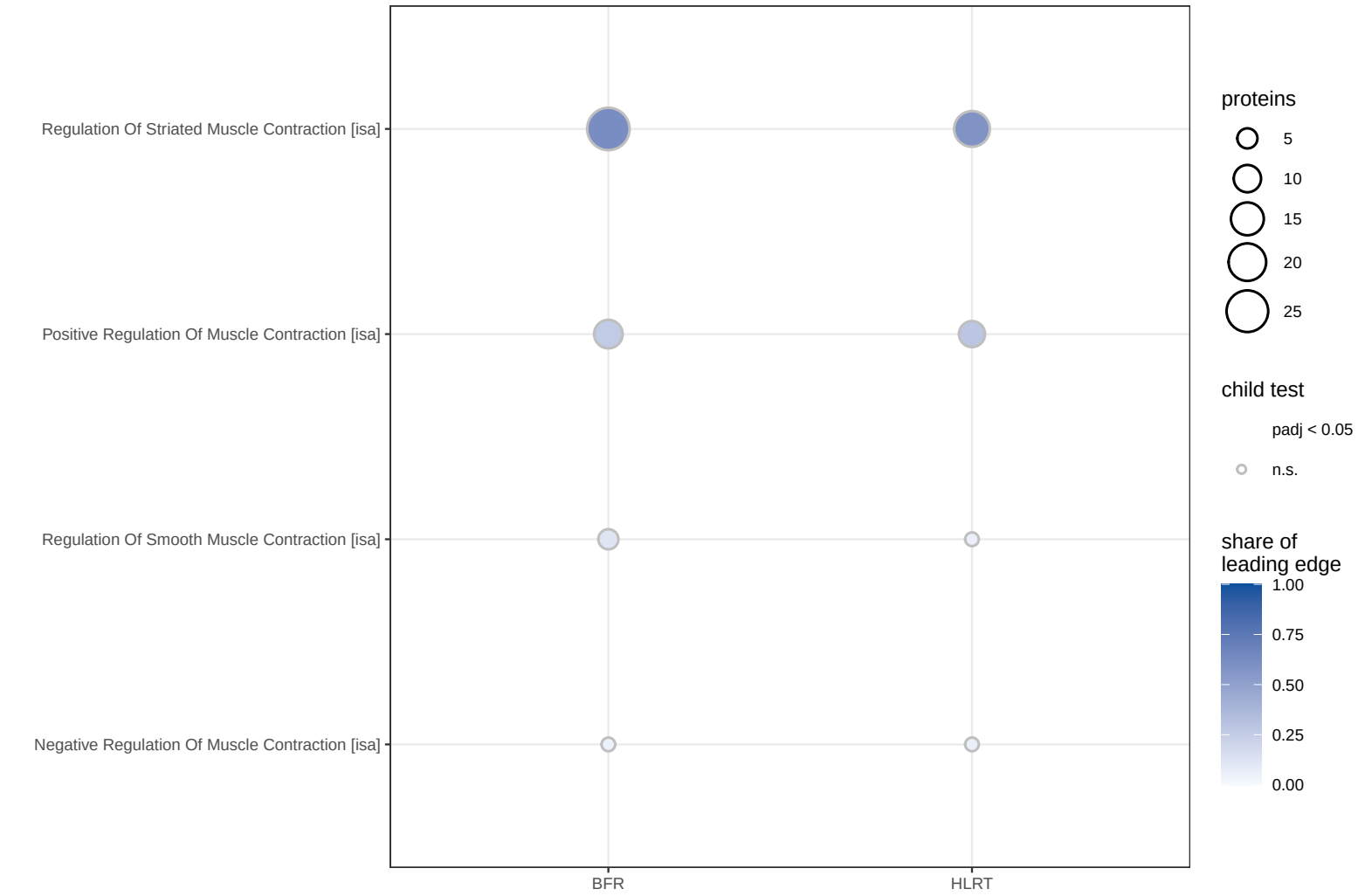
S17. Muscle Organ Development

leading edge: BFR 75, HLRT 60; in significant descendants: BFR 65%, HLRT 70%



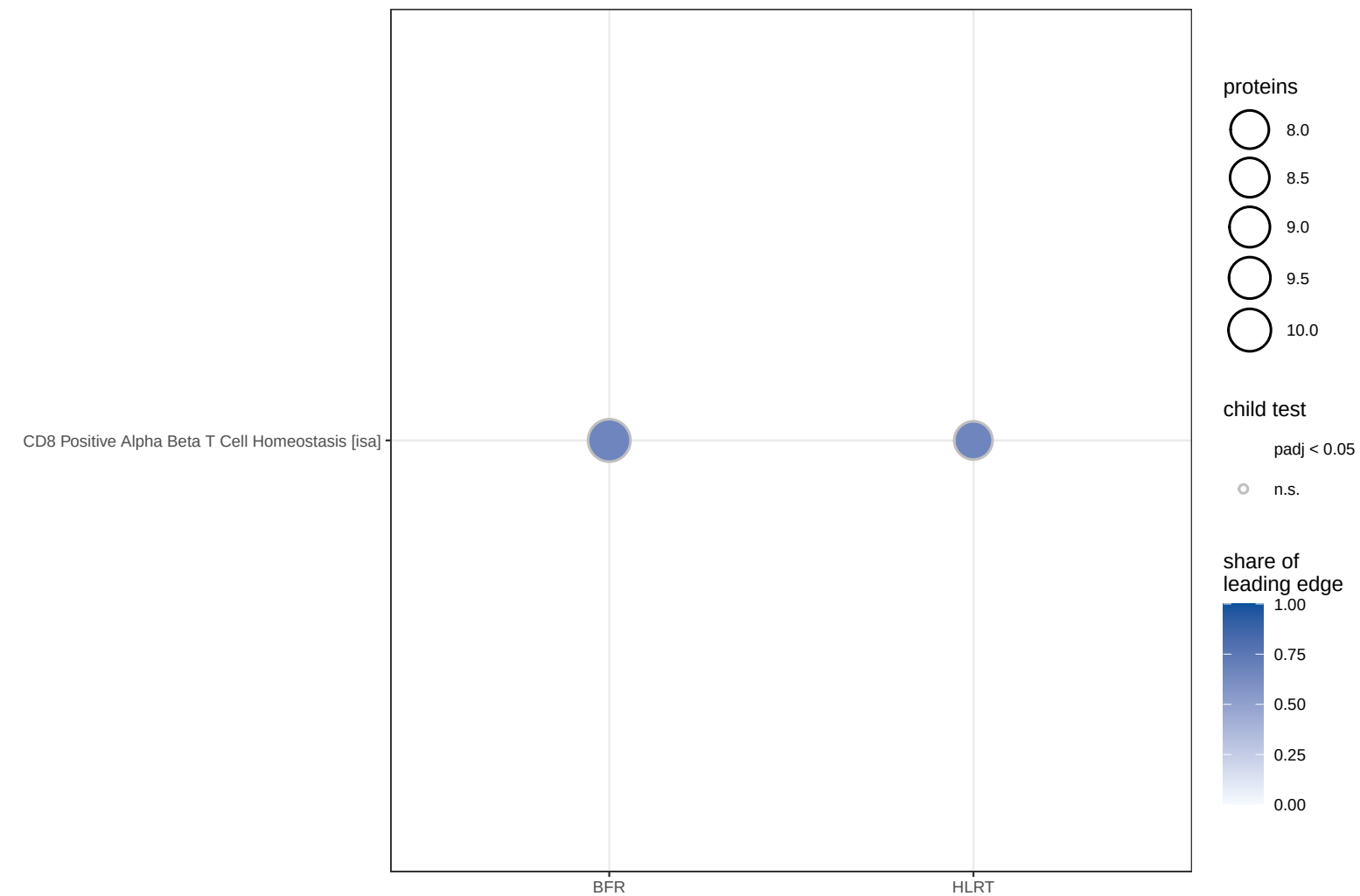
S17. Regulation Of Muscle Contraction

leading edge: BFR 42, HLRT 31; in significant descendants: BFR 62%, HLRT 58%



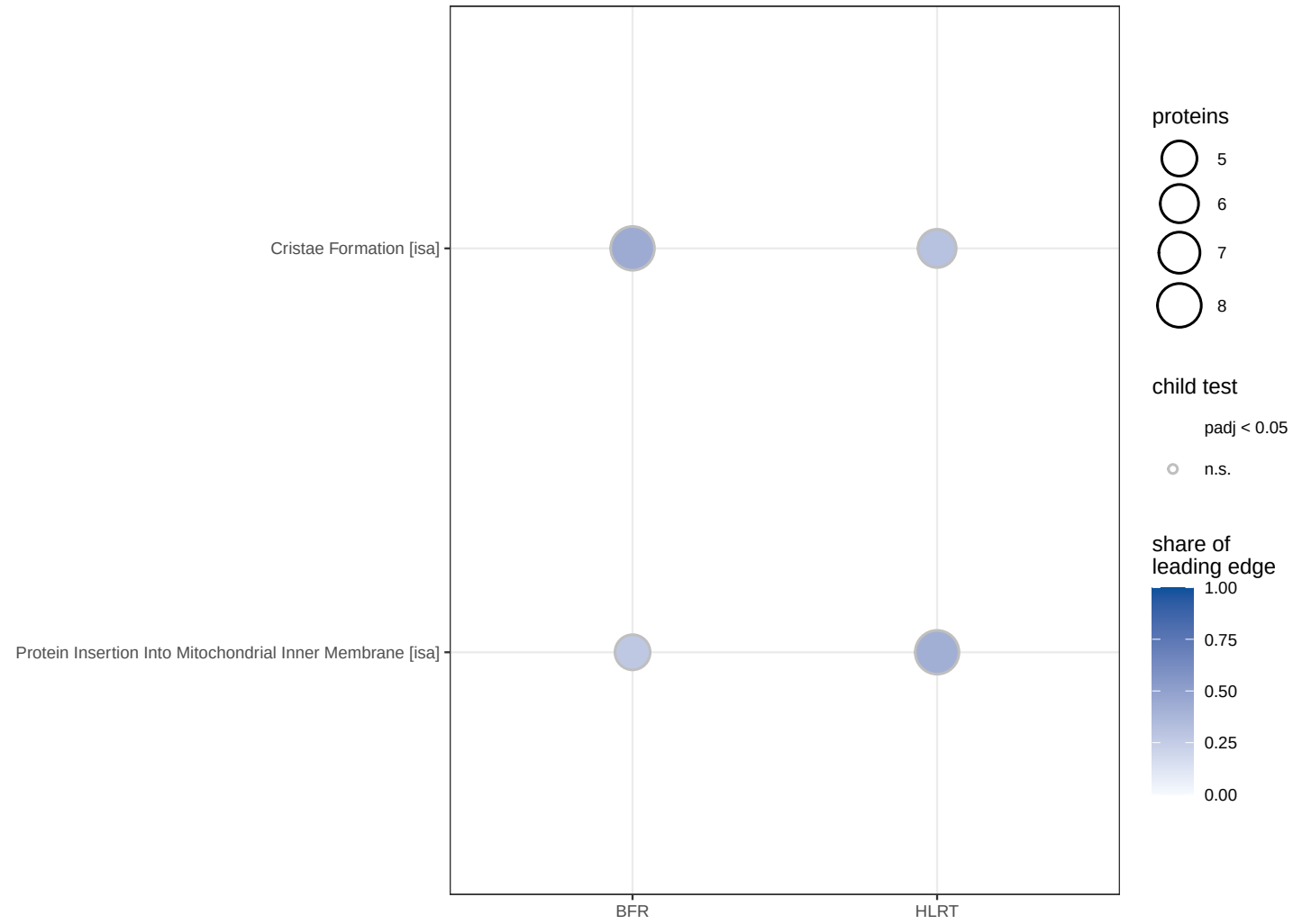
S17. T Cell Homeostasis

leading edge: BFR 15, HLRT 12; in significant descendants: BFR 0%, HLRT 0%



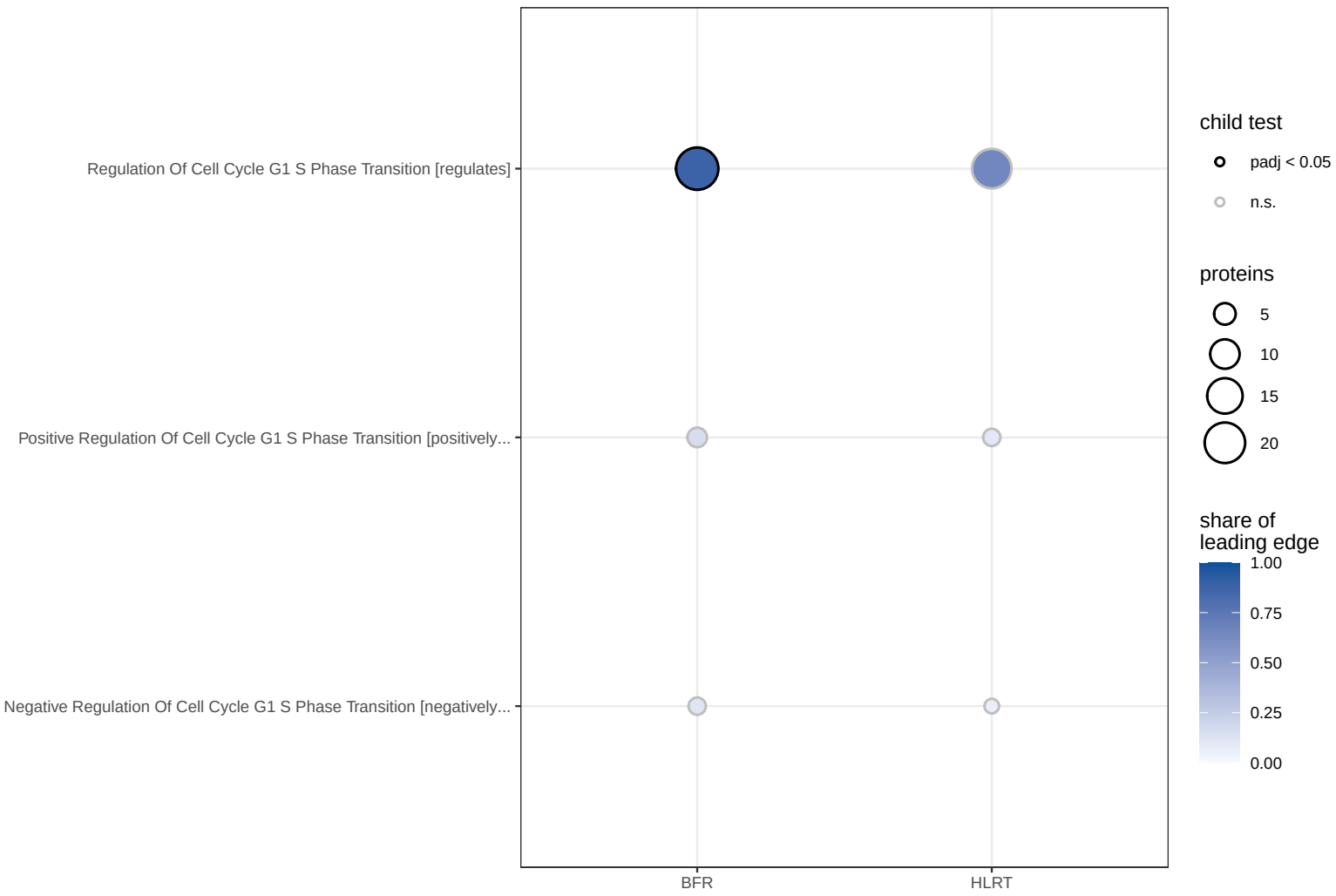
S17. Inner Mitochondrial Membrane Organization

leading edge: BFR 18, HLRT 19; in significant descendants: BFR 0%, HLRT 0%



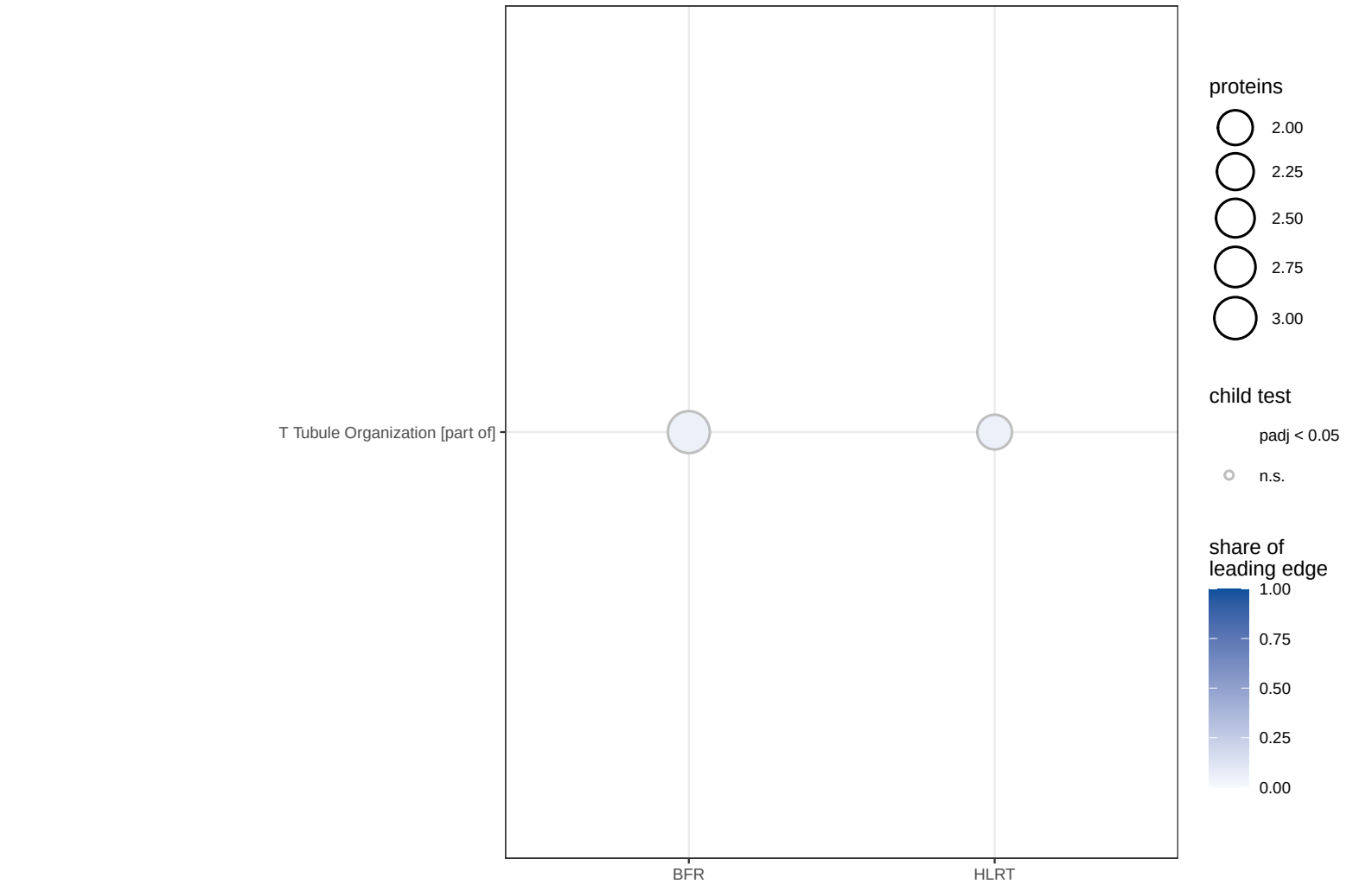
S17. Cell Cycle G1 S Phase Transition

leading edge: BFR 25, HLRT 29; in significant descendants: BFR 88%, HLRT 66%



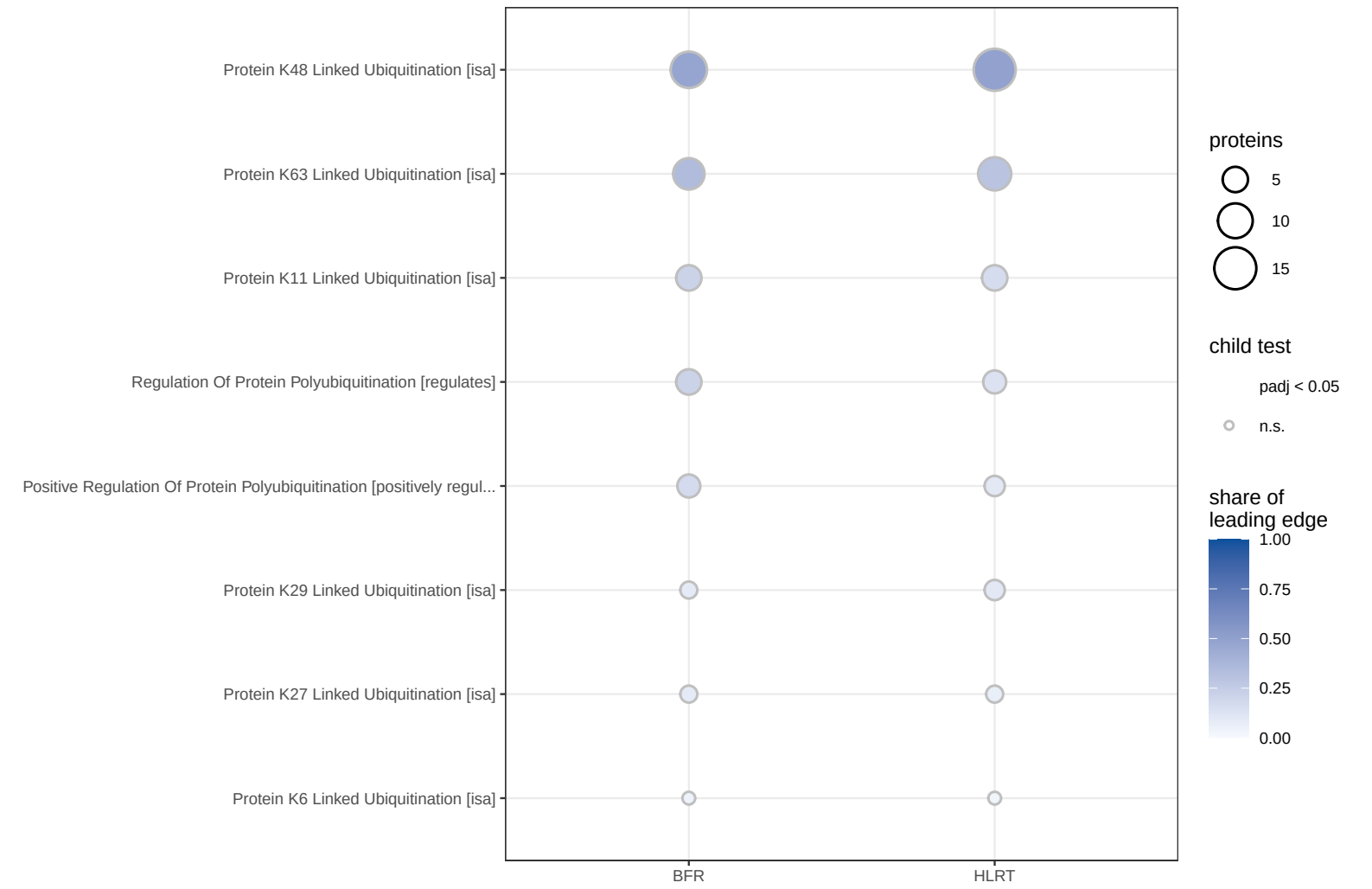
S17. Muscle Cell Development

leading edge: BFR 54, HLRT 35; in significant descendants: BFR 70%, HLRT 80%



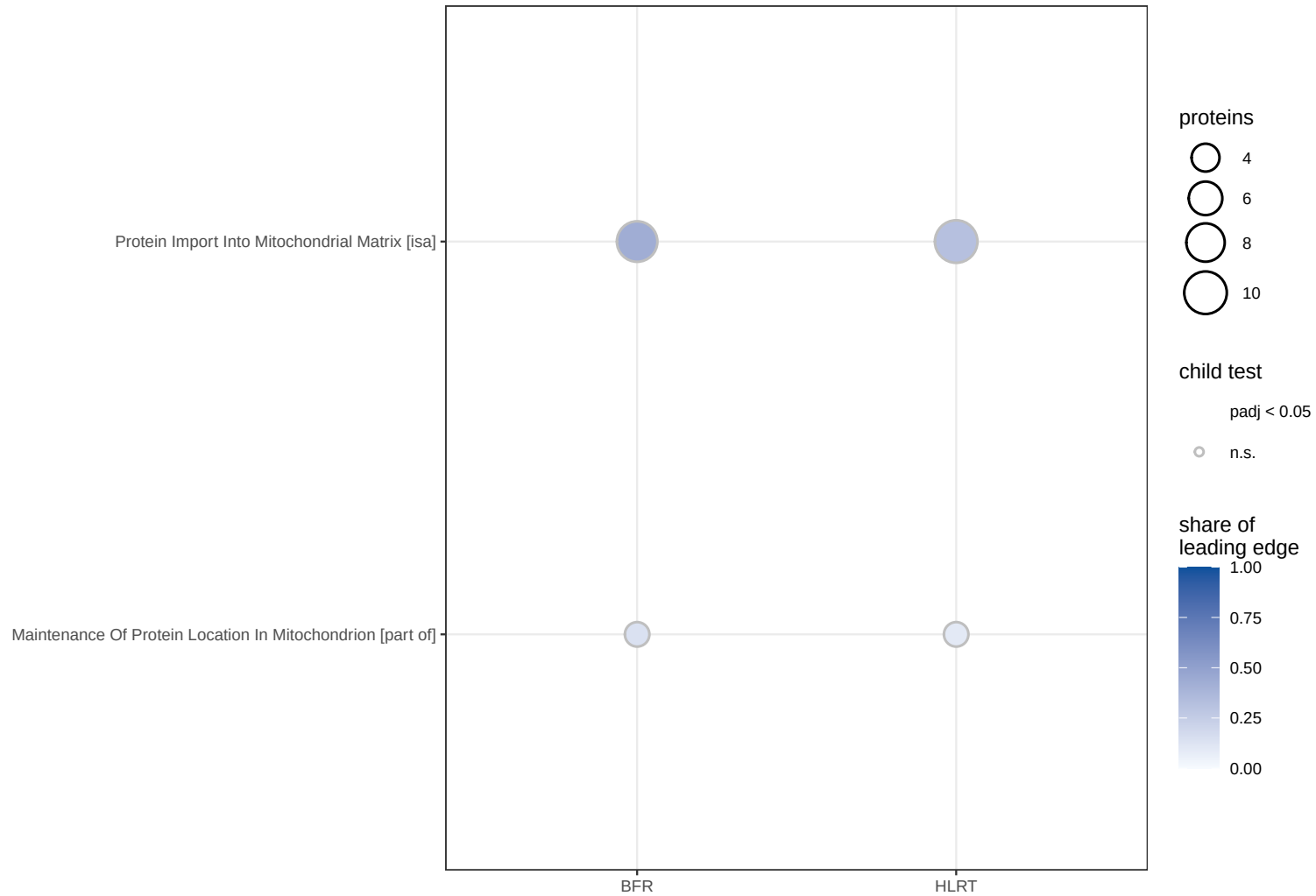
S17. Protein Polyubiquitination

leading edge: BFR 23, HLRT 30; in significant descendants: BFR 61%, HLRT 50%



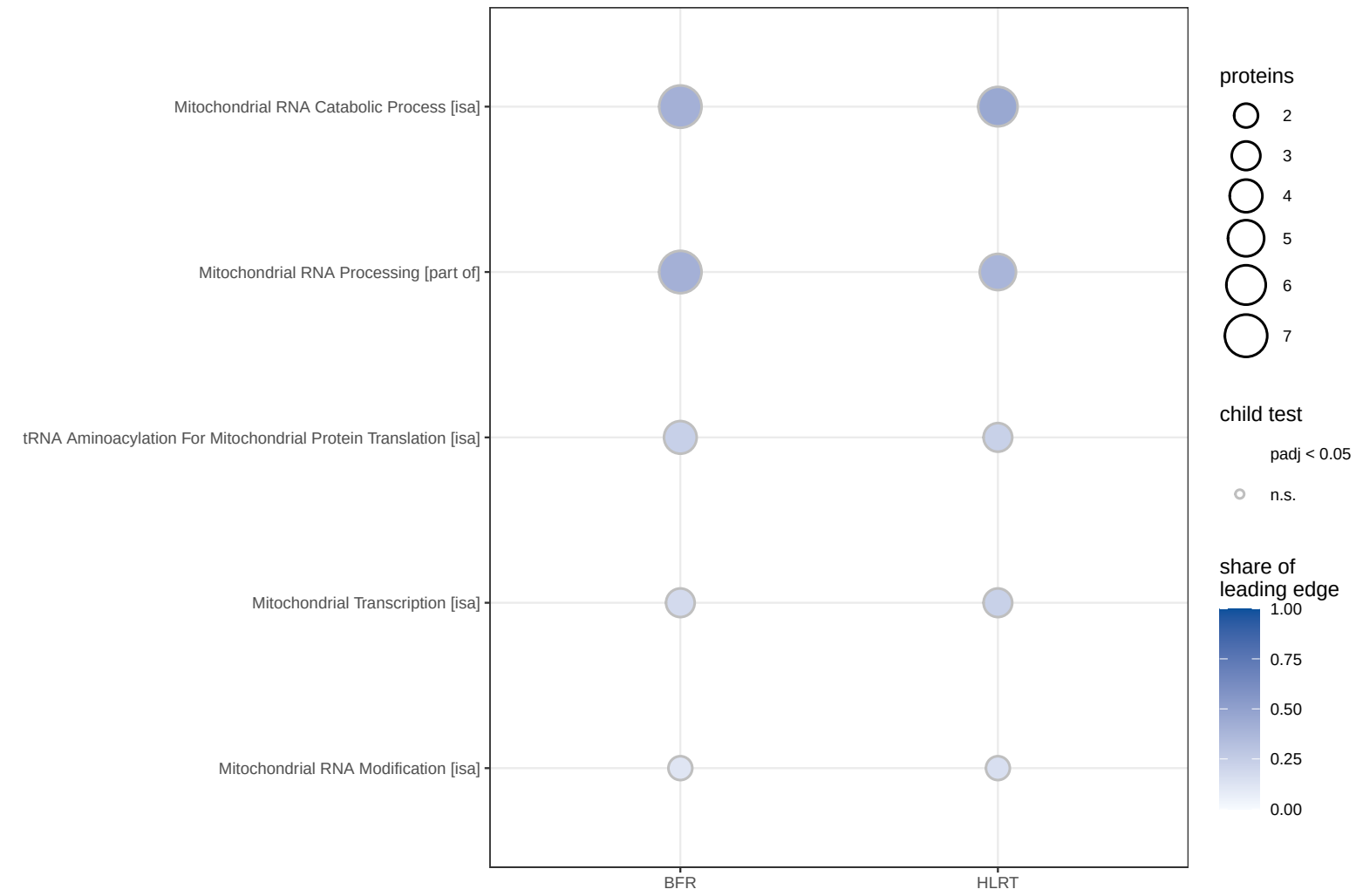
S17. Protein Localization To Mitochondrion

leading edge: BFR 21, HLRT 31; in significant descendants: BFR 43%, HLRT 32%



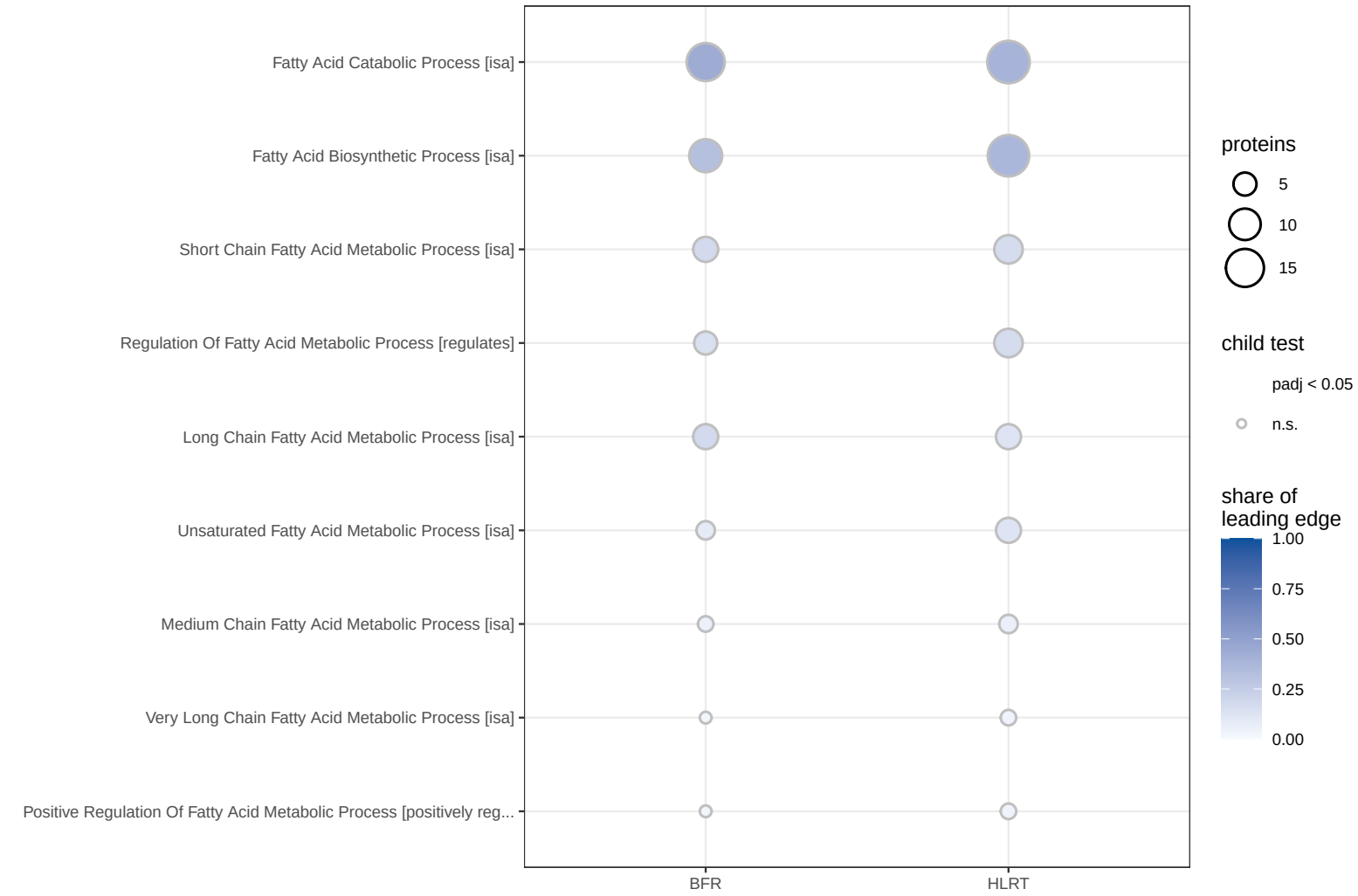
S17. Mitochondrial RNA Metabolic Process

leading edge: BFR 17, HLRT 13; in significant descendants: BFR 0%, HLRT 0%



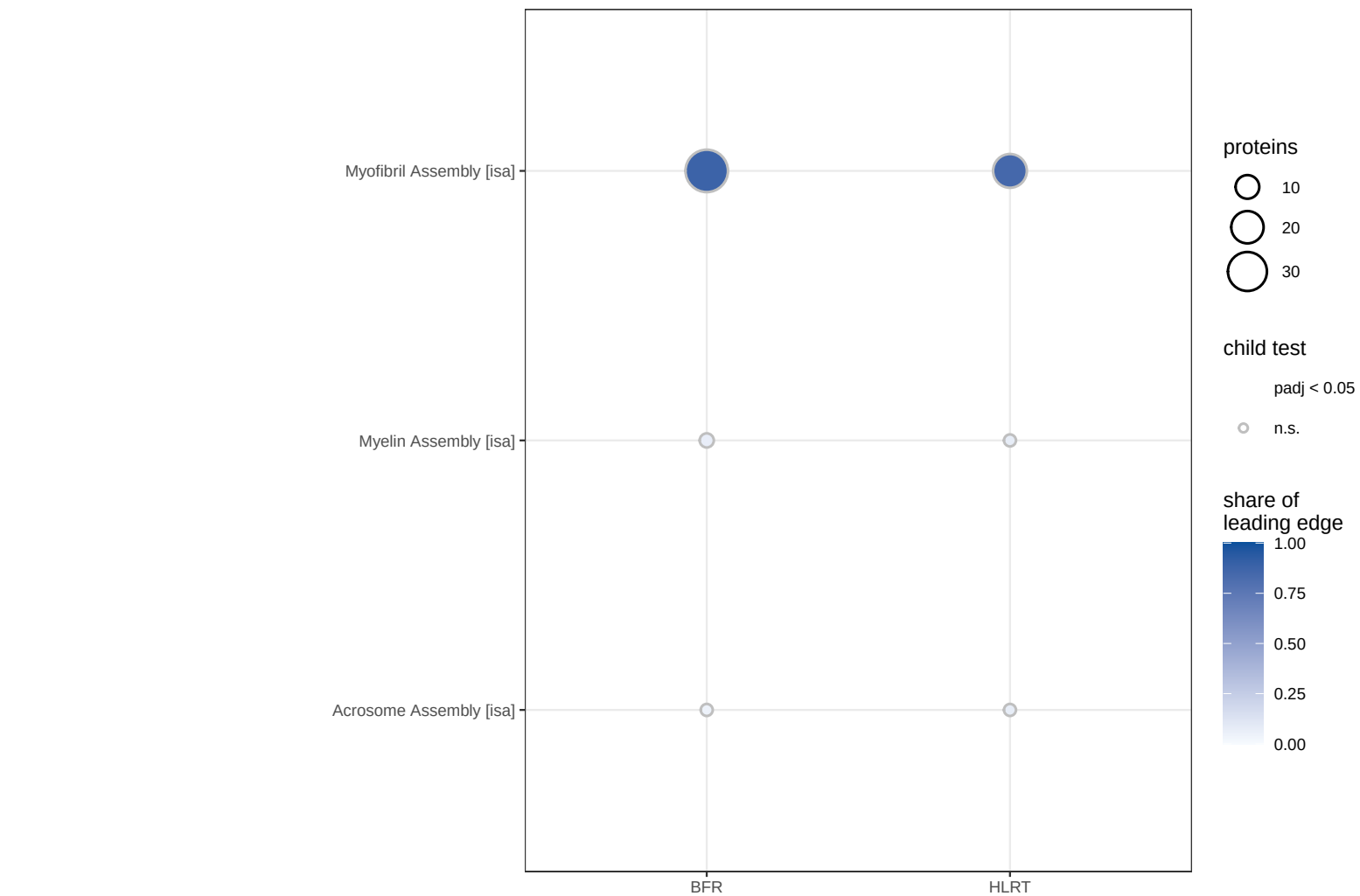
S17. Fatty Acid Metabolic Process

leading edge: BFR 34, HLRT 48; in significant descendants: BFR 44%, HLRT 40%



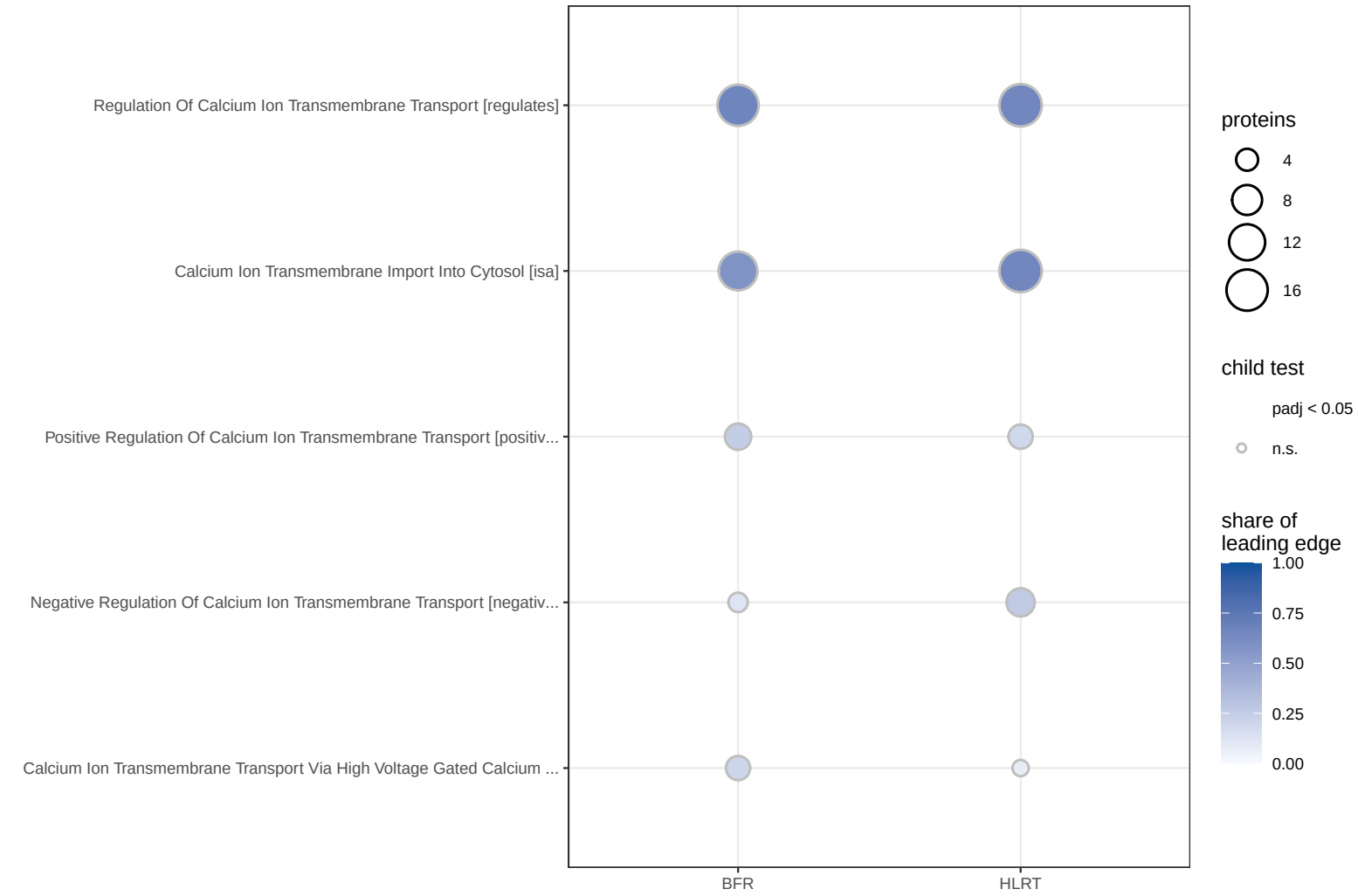
S17. Cellular Component Assembly Involved In Morphogenesis

leading edge: BFR 41, HLRT 26; in significant descendants: BFR 88%, HLRT 85%



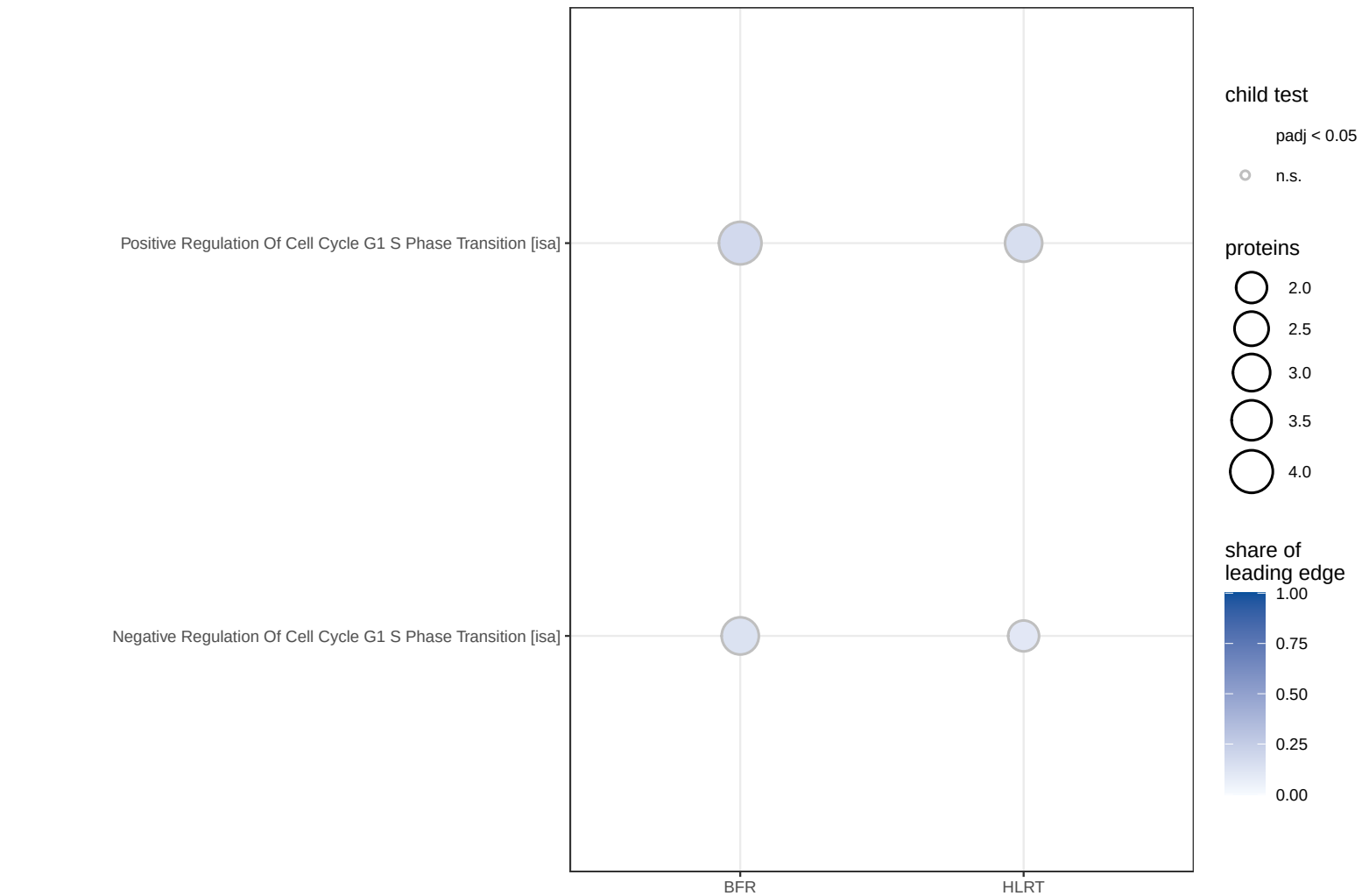
S17. Calcium Ion Transmembrane Transport

leading edge: BFR 24, HLRT 26; in significant descendants: BFR 75%, HLRT 77%



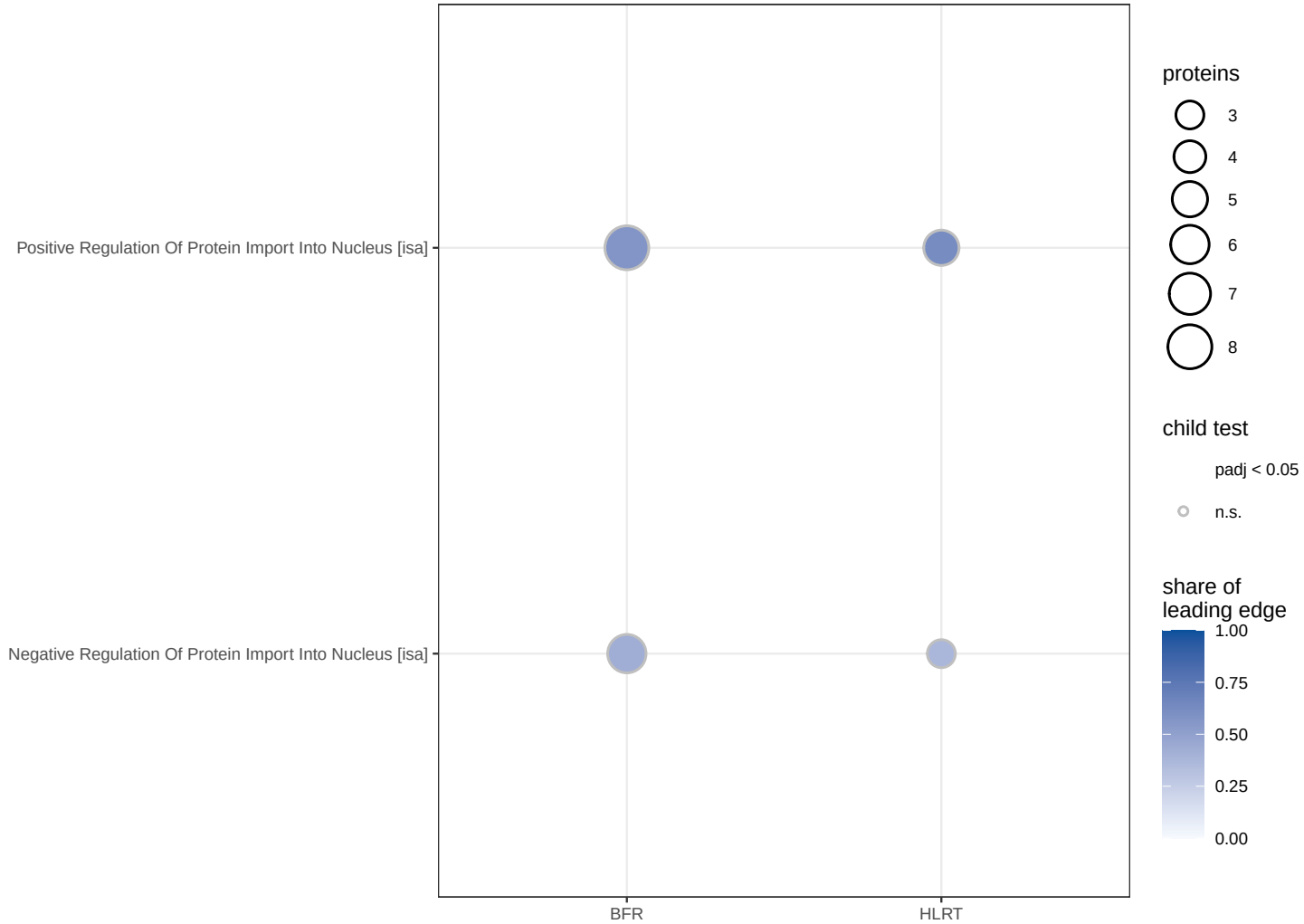
S17. Regulation Of Cell Cycle G1 S Phase Transition

leading edge: BFR 22, HLRT 19; in significant descendants: BFR 0%, HLRT 0%



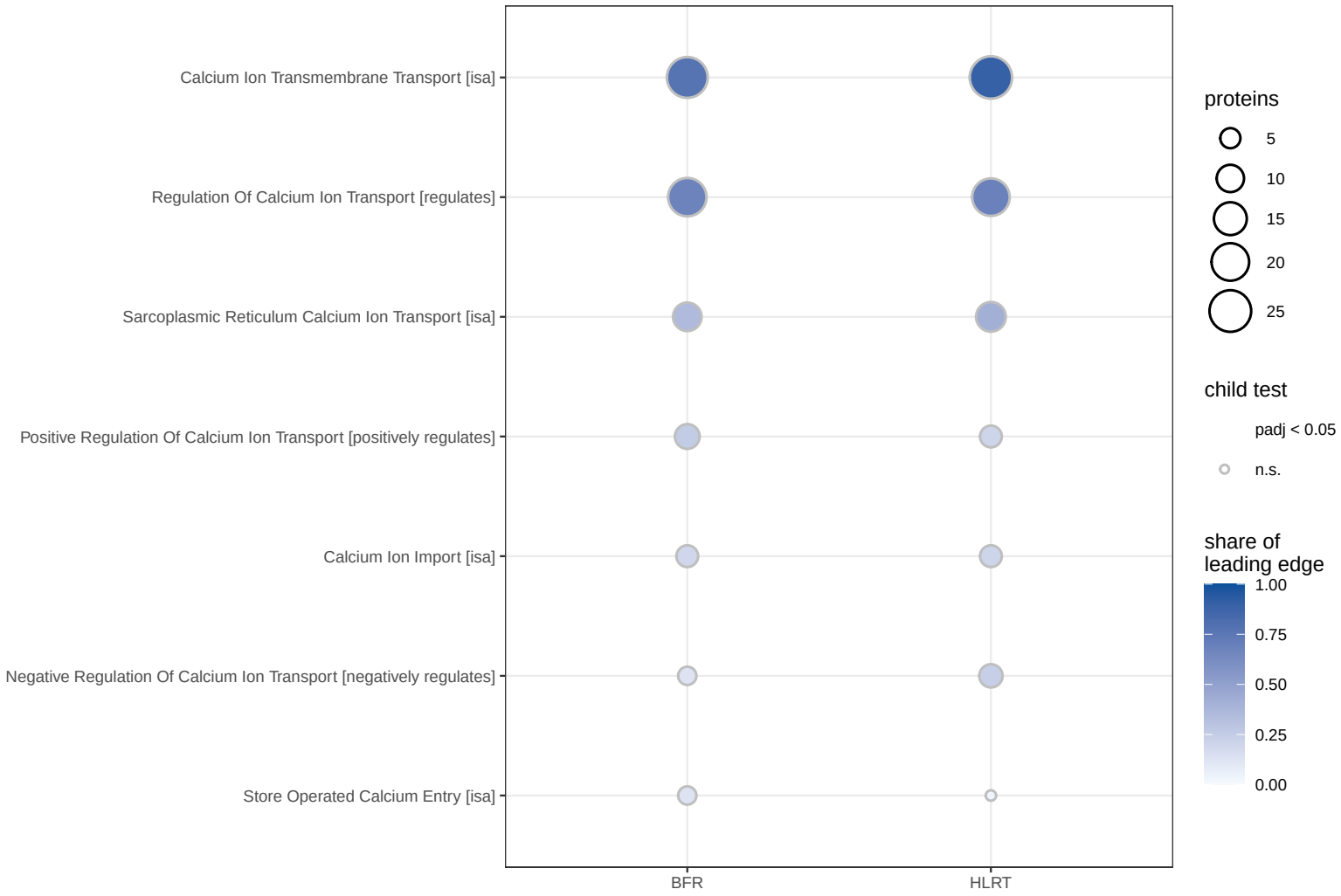
S17. Regulation Of Protein Import Into Nucleus

leading edge: BFR 14, HLRT 8; in significant descendants: BFR 0%, HLRT 0%



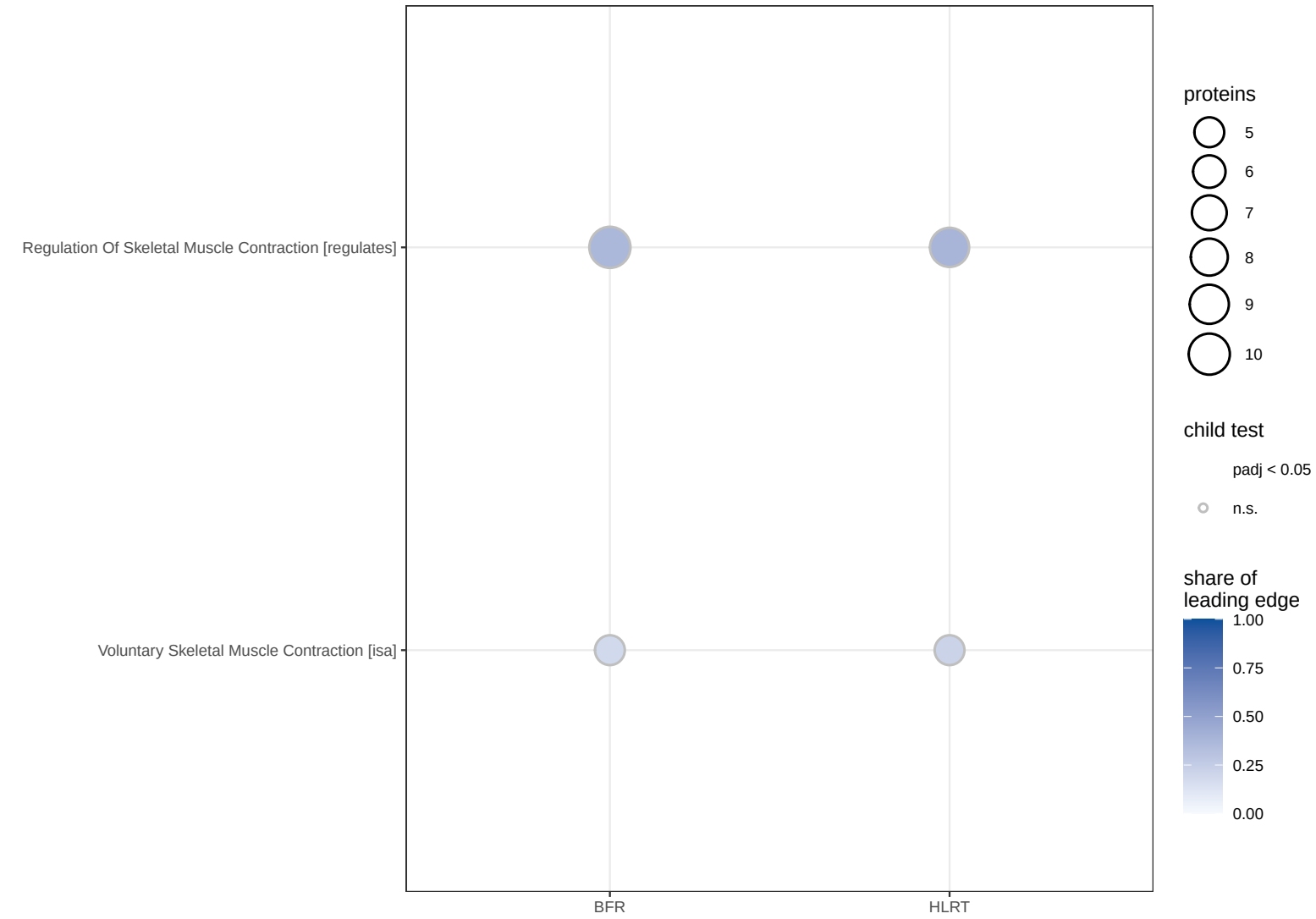
S17. Calcium Ion Transport

leading edge: BFR 31, HLRT 29; in significant descendants: BFR 94%, HLRT 97%



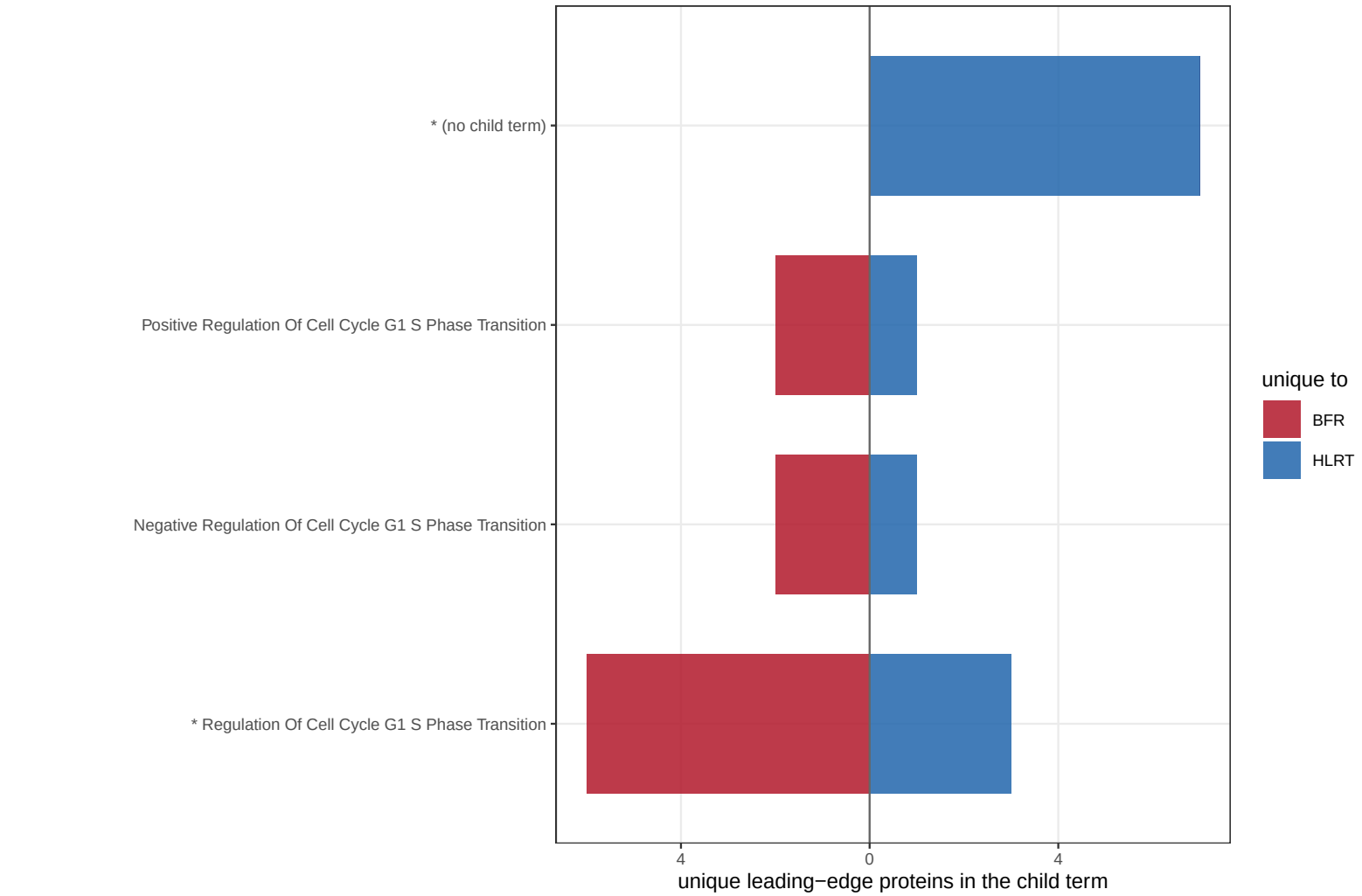
S17. Skeletal Muscle Contraction

leading edge: BFR 27, HLRT 23; in significant descendants: BFR 0%, HLRT 0%



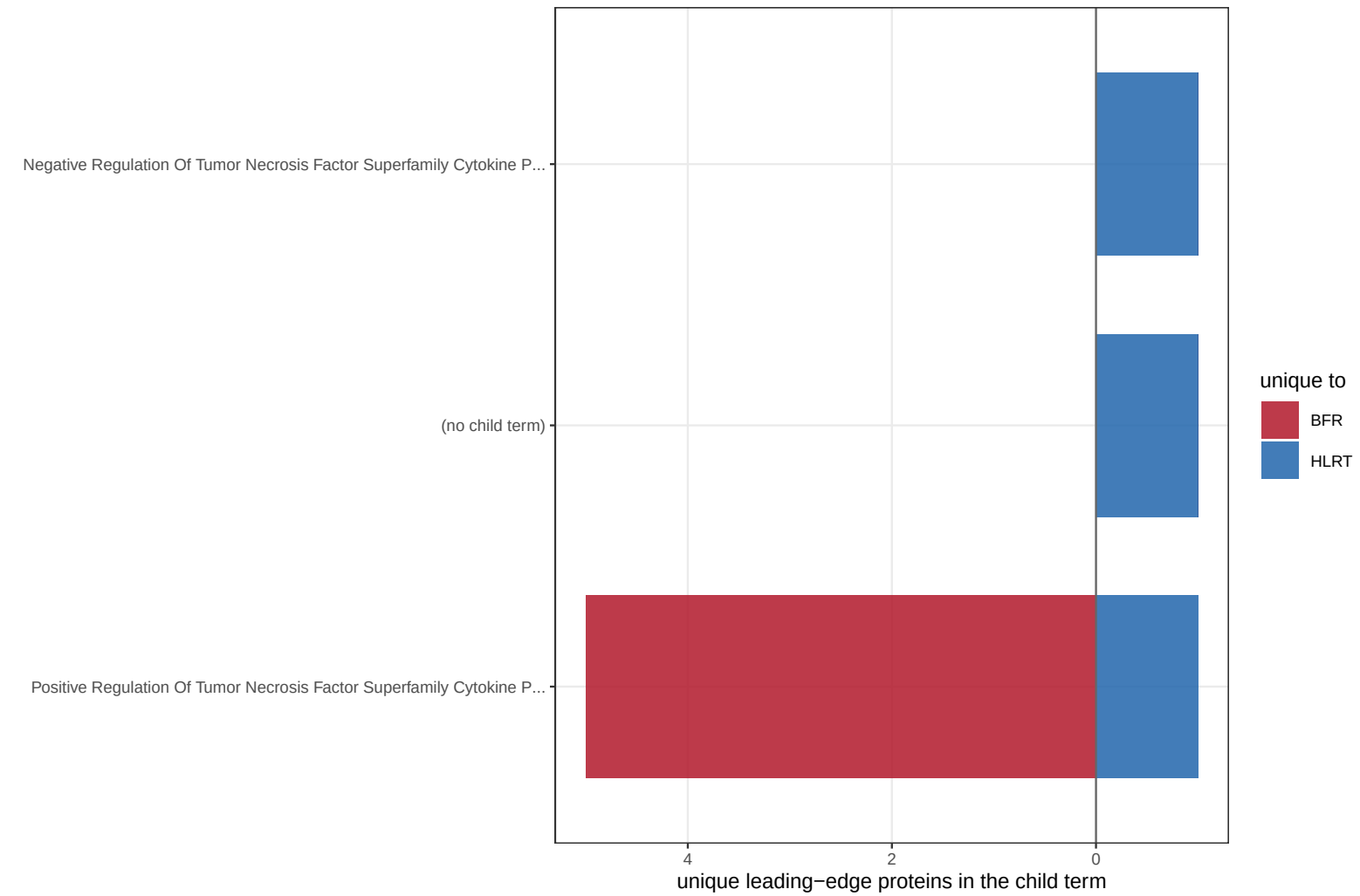
S18. Cell Cycle G1 S Phase Transition

only BFR 6, only HLRT 10, shared 19; divergence 1.87, permutation p 0.0155 (BH 0.372)



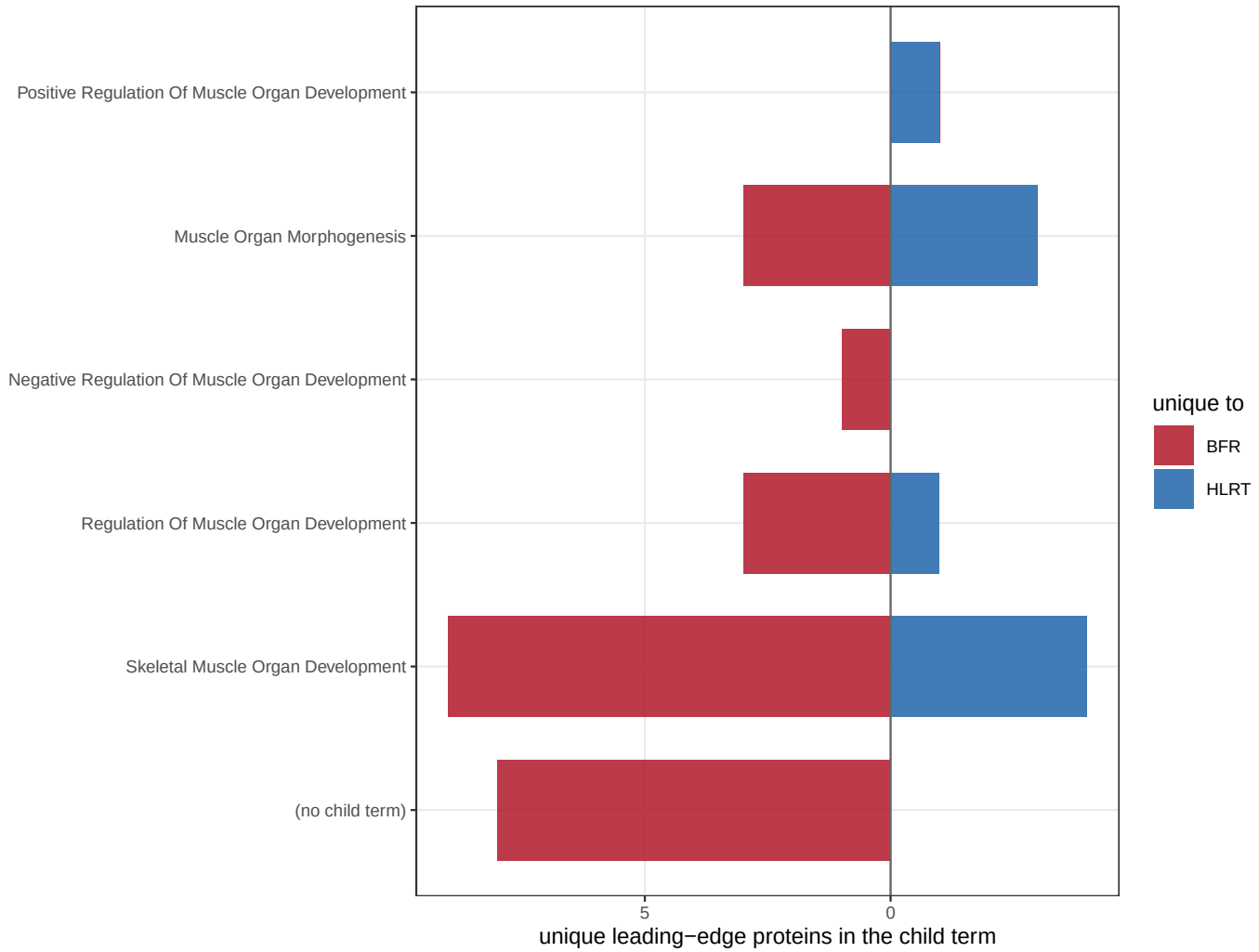
S18. Tumor Necrosis Factor Superfamily Cytokine Production

only BFR 5, only HLRT 3, shared 16; divergence 1.33, permutation p 0.109 (BH 0.913)



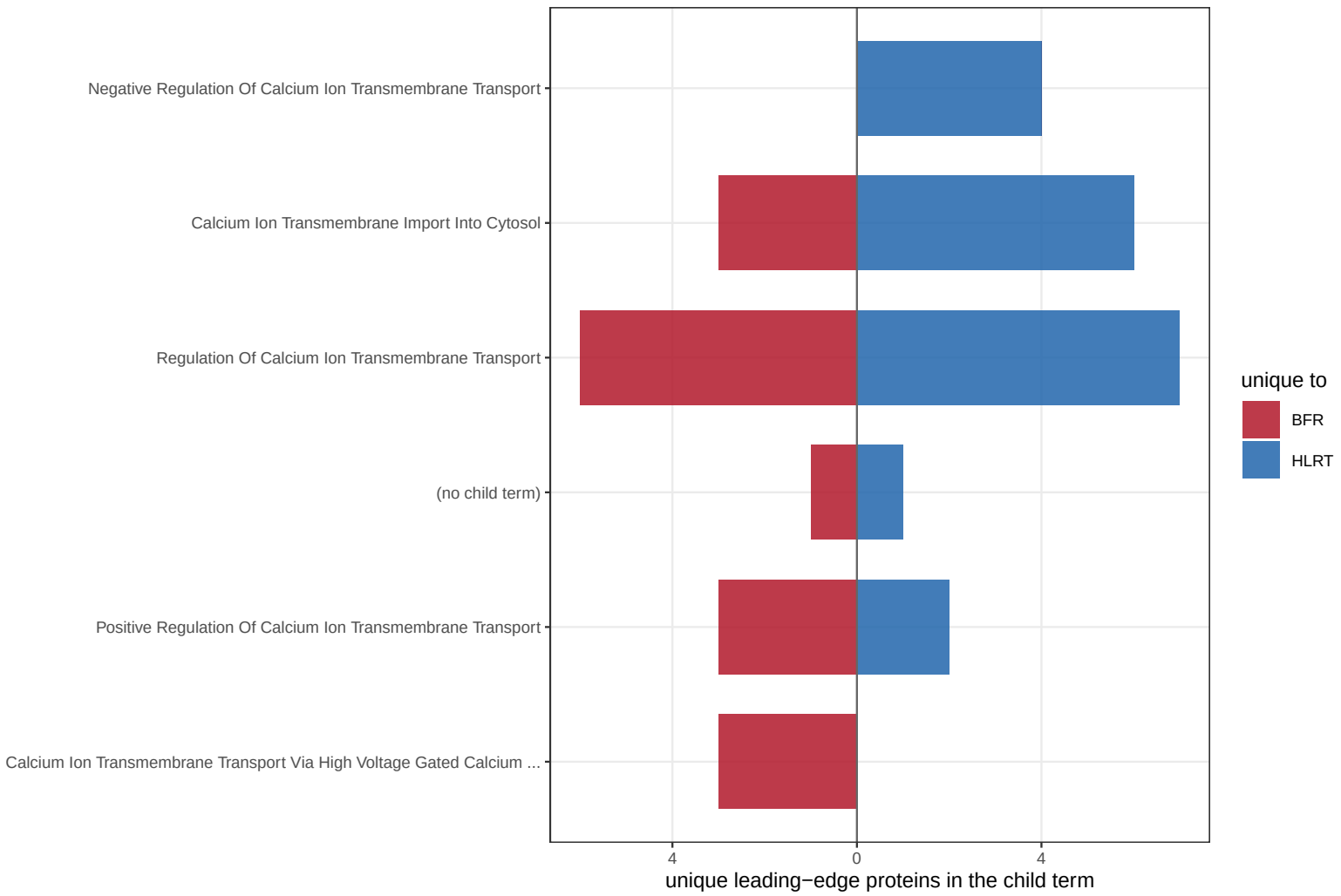
S18. Muscle Organ Development

only BFR 22, only HLRT 7, shared 53; divergence 1.01, permutation p 0.146 (BH 0.913)



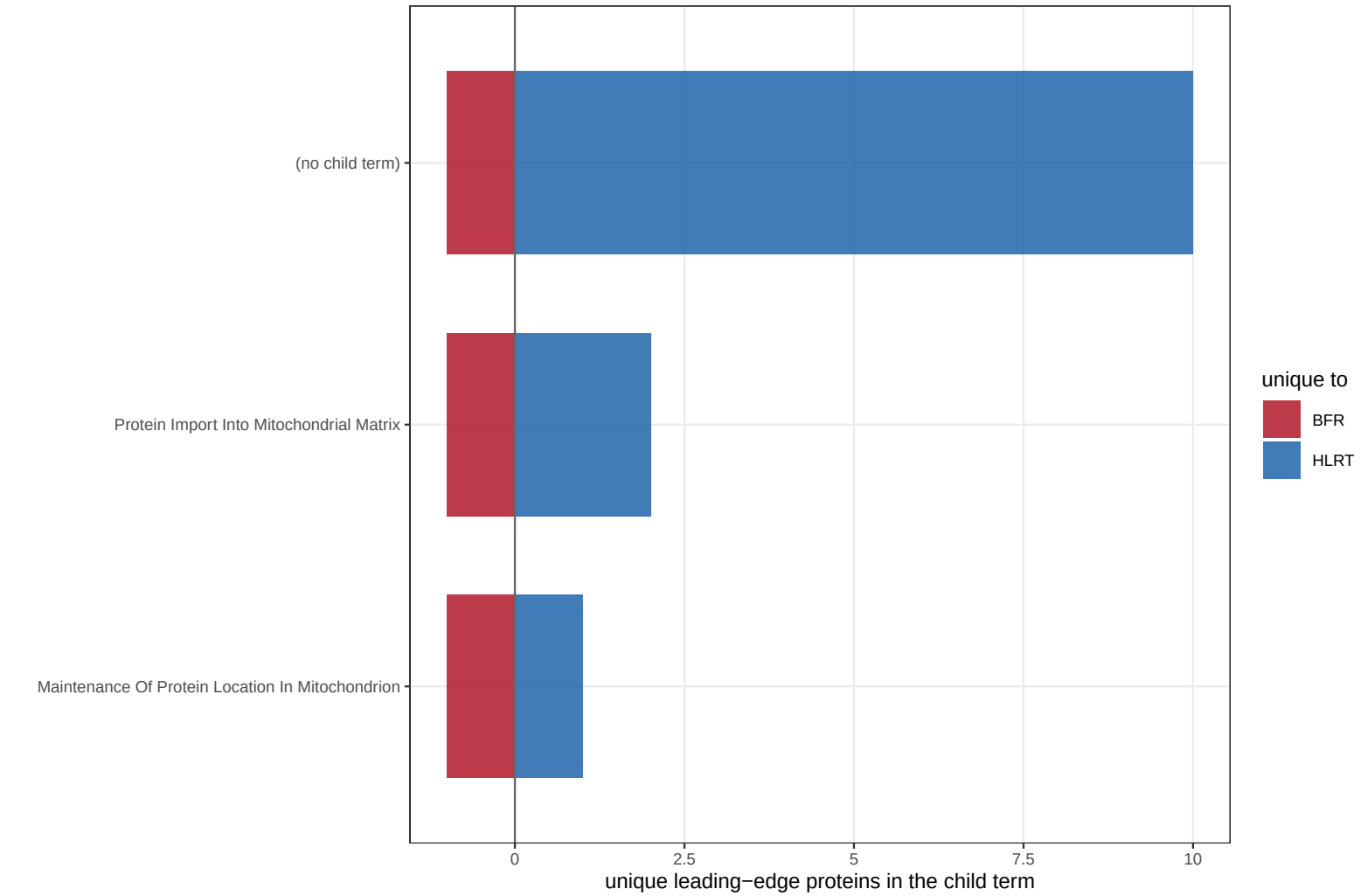
S18. Calcium Ion Transmembrane Transport

only BFR 7, only HLRT 9, shared 17; divergence 1.43, permutation p 0.154 (BH 0.913)



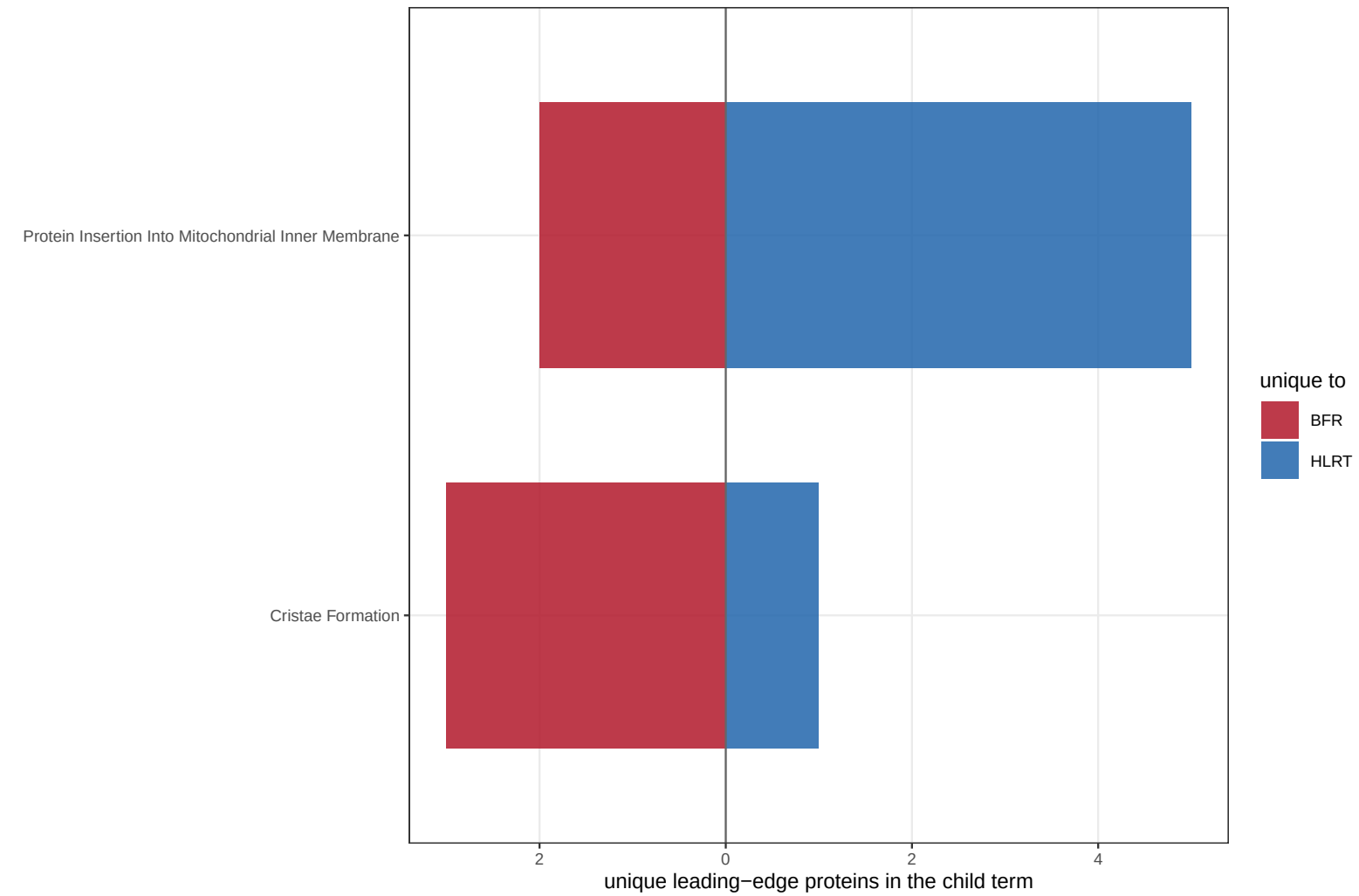
S18. Protein Localization To Mitochondrion

only BFR 3, only HLRT 13, shared 18; divergence 0.87, permutation p 0.195 (BH 0.913)



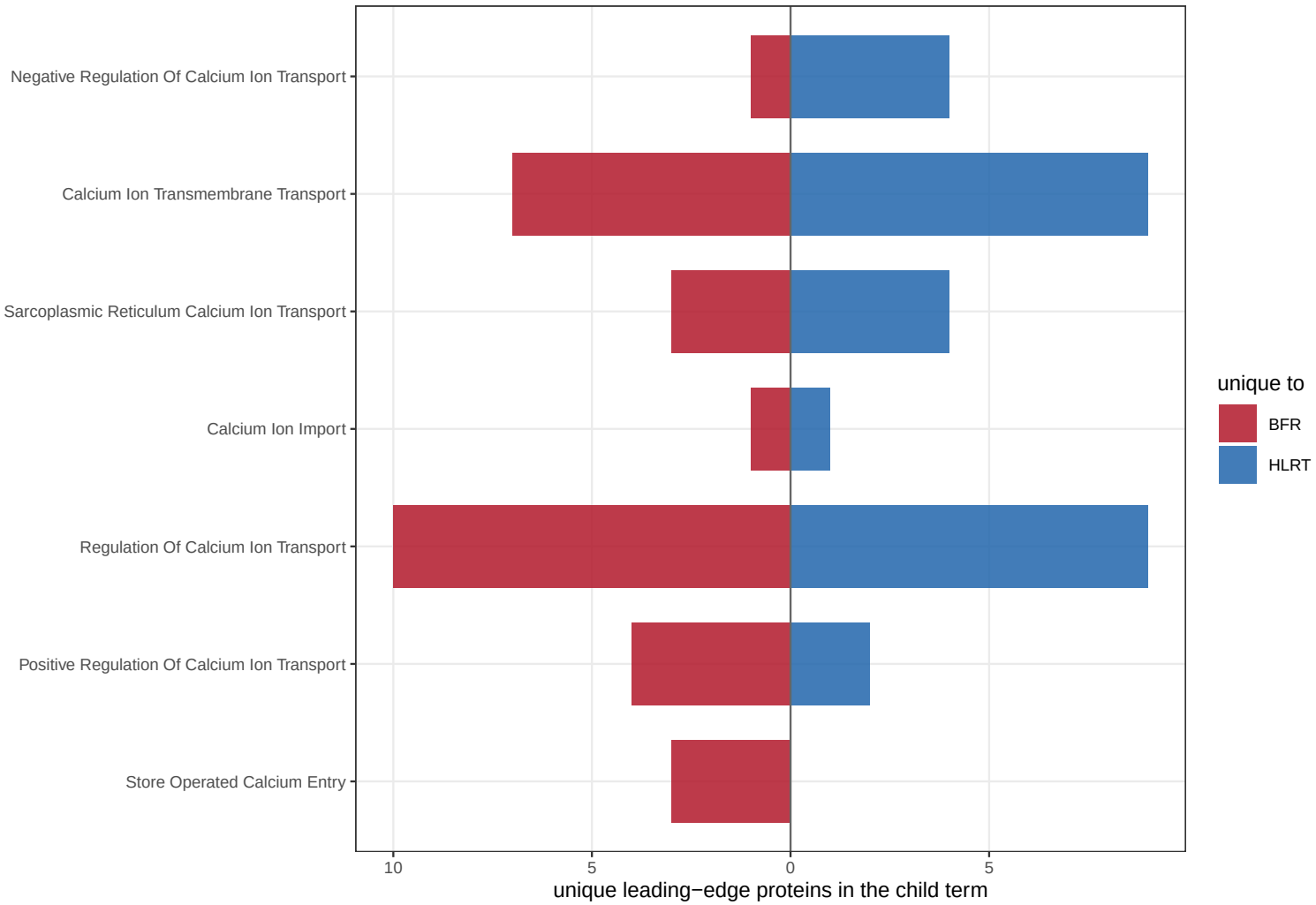
S18. Inner Mitochondrial Membrane Organization

only BFR 5, only HLRT 6, shared 13; divergence 0.87, permutation p 0.238 (BH 0.913)



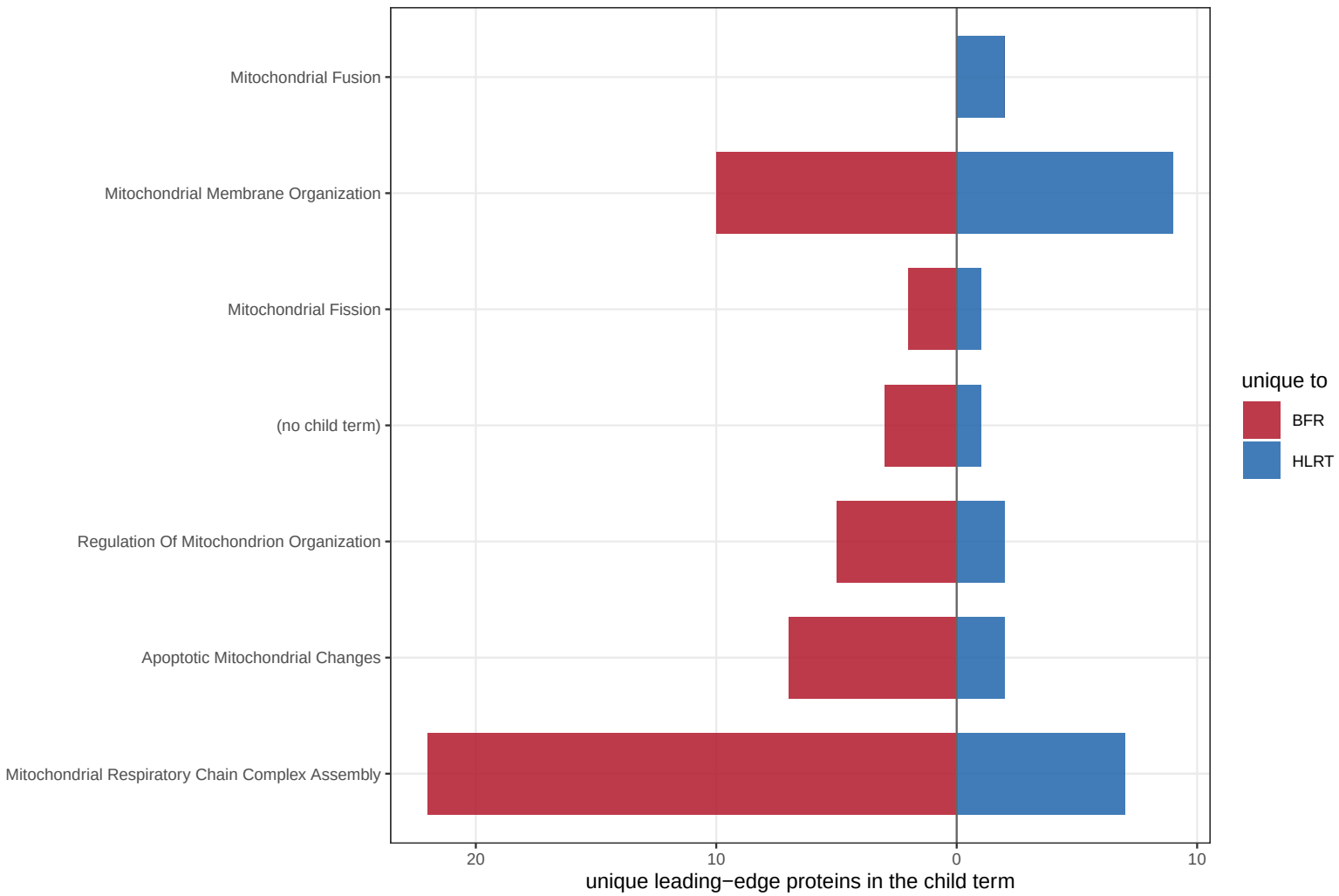
S18. Calcium Ion Transport

only BFR 13, only HLRT 11, shared 18; divergence 1.12, permutation p 0.266 (BH 0.913)



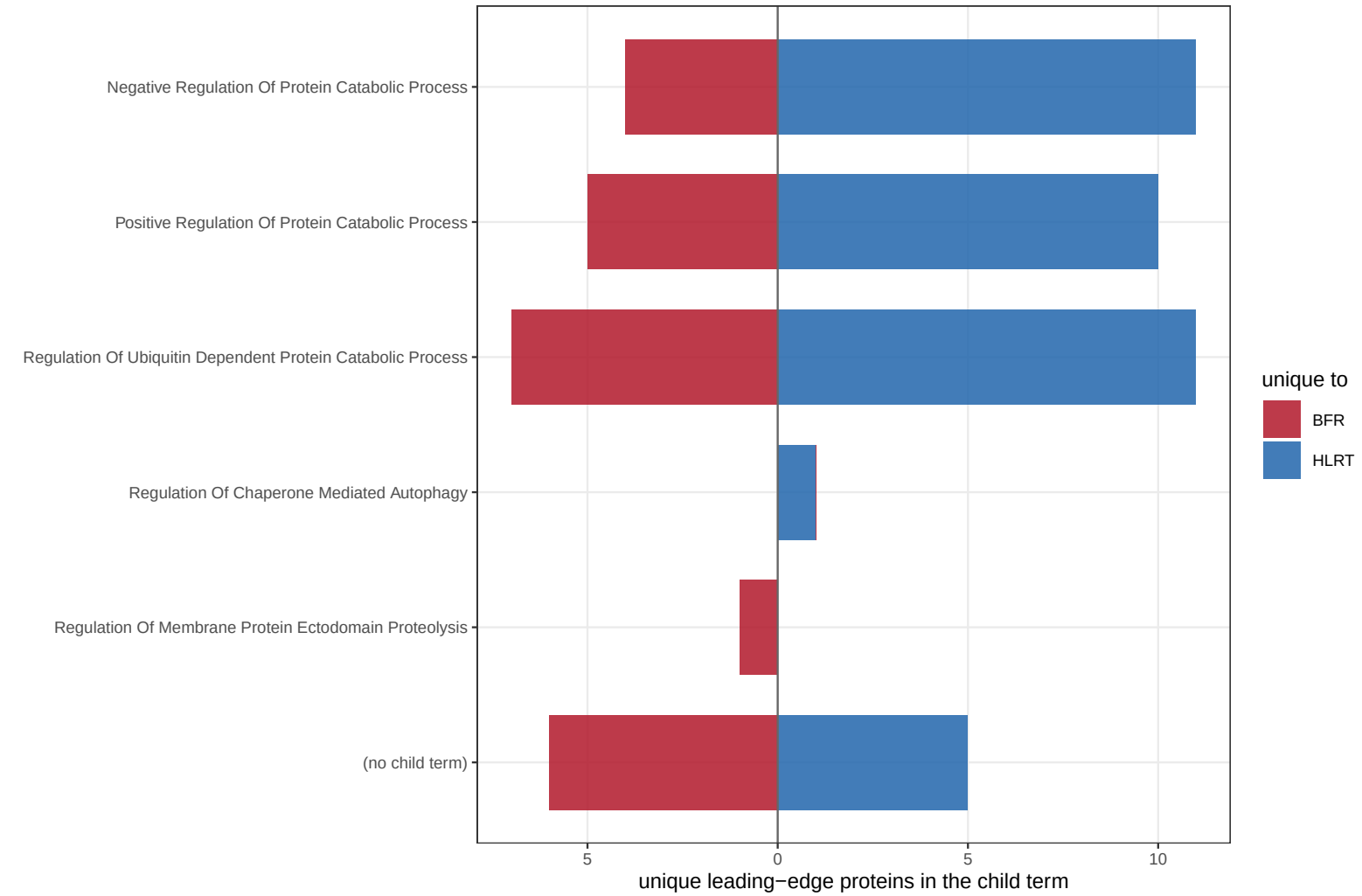
S18. Mitochondrion Organization

only BFR 40, only HLRT 16, shared 66; divergence 0.62, permutation p 0.337 (BH 1)



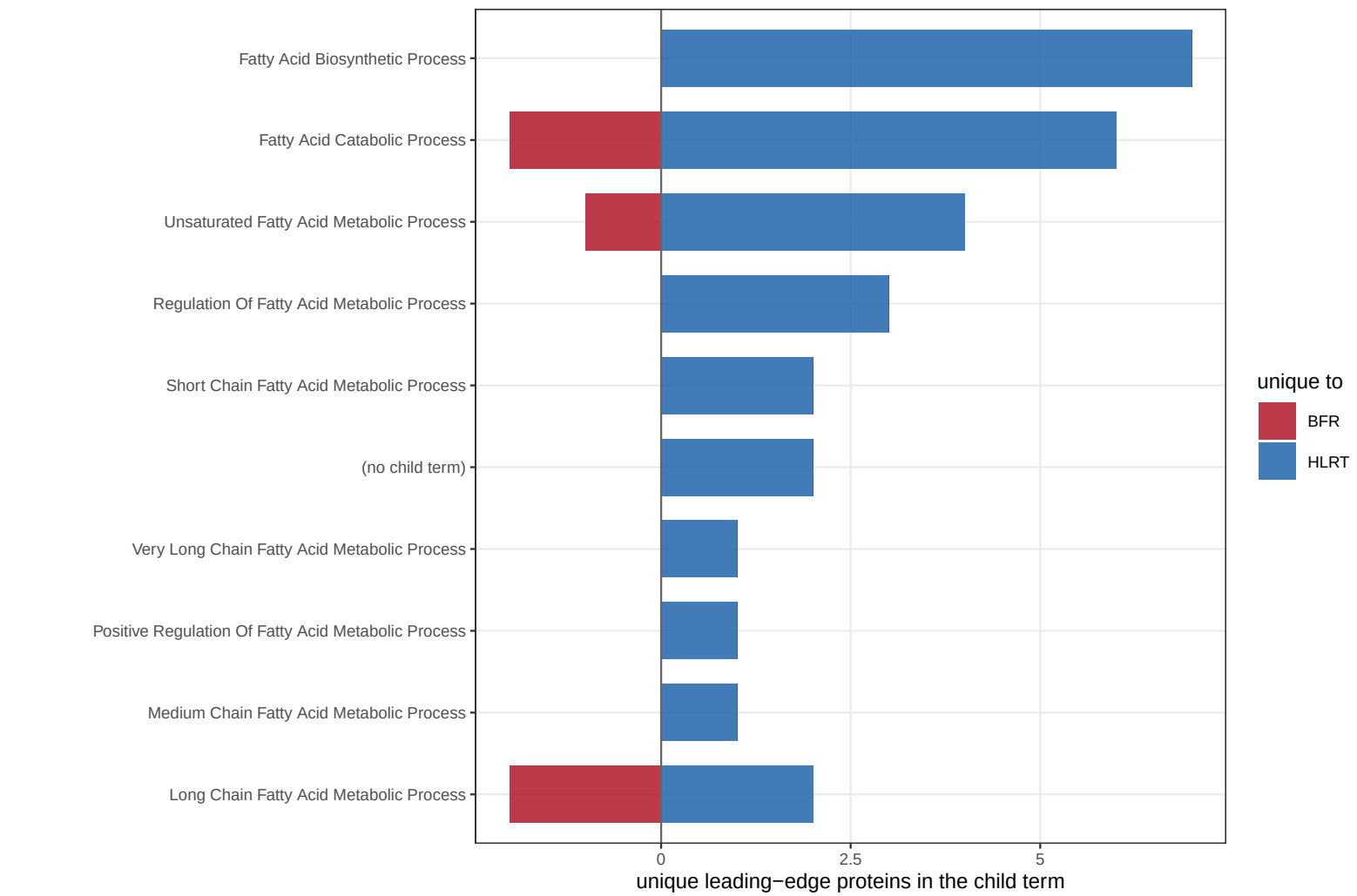
S18. Regulation Of Protein Catabolic Process

only BFR 17, only HLRT 26, shared 26; divergence 0.55, permutation p 0.461 (BH 1)



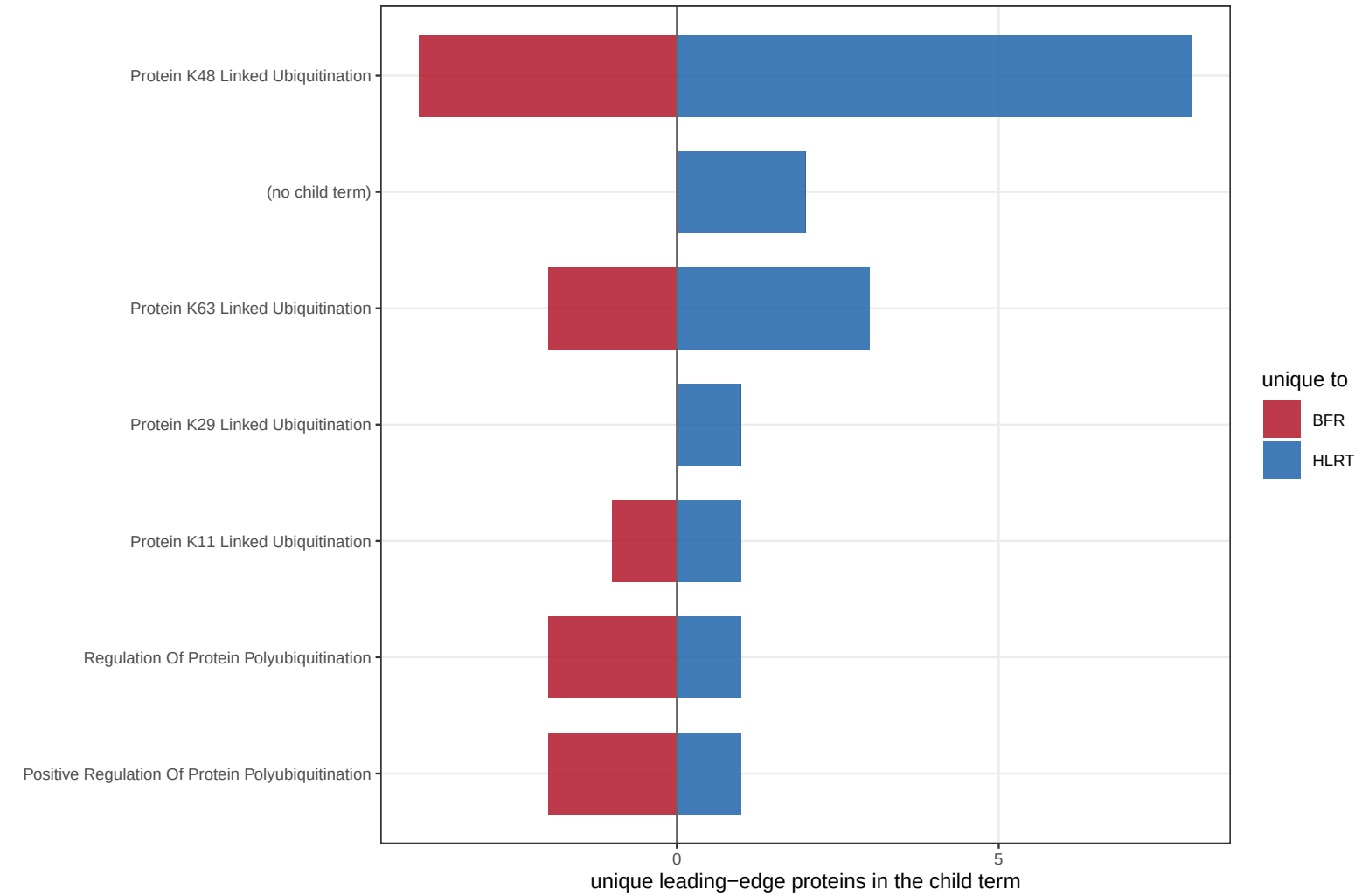
S18. Fatty Acid Metabolic Process

only BFR 4, only HLRT 18, shared 30; divergence 1.53, permutation p 0.532 (BH 1)



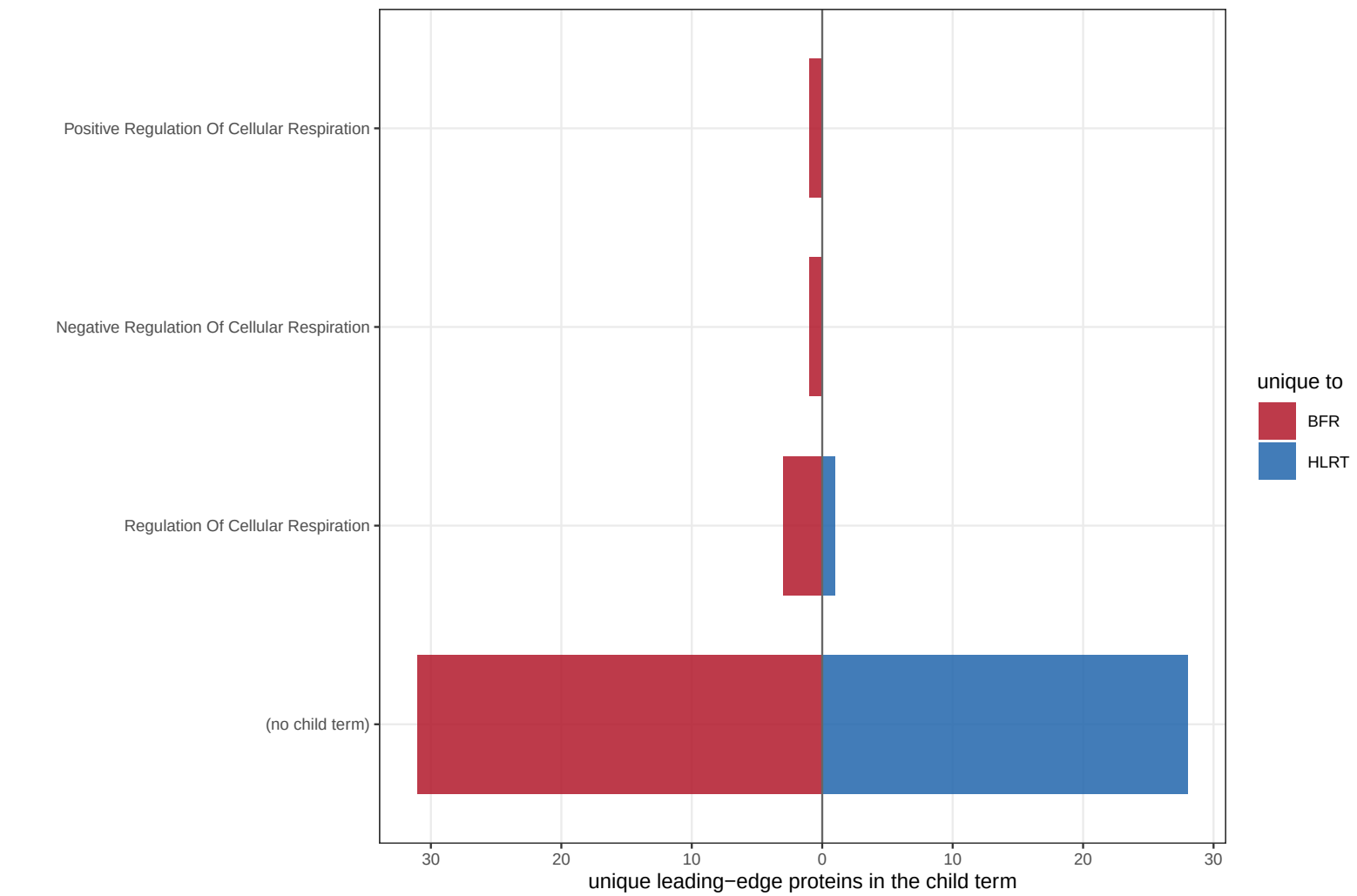
S18. Protein Polyubiquitination

only BFR 6, only HLRT 13, shared 17; divergence 0.99, permutation p 0.554 (BH 1)



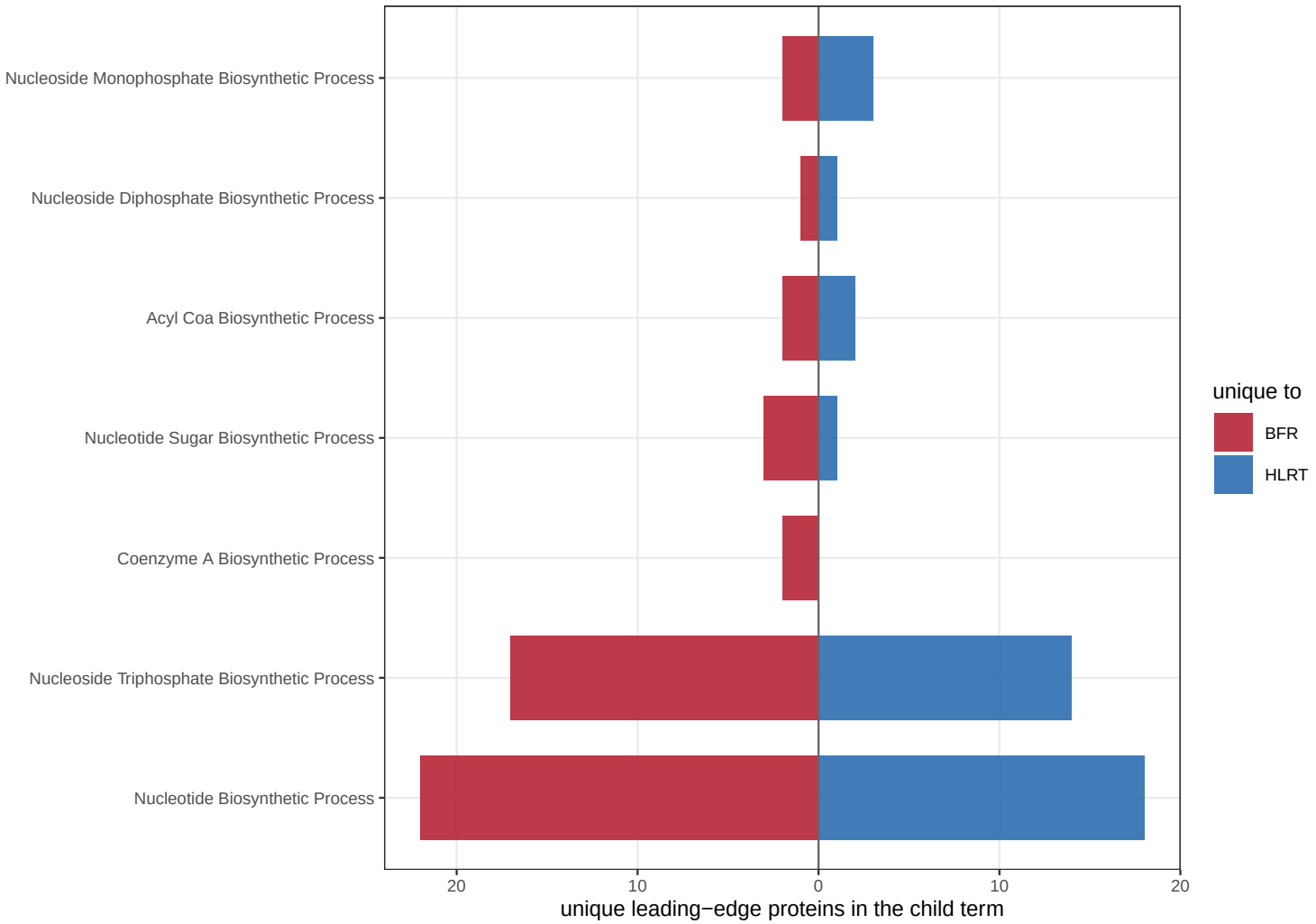
S18. Cellular Respiration

only BFR 34, only HLRT 29, shared 79; divergence 0.17, permutation p 0.606 (BH 1)



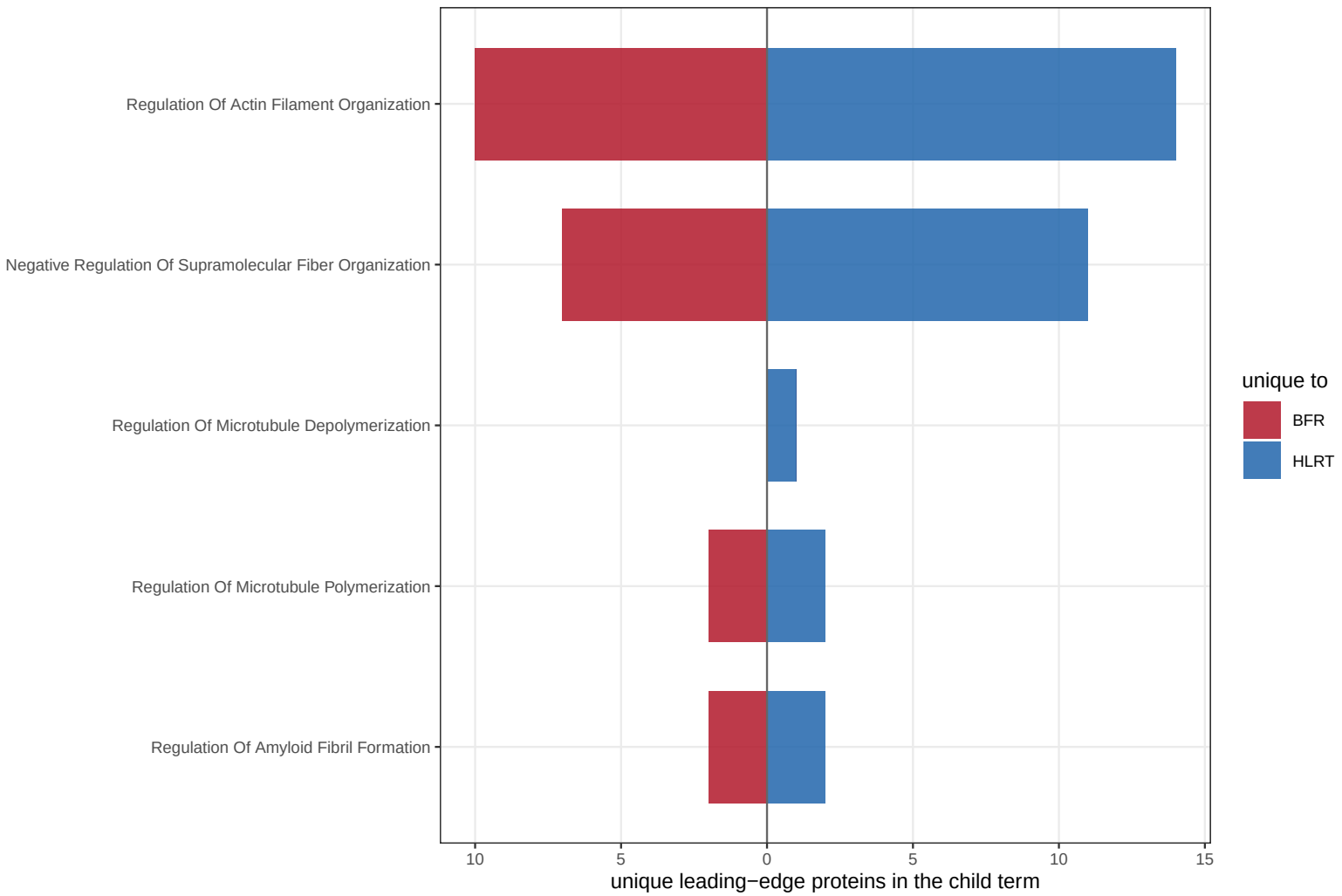
S18. Nucleoside Phosphate Biosynthetic Process

only BFR 29, only HLRT 21, shared 55; divergence 0.42, permutation p 0.641 (BH 1)



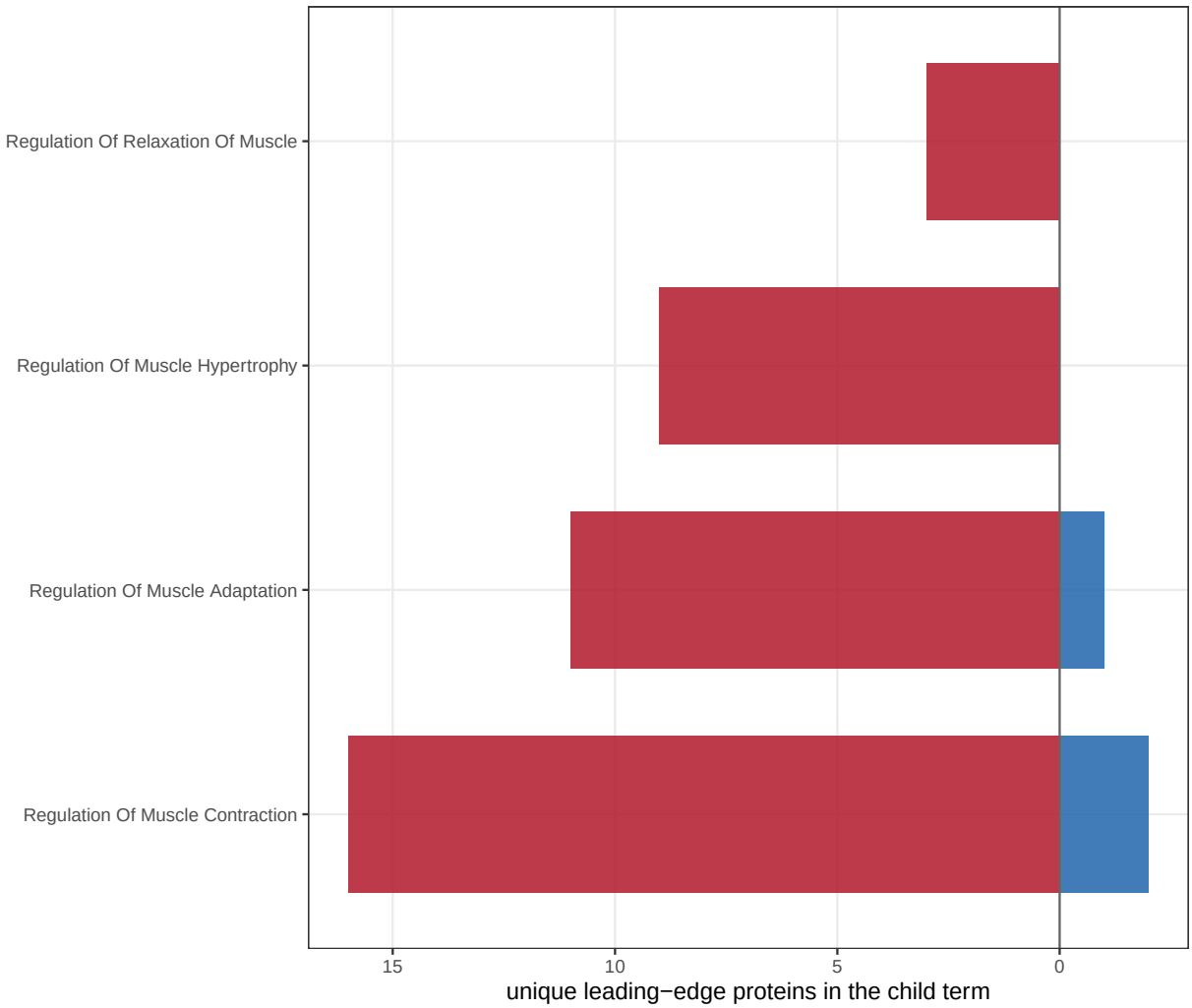
S18. Regulation Of Supramolecular Fiber Organization

only BFR 15, only HLRT 17, shared 23; divergence 0.43, permutation p 0.672 (BH 1)



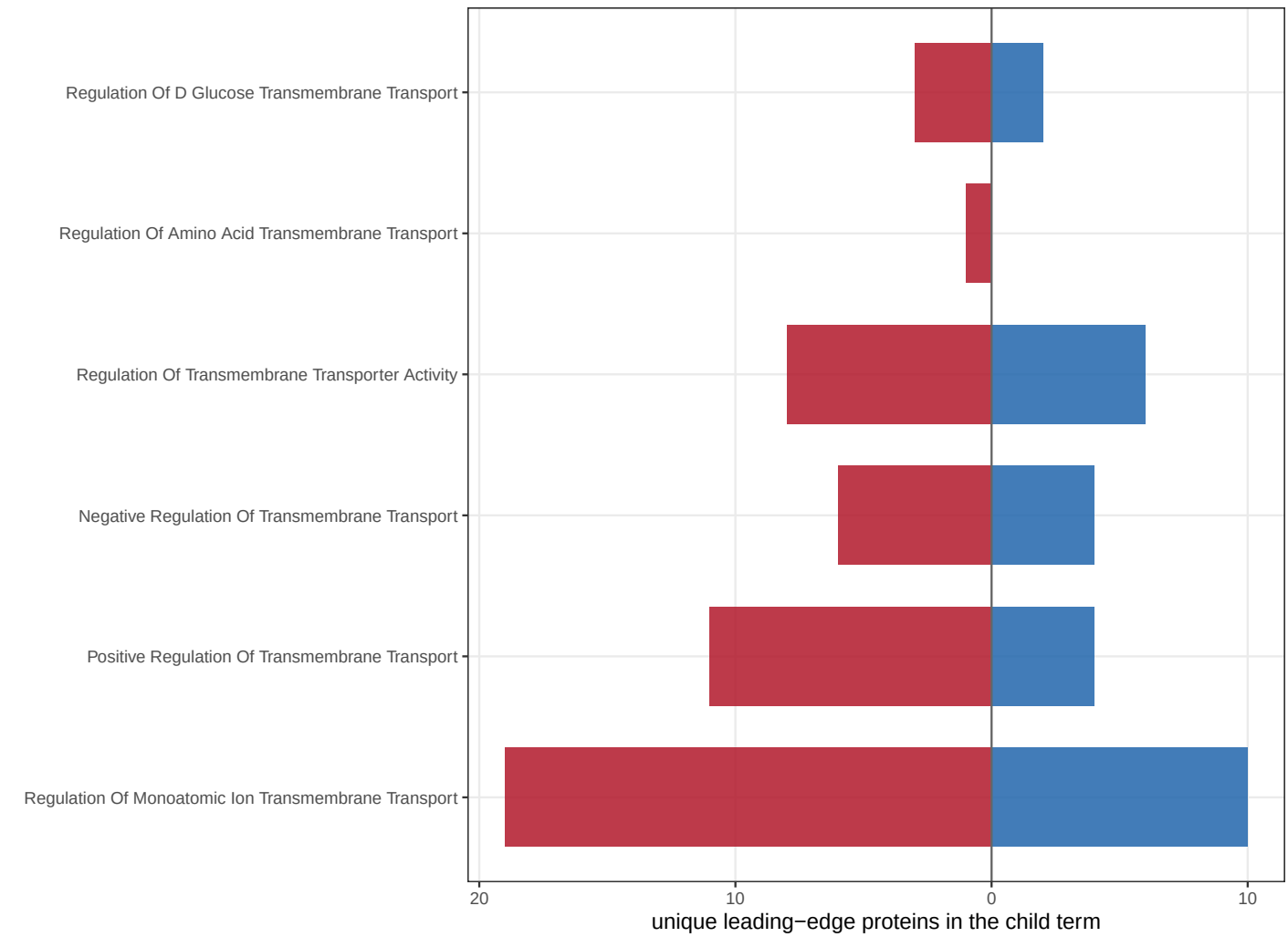
S18. Regulation Of Muscle System Process

only BFR 26, only HLRT 3, shared 29; divergence 0.60, permutation p 0.687 (BH 1)



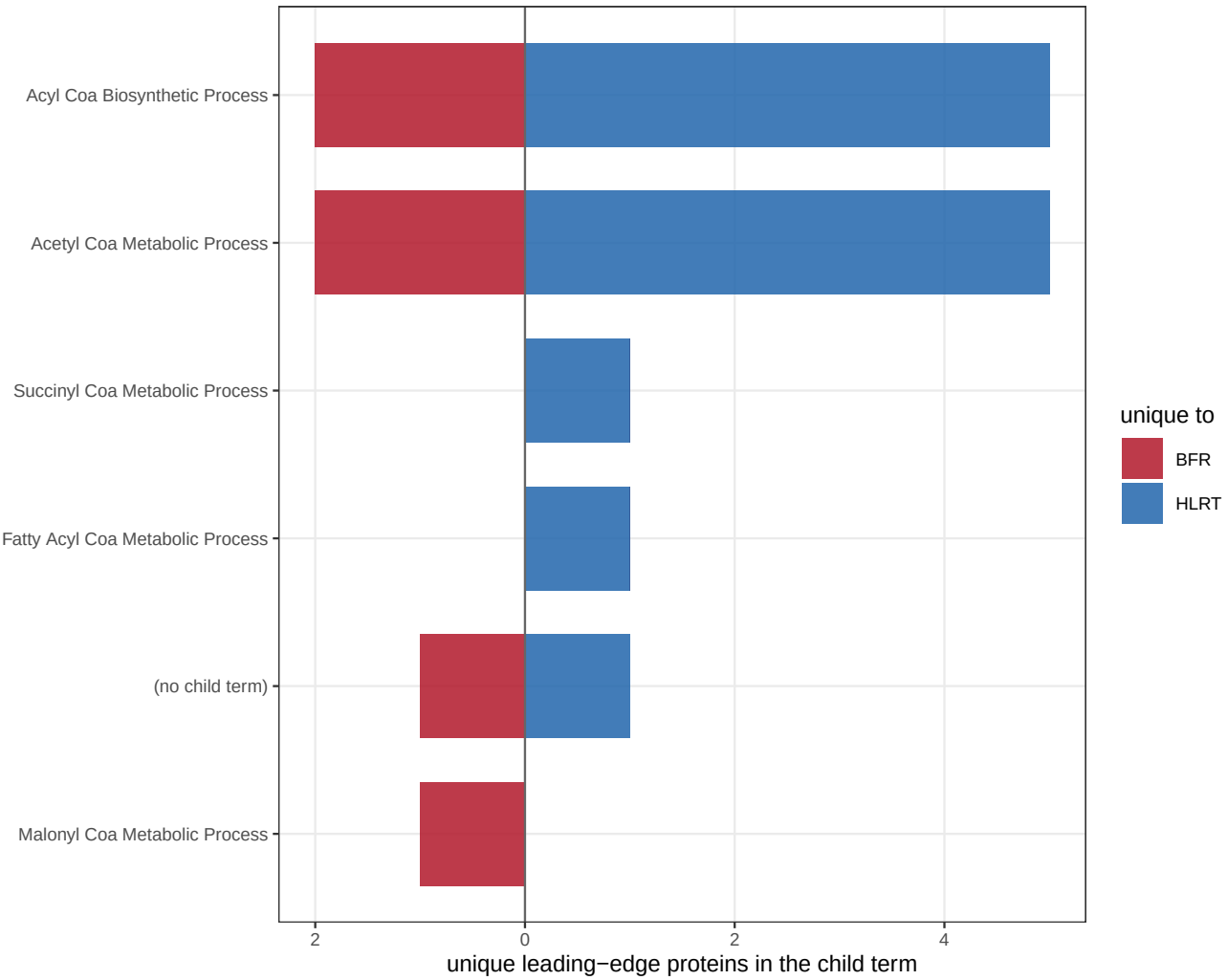
S18. Regulation Of Transmembrane Transport

only BFR 27, only HLRT 18, shared 33; divergence 0.41, permutation p 0.822 (BH 1)



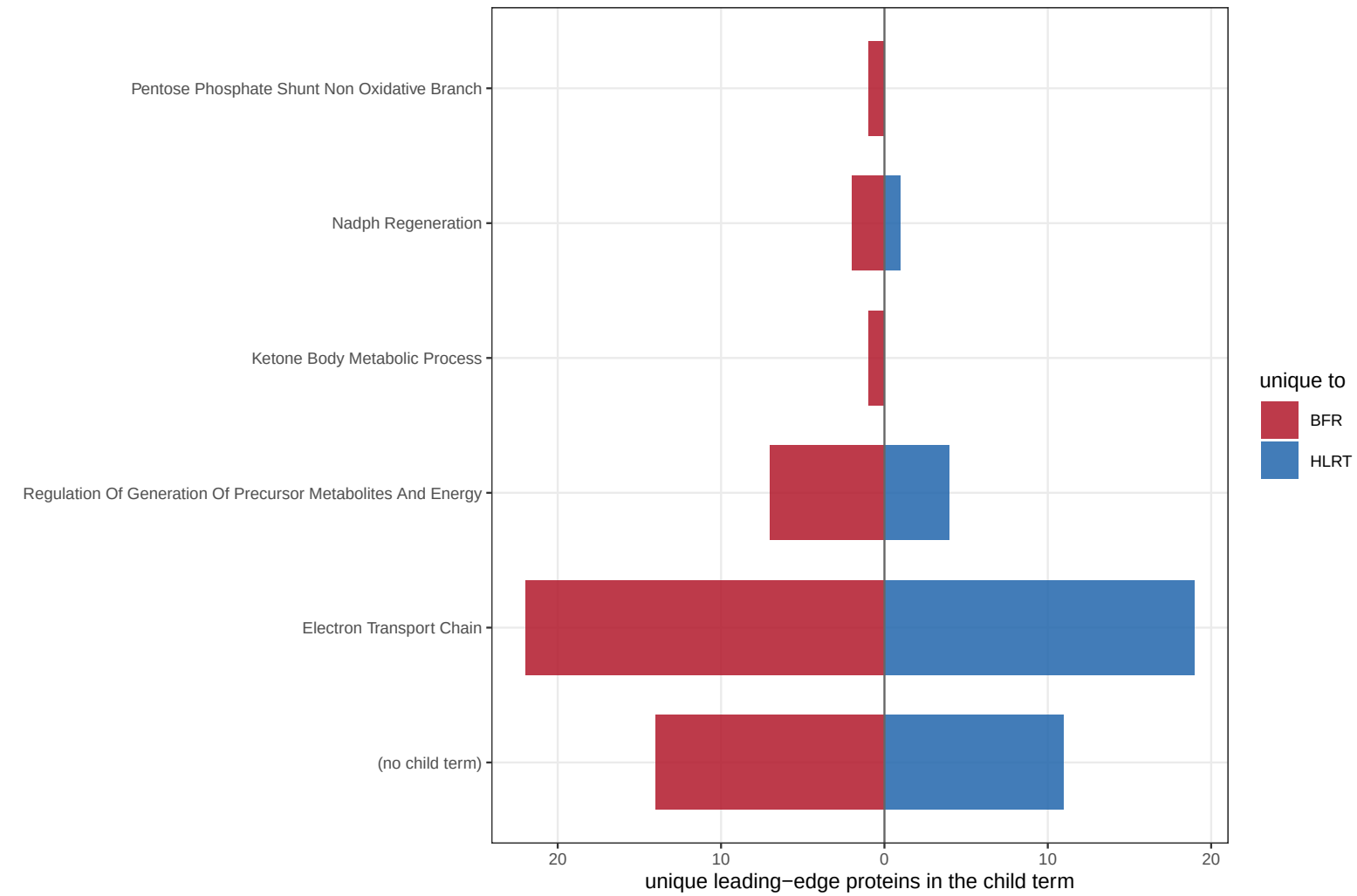
S18. Acyl Coa Metabolic Process

only BFR 3, only HLRT 7, shared 20; divergence 0.90, permutation p 0.896 (BH 1)



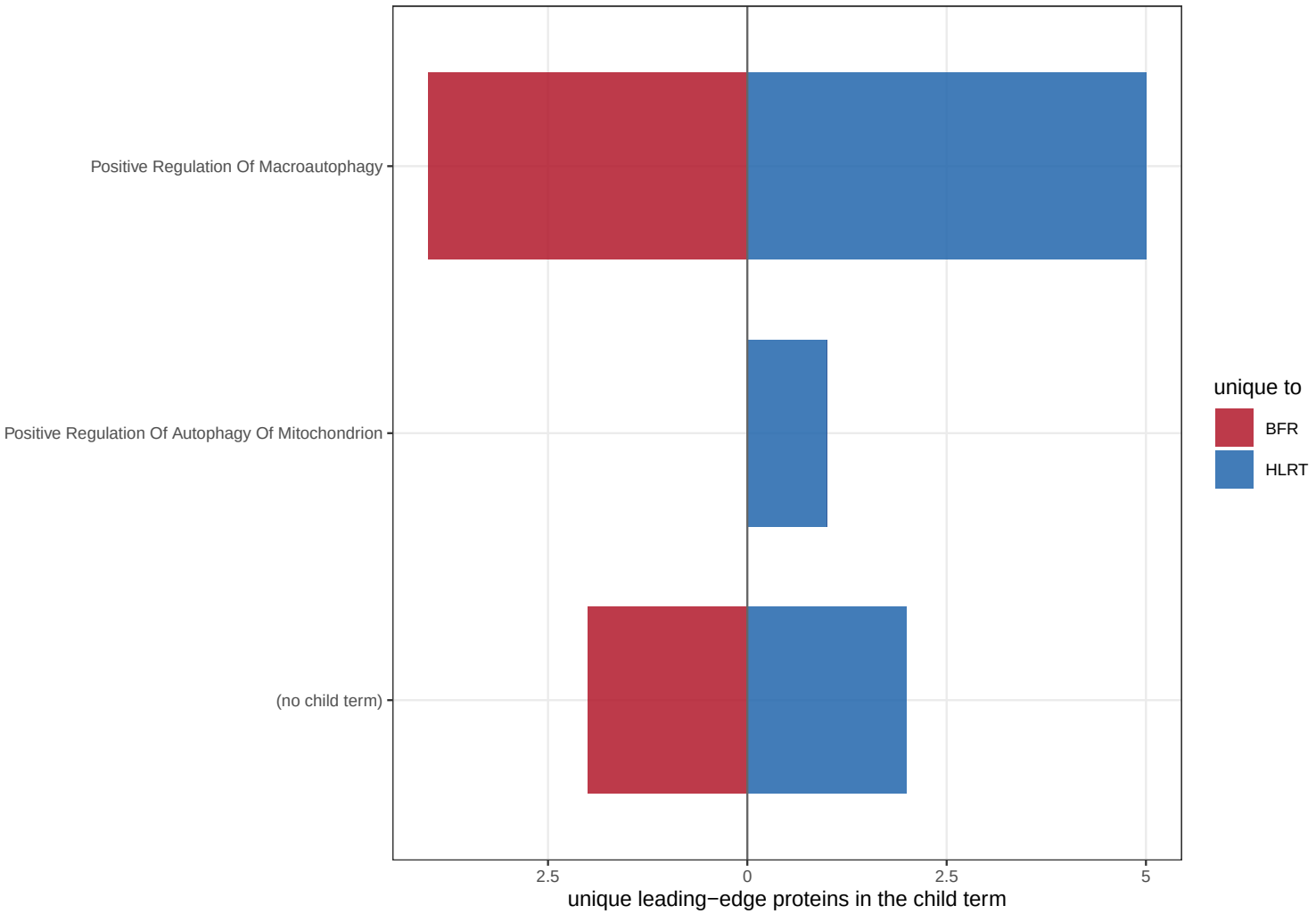
S18. Generation Of Precursor Metabolites And Energy

only BFR 44, only HLRT 35, shared 88; divergence 0.15, permutation p 0.94 (BH 1)



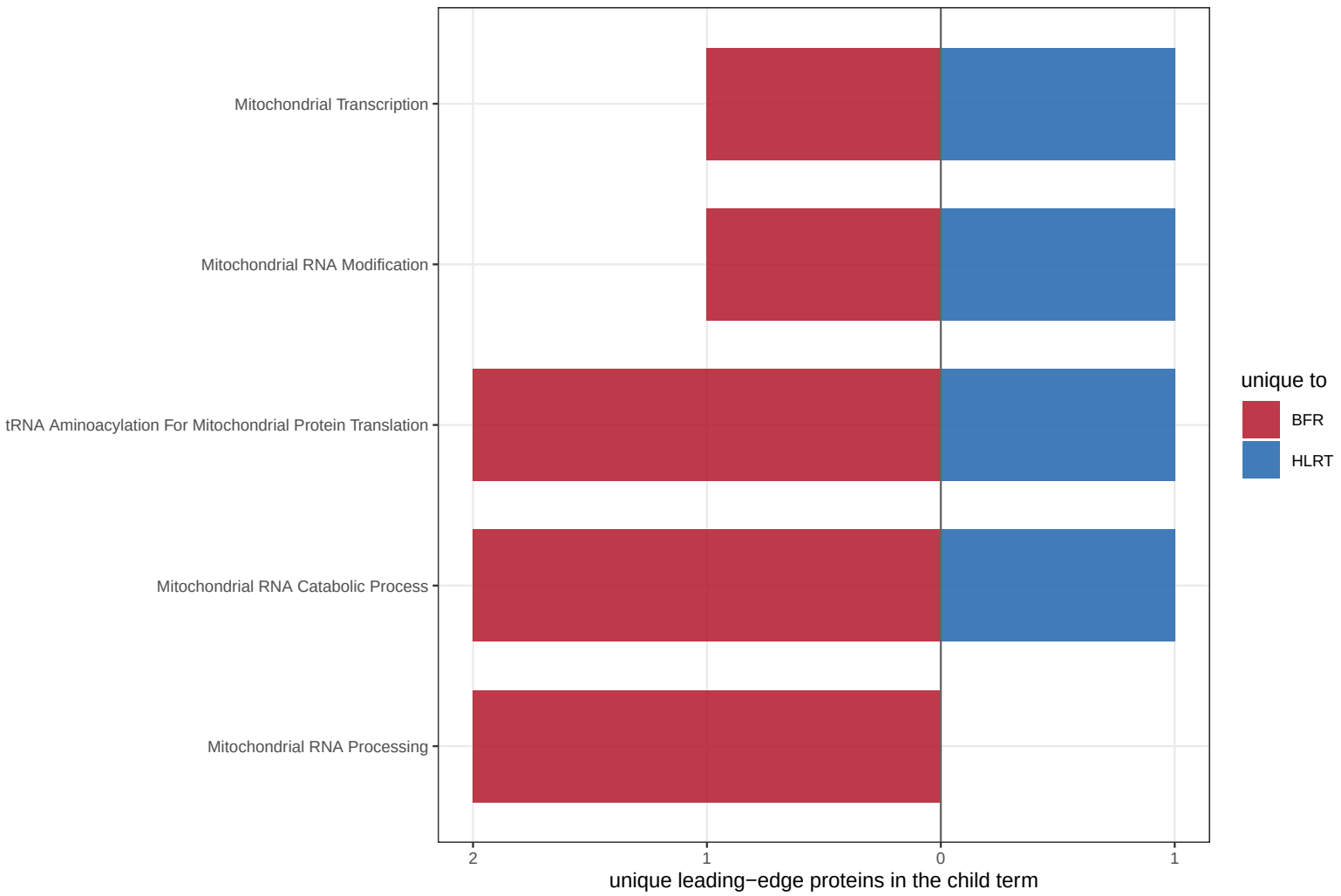
S18. Positive Regulation Of Autophagy

only BFR 6, only HLRT 7, shared 11; divergence 0.24, permutation p 1 (BH 1)



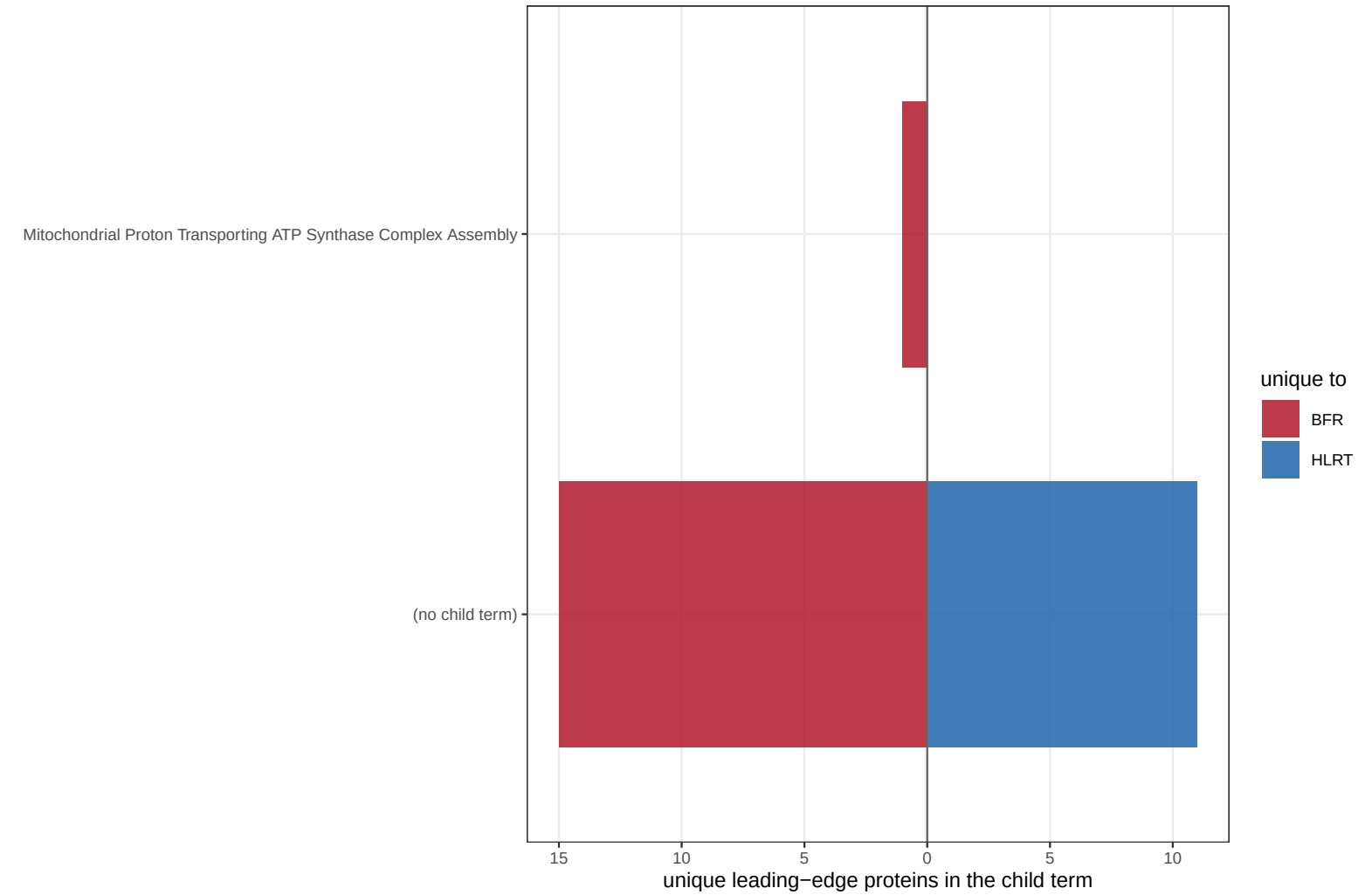
S18. Mitochondrial RNA Metabolic Process

only BFR 7, only HLRT 3, shared 10; divergence 0.76, permutation p 1 (BH 1)



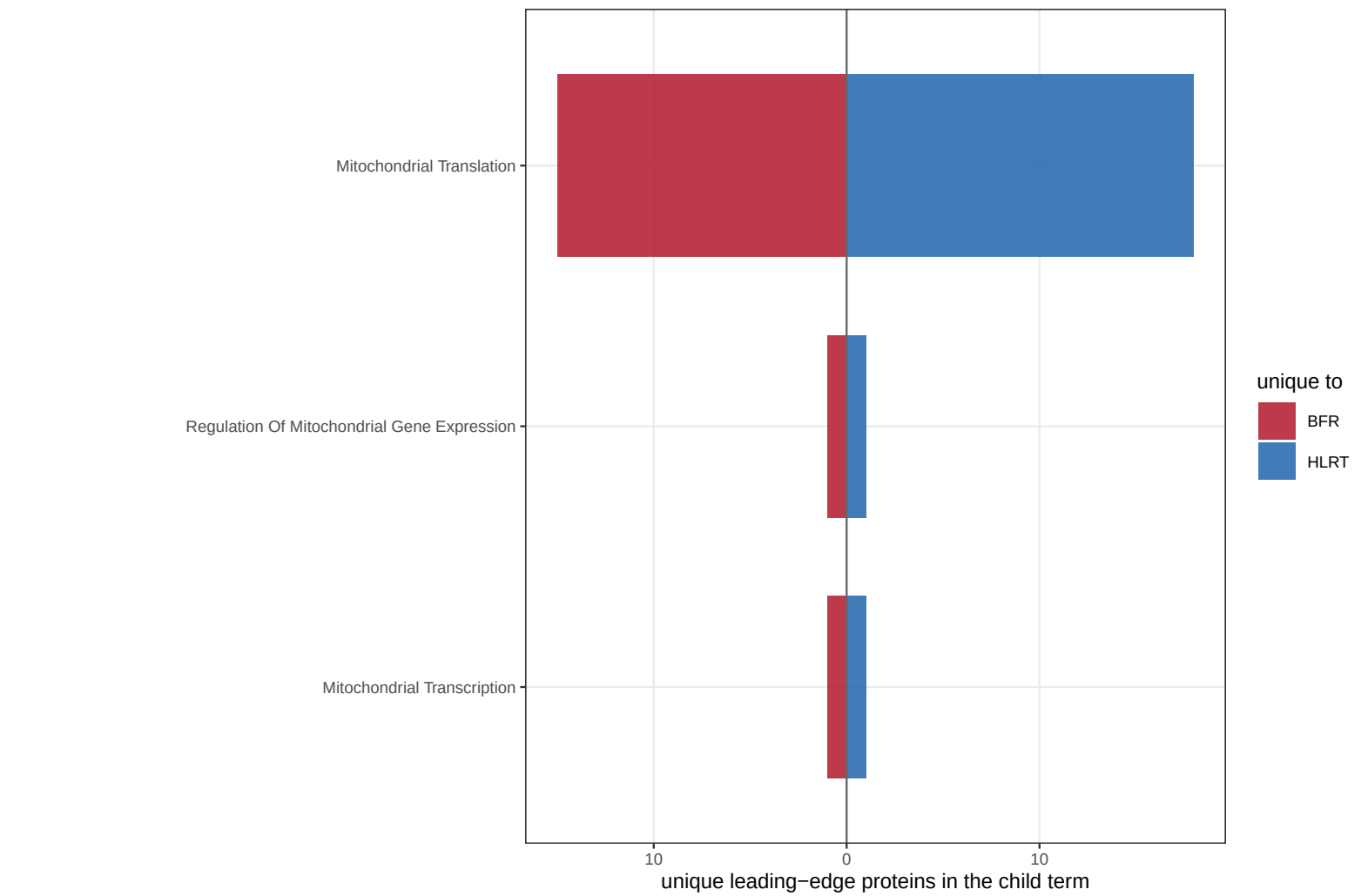
S18. Mitochondrial Respiratory Chain Complex Assembly

only BFR 16, only HLRT 11, shared 29; divergence 0.12, permutation p 1 (BH 1)



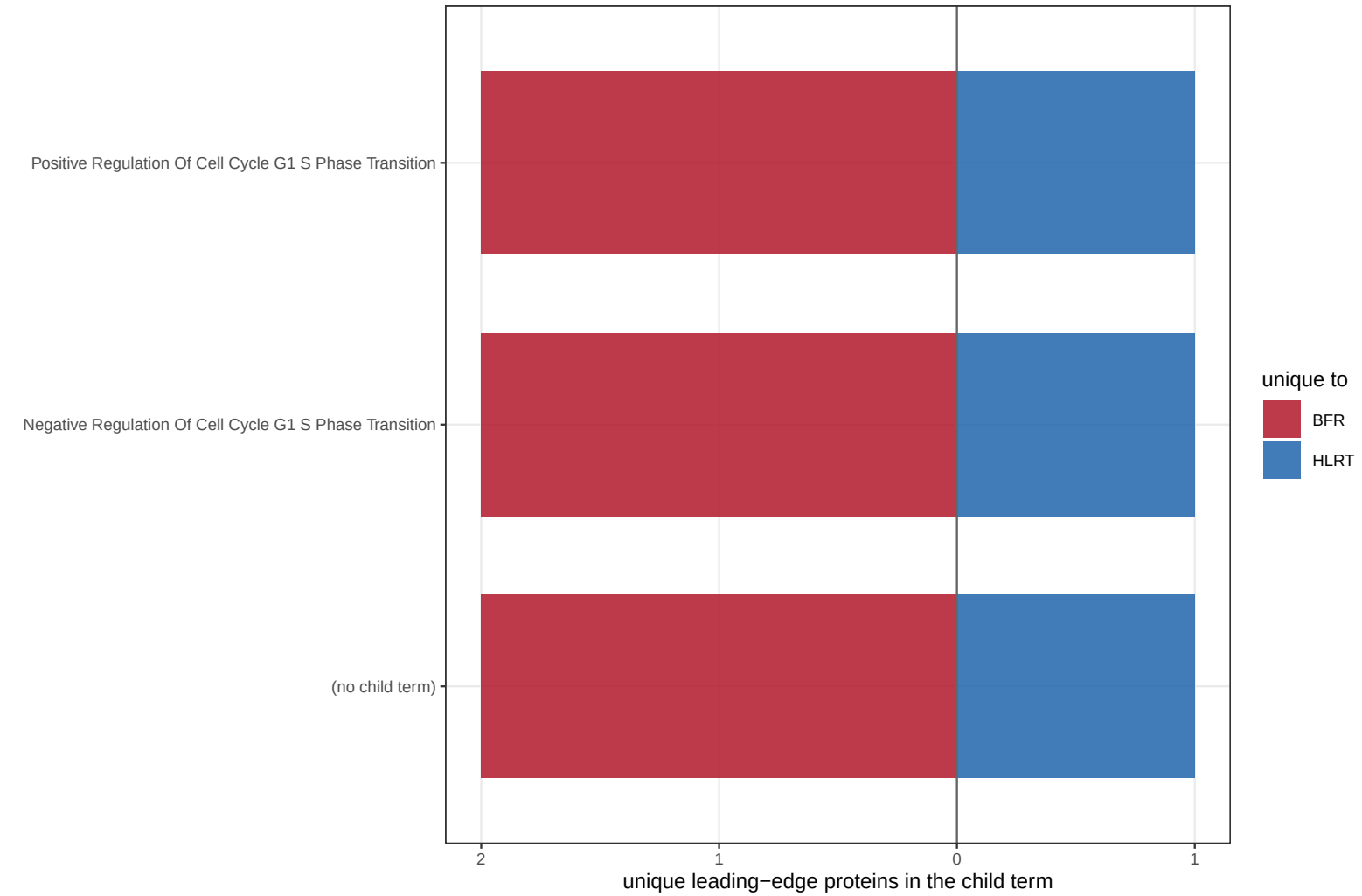
S18. Mitochondrial Gene Expression

only BFR 16, only HLRT 19, shared 51; divergence 0.03, permutation p 1 (BH 1)



S18. Regulation Of Cell Cycle G1 S Phase Transition

only BFR 6, only HLRT 3, shared 16; divergence 0.00, permutation p 1 (BH 1)



S18. Proton Transmembrane Transport

only BFR 13, only HLRT 15, shared 50; divergence 0.13, permutation p 1 (BH 1)

